Supplemental Material: Using identity calls to detect structure in acoustic datasets

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Supplemental Tables

Table S1. Atlantic/Mediterranean sperm whale coda datasets.

The eastern Caribbean subset contains 10,090 codas recorded between 2005 and 2019 by the Dominica Sperm Whale Project (for field and data collection methods, see Gero et al., 2016; Tønnesen et al., 2018) and 1,551 codas (recorded between 1981 and 1995) extracted from sperm whale master tapes in the Watkins Marine Mammal Sound Database (for field and data collection methods, see Moore et al., 1993; Watkins, 1985; Watkins & Moore, 1982). Eastern Caribbean codas were primarily recorded off Dominica (96.1%), but also off St. Lucia (1.8%), Canouan (1.2%), Bequia (0.6%), and Guadeloupe (0.3%). Coda extraction methods for the Watkins Marine Mammal Sound Database (Sayigh et al., 2016) are summarized in Method S1. Field and data collection methods for the Balearic Islands dataset are detailed in Pirotta et al (2011) and Rendell et al. (2014). The Gulf of Mexico and Atlantic Panama codas were recorded on single days in 1992 and 1993, respectively (Weilgart & Whitehead, 1997).

Recording location	**Recording years**	**Number of codas (with 3-9 clicks)**	**Number of repertoires**
Eastern Caribbean	19: 81, 84, 87, 90, 94, 95 20: 05, 07–12, 14–16, 18, 19	11,641	66
Balearic Islands	20: 04–08, 13, 14, 17, 18	1,749	14
Panama	19: 93	313	1
Gulf of Mexico	19: 92	102	1
	Total:	**13,805**	**82**

Table S2. Pacific sperm whale coda datasets.

A majority of the Pacific codas (18,429) were recorded by our lab between 1985 and 2014 (for field and data collection methods, see Cantor et al., 2016; Rendell & Whitehead, 2003; Weilgart & Whitehead, 1997). Fifty-two codas were recorded in 1978 and extracted from a single sperm whale master tape in the Watkins Marine Mammal Sound Database (Sayigh et al., 2016; Method S1).

Recording location	**Recording years**	**Number of codas (with 3-9 clicks)**	**Number of repertoires**
Galápagos Islands	19: 78, 85, 87, 89, 91, 95, 99 20: 13, 14	10,108	70
Chile	19: 93 20: 00	5,688	14
Ecuador	19: 85, 91, 93	773	8
Peru	19: 93	657	6
Jarvis Island	19: 92	510	3
Baker Island	19: 92	269	1
Tonga	19: 92	151	1
Easter Island	19: 93	90	1
Panama	19: 92	191	1
New Zealand	19: 93	44	1
	Total:	**18,481**	**106**

Table S3. Baseline dendrogram metrics for each dataset using the default parameters.

Dataset	**Number of call types**	**Number of ID call types**	**Number of ID clades**	**Tree ID call proportion**	**Average similarity of trial dendrograms to baseline (± S.D.)**[1]
Sperm whales (Atlantic/ Mediterranean)	56	10	3	0.460	0.992 ± 0.017
Sperm whales (Pacific)	63	19	5	0.378	0.972 ± 0.037
Wrens	9	4	2	0.503	0.989 ± 0.032
Crickets	4	4	2	0.986	1.000 ± 0.000

[1] The average similarity of trial dendrograms to the baseline dendrogram was only calculated for trials that produced a dendrogram (i.e. those trials for which identity calls and identity clades were detected). The number of trials that did not produce a dendrogram for each dataset are: 0 (Atlantic/Mediterranean sperm whales), 0 (Pacific sperm whales), 3 (wrens), and 2 (crickets).

Table S4. Atlantic/Mediterranean sperm whale option/parameter trial results.

For each trial, one option/parameter value (the written parameter in each row) was varied while the others were kept at the default/baseline values (top row). Trials 18 and 19 had identical parameters to the baseline.

Trial	**Initialization**	**Criterion**	**Linkage**	***critfact***	***minrep***	**Number of call types**	**Number of ID call types**	**Tree ID call proportion**	**Number of ID clades**	**Similarity to baseline**
Baseline	*k*-means	BIC	average	14	6	56	10	0.460	3	1.00
1	random.post					60	10	0.463	3	0.996
2	random.clas					59	9	0.453	3	0.996
3		AICc				60	13	0.470	3	1.00
4		AIC				62	13	0.460	3	0.996
5		ICL				55	9	0.491	3	0.982
6			single			56	9	0.592	3	0.929
7			complete			56	7	0.677	2	0.964
8				6		56	14	0.712	3	1.00
9				10		56	11	0.484	3	1.00
10				18		56	9	0.453	3	1.00
11				22		56	8	0.451	3	1.00
12				26		56	7	0.398	3	1.00
13					4	56	10	0.460	3	1.00
14					8	56	7	0.671	2	0.996
15					10	56	7	0.671	2	0.996
16					12	56	7	0.671	2	0.996
17					14	56	7	0.672	2	0.996
18						56	9	0.460	3	0.996
19						57	11	0.471	3	0.996

Table S5. Pacific sperm whale option/parameter trial results.

For each trial, one option/parameter value (the written parameter in each row) was varied while the others were kept at the default/baseline values (top row). Trials 18 and 19 had identical parameters to the baseline.

Trial	**Initialization**	**Criterion**	**Linkage**	***critfact***	***minrep***	**Number of call types**	**Number of ID call types**	**Tree ID call proportion**	**Number of ID clades**	**Similarity to baseline**
Baseline	*k*-means	BIC	average	14	6	63	19	0.378	5	1.00
1	random.post					65	20	0.288	6	0.962
2	random.clas					63	21	0.399	5	0.991
3		AICc				65	19	0.374	6	0.958
4		AIC				69	18	0.381	5	0.988
5		ICL				61	16	0.329	5	0.962
6			single			63	18	0.308	5	0.886
7			complete			63	13	0.319	5	0.872
8				6		63	26	0.462	7	0.939
9				10		63	22	0.392	6	0.960
10				18		63	17	0.327	5	1.00
11				22		63	17	0.327	5	1.00
12				26		63	14	0.264	5	1.00
13					4	63	19	0.378	5	1.00
14					8	63	16	0.374	4	1.00
15					10	63	16	0.374	4	1.00
16					12	63	17	0.389	4	1.00
17					14	63	14	0.366	3	0.986
18						64	18	0.361	5	0.983
19						64	20	0.311	6	0.957

Table S6. Wren option/parameter trial results.

For each trial, one option/parameter value (the written parameter in each row) was varied while the others were kept at the default/baseline values (top row). Trials 18 and 19 had identical parameters to the baseline.

Trial	**Initialization**	**Criterion**	**Linkage**	***critfact***	***minrep***	**Number of call types**	**Number of ID call types**	**Tree ID call proportion**	**Number of ID clades**	**Similarity to baseline**
Baseline	*k*-means	AICc	average	14	6	9	4	0.503	2	1.00
1	random.post					6	2	0.307	2	0.905
2	random.clas					6	0	0.000	0	NA
3		AIC				10	3	0.393	2	1.00
4		BIC				3	0	0.000	0	NA
5		ICL				2	0	0.000	0	NA
6			single			9	4	0.503	2	1.00
7			complete			9	1	0.624	1	0.905
8				6		9	4	0.519	2	1.00
9				10		9	3	0.440	2	1.00
10				18		9	4	0.503	2	1.00
11				22		9	3	0.305	2	1.00
12				26		9	2	0.204	1	1.00
13					3	9	3	0.44	2	1.00
14					5	9	3	0.44	2	1.00
15					7	9	4	0.503	2	1.00
16					9	9	4	0.503	2	1.00
17					11	9	4	0.503	2	1.00
18						8	3	0.459	2	1.00
19						9	4	0.462	2	1.00

Table S7. Cricket option/parameter trial results.

For each trial, one option/parameter value (the written parameter in each row) was varied while the others were kept at the default/baseline values (see top row). Trials 18 and 19 had identical parameters to the baseline.

Trial	**Initialization**	**Criterion**	**Linkage**	***critfact***	***minrep***	**Number of call types**	**Number of ID call types**	**Tree ID call proportion**	**Number of ID clades**	**Similarity to baseline**
Baseline	*k*-means	AICc	average	14	6	4	4	0.986	2	1.00
1	random.post					3	2	0.793	2	1.00
2	random.clas					3	2	0.77	2	1.00
3		AIC				5	3	0.891	2	1.00
4		BIC				2	1	0.901	1	1.00
5		ICL				2	1	0.901	1	1.00
6			single			4	4	0.986	2	1.00
7			complete			4	4	0.986	2	1.00
8				6		4	4	0.986	2	1.00
9				10		4	4	0.986	2	1.00
10				18		4	4	0.986	2	1.00
11				22		4	4	0.986	2	1.00
12				26		4	4	0.986	2	1.00
13					3	4	4	0.986	2	1.00
14					5	4	4	0.986	2	1.00
15					7	4	4	0.986	2	1.00
16					9	4	0	0.000	0	NA
17					11	4	0	0.000	0	NA
18						4	4	0.985	2	1.00
19						4	4	0.985	2	1.00

Table S8. Identity clades for baseline Atlantic/Mediterranean sperm whale dendrogram.

Identity clade name	**Number of repertoires**	**Average repertoire/clade correlation (± S.D.)[2]**	**Number of identity calls**	**Clade identity call proportion**
EC1	59	0.652 ± 0.178	2	0.373
Mediterranean	14	0.849 ± 0.097	3	0.699
EC2	7	0.917 ± 0.161	5	0.717

[2]Calculated as the average correlation of each repertoire's type usage with the clade's type usage (i.e. the median type usage of all repertoires in the clade)

Table S9. Identity clades for baseline Pacific sperm whale dendrogram.

Identity clade name	**Number of repertoires**	**Average repertoire/clade correlation (± S.D.)[3]**	**Number of identity calls**	**Clade identity call proportion**
Putative fifth clan	6	0.849 ± 0.088	4	0.563
Four-Plus	18	0.505 ± 0.286	2	0.216
Short	32	0.495 ± 0.236	3	0.321
Plus-One	13	0.872 ± 0.114	3	0.545
Regular	37	0.781 ± 0.237	7	0.419

[3]Calculated as the average correlation of each repertoire's type usage with the clade's type usage (i.e. the median type usage of all repertoires in the clade)

Table S10. Identity clades for baseline wren dendrogram.

Identity clade name	**Number of repertoires**	**Average repertoire/clade correlation (± S.D.)**[4]	**Number of identity calls**	**Clade identity call proportion**
H. l. hilaris	12	0.844 ± 0.228	1	0.548
H. l. leucophrys	29	0.632 ± 0.288	3	0.485

[4]Calculated as the average correlation of each repertoire's type usage with the clade's type usage (i.e. the median type usage of all repertoires in the clade)

Table S11. Identity clades for baseline cricket dendrogram.

Identity clade name	**Number of repertoires**	**Average repertoire/clade correlation (± S.D.)[5]**	**Number of identity calls**	**Clade identity call proportion**
T. oceanicus	8	0.991 ± 0.013	2	0.993
T. commodus	8	0.987 ± 0.021	2	0.978

[5]Calculated as the average correlation of each repertoire's type usage with the clade's type usage (i.e. the median type usage of all repertoires in the clade)

Table S12. Atlantic/Mediterranean sperm whale coda types from IDcall and past work.

Method S3 contains some key details for interpreting this table. Type names refer to the overarching rhythmic pattern of clicks in each coda regardless of duration, which is why multiple numeric codes can be linked to one type. 'R' stands for 'Regular' (i.e. all inter-click intervals (ICIs) are approximately equal), '+' denotes an extended pause between clicks, 'D' stands for 'Decreasing' (i.e. ICIs become shorter throughout the coda), and 'I' stands for 'Increasing' (i.e. ICIs become longer throughout the coda). To name the coda types detected by IDcall, the centroid ICI values for each type were divided by the smallest ICI and the resultant values were inspected to determine the overall rhythmic pattern of the coda. Values within ~0.3 of each other were considered the same for naming purposes. Bolded numeric codes are identity codas and are coloured by clan (see Fig. 1 for colours). The final four columns show coda types detected in past work using different methods that *rhythmically* match coda types found using IDcall (see Method S3 and the publications themselves for additional details on how the coda types were delineated and named). The final row shows the number of coda types from each study that rhythmically match coda types in the present study over the total number of 3- to 9-click coda types in each study.

Number of clicks	**Numeric code(s)**	**Type name**	**Gero, Bøttcher, et al. (2016)**	**Gero, Whitehead, et al. (2016)**	**Pavan et al. (2000)**	**Drouot et al. (2004)[6]**
3	31, 34	3R	3R	3R		3reg
	33	2+1				
	32	1+2	3D	3D		
4	**41**, **42**, 46	3+1			3+1	3+1
	43	1+3	$1+3_1$, $1+3_2$	$1+3_1$, $1+3_2$		
	44, 47	4R	4R1, 4R2	4R1, 4R2		4reg
	45	4D	4D	4D		
5	**53**, 58, 59	1+1+3	1+1+3	1+1+3		

[6] We compared coda types detected using IDcall to those recorded by Drouot et al. (2004) in the southwestern basin of the Mediterranean Sea since the codas included in the present study were recorded off the Balearic Islands. The Drouot et al. (2004) study also includes codas recorded in the Tyrrhenian and Ionian Seas.

	51, 54, 56, 57	5R	5R1, 5R2, 5R3	5R1, 5R2, 5R3		5reg
	52	4+1			4+1	
	55	5D				
6	61, 68, 69	6R	6R			6reg
	62, 63, 64	1+1+4		6D		
	65	4+1+1				
	66	2+1+1+1+1				
	67, 610	5+1	6I	6I		
7	71, 77	7R	7R			7reg
	72	1+1+5				
	73	7D	7D	7D		
	74	5+1++1				
	75	2+1+1+1+1+1				
	76	6+1	7I	7I		
8	81	6+1+1				
	82, 87, 89	8R				
	83	4+4				
	84, 810	7+1	8I	8I		
	85	1+7		8D		
	86, 88	1+1+6				
9	91	6+1+1+1				
	92	1++1++7				
	93, 94	9R	9R			
	95, 96, 99	8+1	9I	9I		
	97	1+8		9D		
	98	9I				
Type matches/total types found:			**19/20**[7]	**19/20**[8]	**2/2**	**6/7**[9]

[7] Missing type: 2+3
[8] Missing type: 2+3
[9] Missing type: 3++1. There were additional coda types with "undefined patterns" (3var, 4var, 5var, 6var) detected in the southwestern basin of the Mediterranean Sea, but we were unable to determine if these types were also detected by IDcall because the type patterns/durations were not described in Drouot et al. (2004). Those types are not included in the total coda type count.

Table S13. Pacific sperm whale coda types from IDcall and past work.

Method S3 contains some key details for interpreting this table. Type names refer to the overarching rhythmic pattern of clicks in each coda regardless of duration, which is why multiple numeric codes can be linked to one type. ‘R’ stands for ‘Regular’ '(i.e. all inter-click intervals (ICIs) are approximately equal), ‘+’ denotes an extended pause between clicks, ‘D’ stands for ‘Decreasing’ (i.e. ICIs become shorter throughout the coda), and ‘I’ stands for ‘Increasing’ (i.e. ICIs become longer throughout the coda). To name the coda types detected by IDcall, the centroid ICI values for each type were divided by the smallest ICI and the resultant values were inspected to determine the overall rhythmic pattern of the coda. Values within ~0.3 of each other were considered the same for naming purposes. Bolded numeric codes are identity codas and are coloured by clan (see Fig. 2 for colours). The final three columns show coda types detected in past work using different methods that *rhythmically* match coda types found using IDcall (see Method S3 and the publications themselves for additional details on how the coda types were delineated and named). The final row shows the number of coda types from each study that rhythmically match coda types in the present study over the total number of 3- to 9-click coda types in each study.

Number of clicks	**Numeric code(s)**	**Type name**	**Weilgart & Whitehead (1997)**	**Rendell & Whitehead (2003)**	**Cantor et al. (2016)**
3	32, 33, 35, 38	2+1	3a, 2+1	2+1	2+1
	34, **37**	1+2	1+2	1+2	1+2
	31, 36, 39	3R	3R, 3b	3R	3R
4	46	2+1+1			
	47	4I	4L		
	42, 49	3+1	3+1	3+1	3+1
	41, 43	4R	4R	4R	4R
	44, **48**	1+1+2		2+2	
	45	1+3	4Var example	1+3	
	410	2++1+1			
5	**55**	4+1	4+1	4+1	4+1

	52, 53, 54, **58**, 59	5R	5R	5R	5R
	56	5I			
	51, 57	3+1+1			
6	61	2+1+1+1+1	2+4		2+4
	62, **64**	4+1++1	4+1++++1	4+1+1	4+1+1
	63, **69**	5+1	5+1, 6Var example	5+1	5+1
	65, 66, 67	6R	6R	6R	6R1, 6R2
	68, 610	6I			6I
7	**74**, 76, 79	7R	7R, 7Var example	7R	7R1, 7R2
	71, 78	7I			
	72	1+6		1+6	
	73	4+1++1++1		4+3	
	75	6+1	6+1	6+1	
	77	5+1++1	5+1++1	5+2A, 5+2B	
	710	3+1+3			
8	**82**, **84**, 86, **88**	8R	8R, 8L	8RA, 8RB	8R1, 8R2
	81, 83, **85**	8I	8Var example	8S	
	87	6+1+1		6+1+1	
9	**91**, 94, 95, 97	9I			
	92, **93**, 96	9R			9R1, 9R2, 9R3
Type matches/total types found:			**18/22[10]**	**25/33[11]**	**18/20[12]**

[10] 'Var' coda types are not included in the total coda type count because they contain all coda type clusters that contained less than 50 codas, and likely represent more than one type. Missing types: 3++1, 4++1, 2+1+1+1, and 9.
[11] Rendell & Whitehead (2003) did not analyze 9-click codas. Missing types: 4A, 3++1, 4++1, 3+4, 7+1, 8A, and 7++1.
[12] Missing types: 1+2+1, 1+3+1

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Supplemental Figures

Figure S1. Map of sperm whale recording locations.

Points on the map show the approximate location of recordings. Pacific abbreviations are: BAK = Baker Island, CHI = Chile, EAS = Easter Island, ECU = Ecuador, GAL = Galápagos Islands, JAR = Jarvis Island, NEW = New Zealand, PacPAN = Pacific coast of Panama, PER = Peru, and TON = Tonga. Atlantic/Mediterranean abbreviations are: AtPAN = Atlantic coast of Panama, BAL = Balearic Islands, CAR = eastern Caribbean, and GOM = Gulf of Mexico.

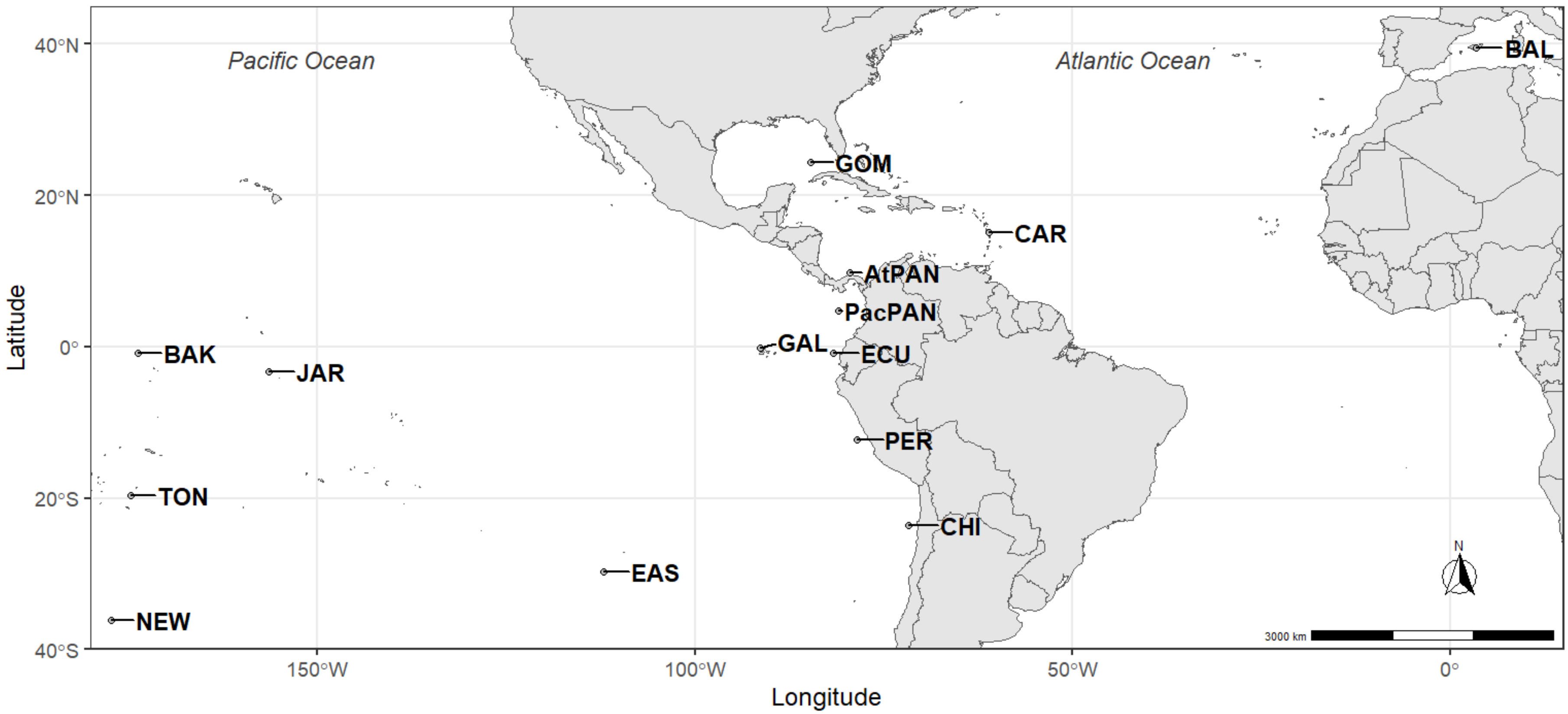

Figure S2. Baseline dendrogram with clans and all coda types for Atlantic/Mediterranean sperm whales.

Average linkage hierarchical clustering dendrogram (top) depicts similarity among sperm whale coda repertoires recorded in the Atlantic/Mediterranean (see Fig. 1 for identity clade colours). Heat map (bottom) depicts coda type usage (rows) for each repertoire (columns) (see Fig. S1 for abbreviations and Fig. 1 for detailed figure explanation).

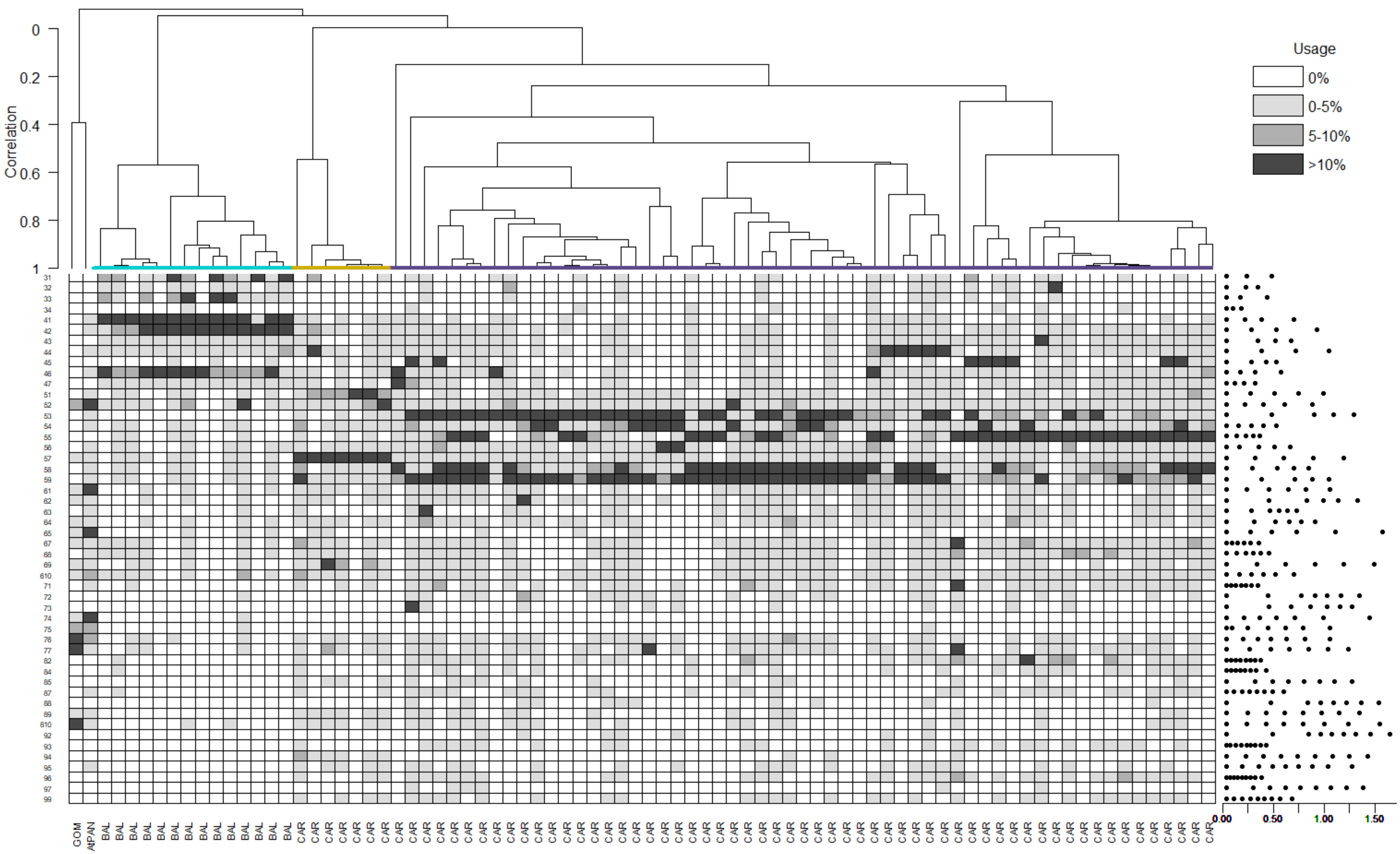
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Figure S3. Baseline dendrogram with clans and all coda types for Pacific sperm whales.

Average linkage hierarchical clustering dendrogram (top) depicts similarity among sperm whale coda repertoires recorded in the Pacific (see Fig. 2 for identity clade colours). Heat map (bottom) depicts coda type usage (rows) for each repertoire (columns) (see Fig. S1 for abbreviations and Fig. 2 for detailed figure explanation).

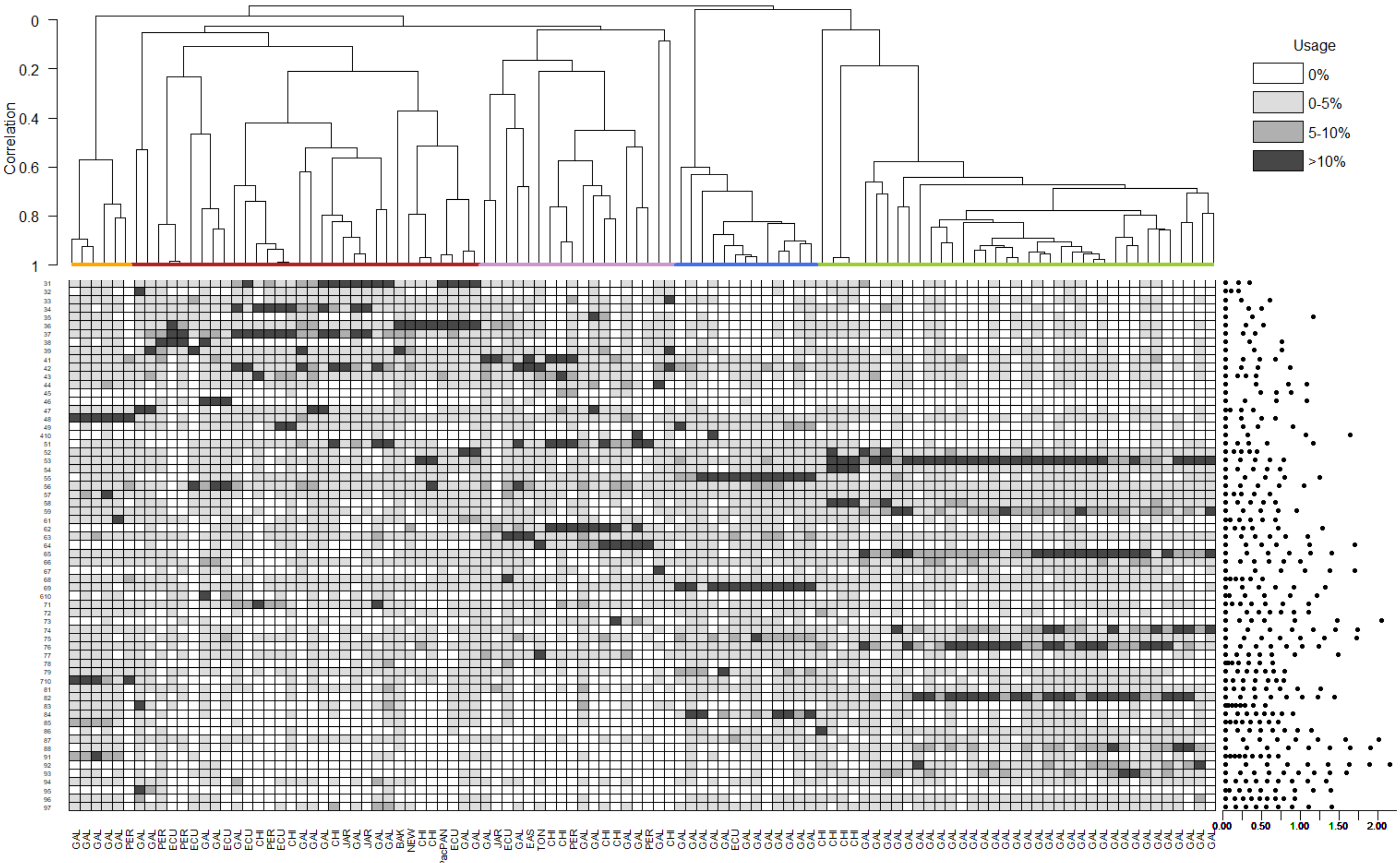
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Figure S4. Trial dendrogram with clans and identity codas for Pacific sperm whales showing the Short clan divided into two identity clades.

Average linkage hierarchical clustering dendrogram (top) depicts similarity among sperm whale coda repertoires recorded in the Pacific. Coloured identity clades correspond to a putative new clan (orange), Regular clan (green), Plus-One clan (blue), Four-Plus clan (pink), and two clans (in shades of red) that were considered the Short clan in the baseline dendrogram (Fig. 2). Heat map (bottom) depicts coda type usage (rows) for each repertoire (columns) (see Fig. S1 for abbreviations and Fig. 2 for detailed figure explanation). Dendrogram was created using default parameter values except for *critfact*, which was set to 10. Note that the identity coda type codes here (i.e. the heat map row labels) do not match those in Fig. 2/Fig. S3/Table S12.

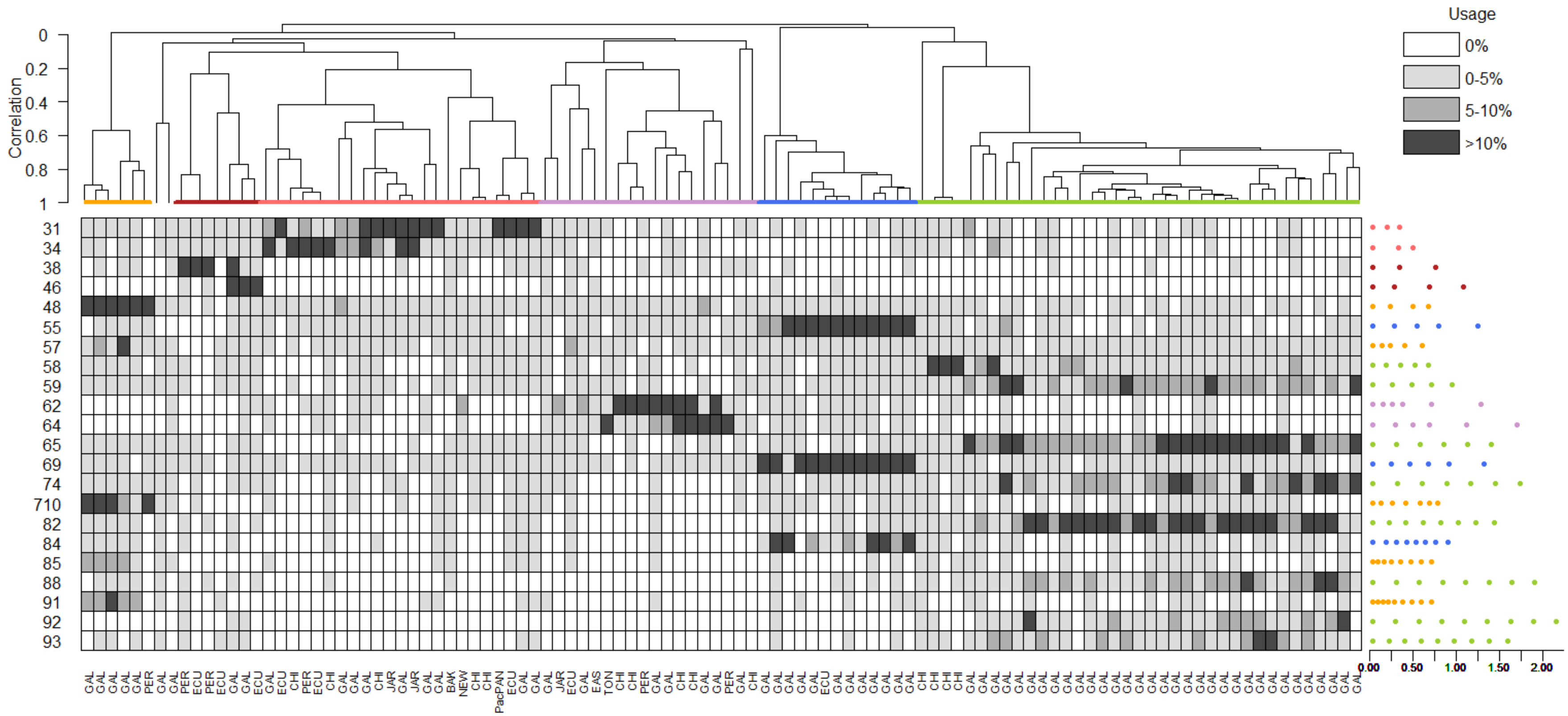
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Figure S5. Trial dendrogram with clans and identity codas for Pacific sperm whales showing the Short clan divided into three identity clades.

Average linkage hierarchical clustering dendrogram (top) depicts similarity among sperm whale coda repertoires recorded in the Pacific. Coloured identity clades correspond to a putative new clan (orange), Regular clan (green), Plus-One clan (blue), Four-Plus clan (pink), and three clans (in shades of red) that were considered the Short clan in the baseline dendrogram (Fig. 2). Heat map (bottom) depicts coda type usage (rows) for each repertoire (columns) (see Fig. S1 for abbreviations and Fig. 2 for detailed figure explanation). Dendrogram was created using default parameter values except for *critfact*, which was set to 6. Note that the identity coda type codes here (i.e. the heat map row labels) do not match those in Fig. 2/Fig. S3/Table S13.

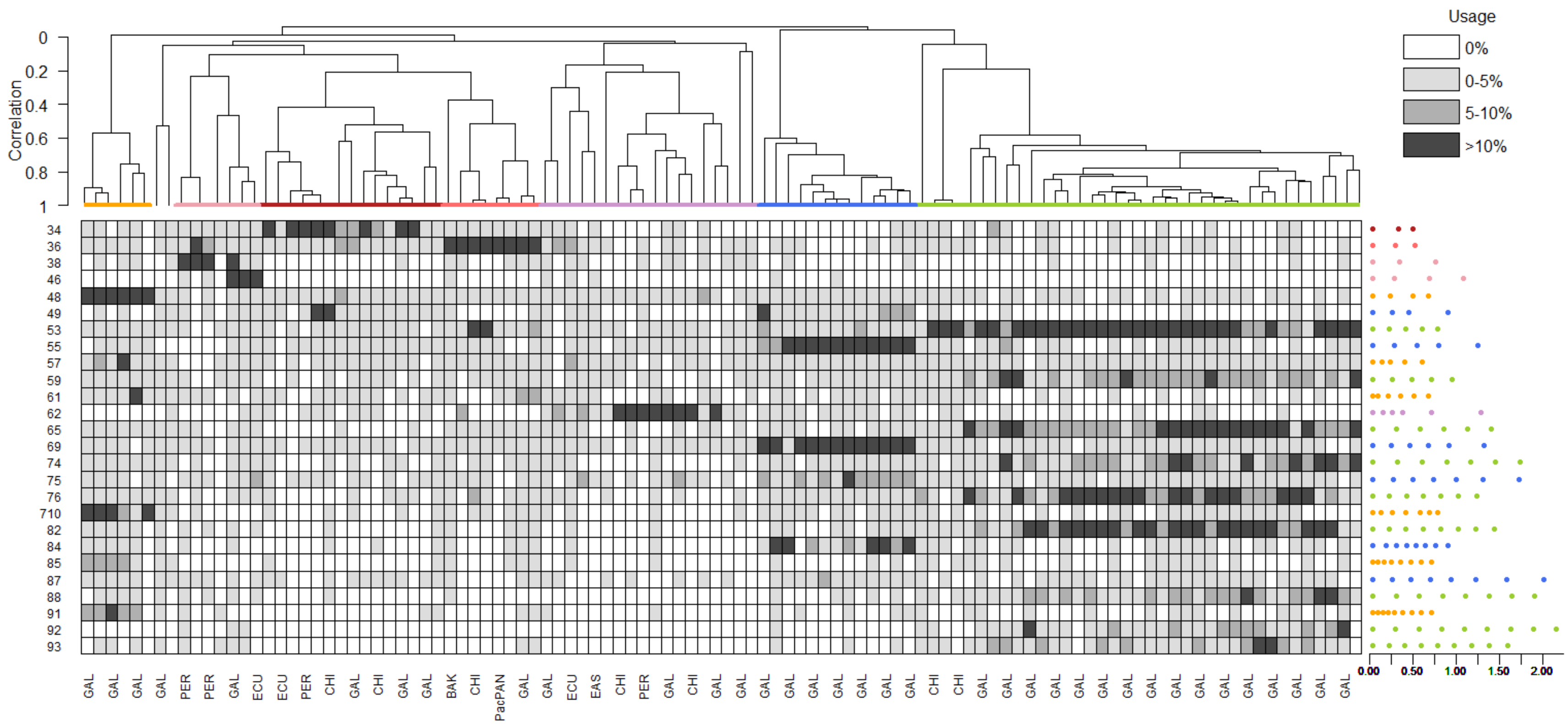
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Figure S6. Baseline dendrogram with subspecies and all song types for wrens.

Average linkage hierarchical clustering dendrogram (top) depicts similarity among repertoires of song frequency vectors of male wrens (see Fig. 3 for identity clade colours). Heat map (bottom) depicts song type usage (rows) for each male (columns) (see Fig. 3 for abbreviations and detailed figure explanation).

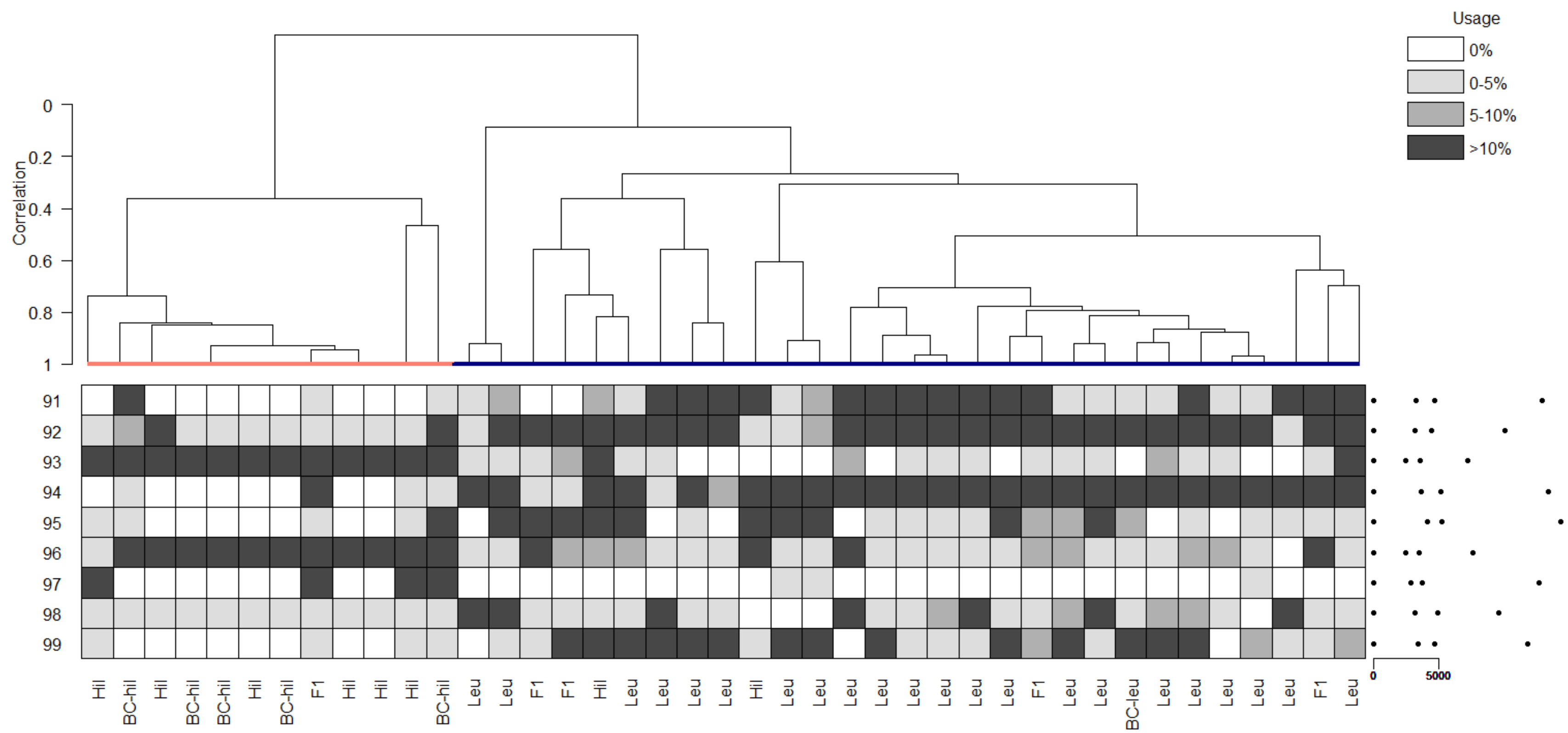

Figure S7. Baseline dendrogram with species and all song types for crickets.

Average linkage hierarchical clustering dendrogram (top) depicts similarity among repertoires of song interval vectors of male crickets from 16 sites (see Fig. 4 for identity clade colours). Heat map (bottom) depicts song type usage (rows) for each field site (columns) (see Fig. 4 for abbreviations and detailed figure explanation).

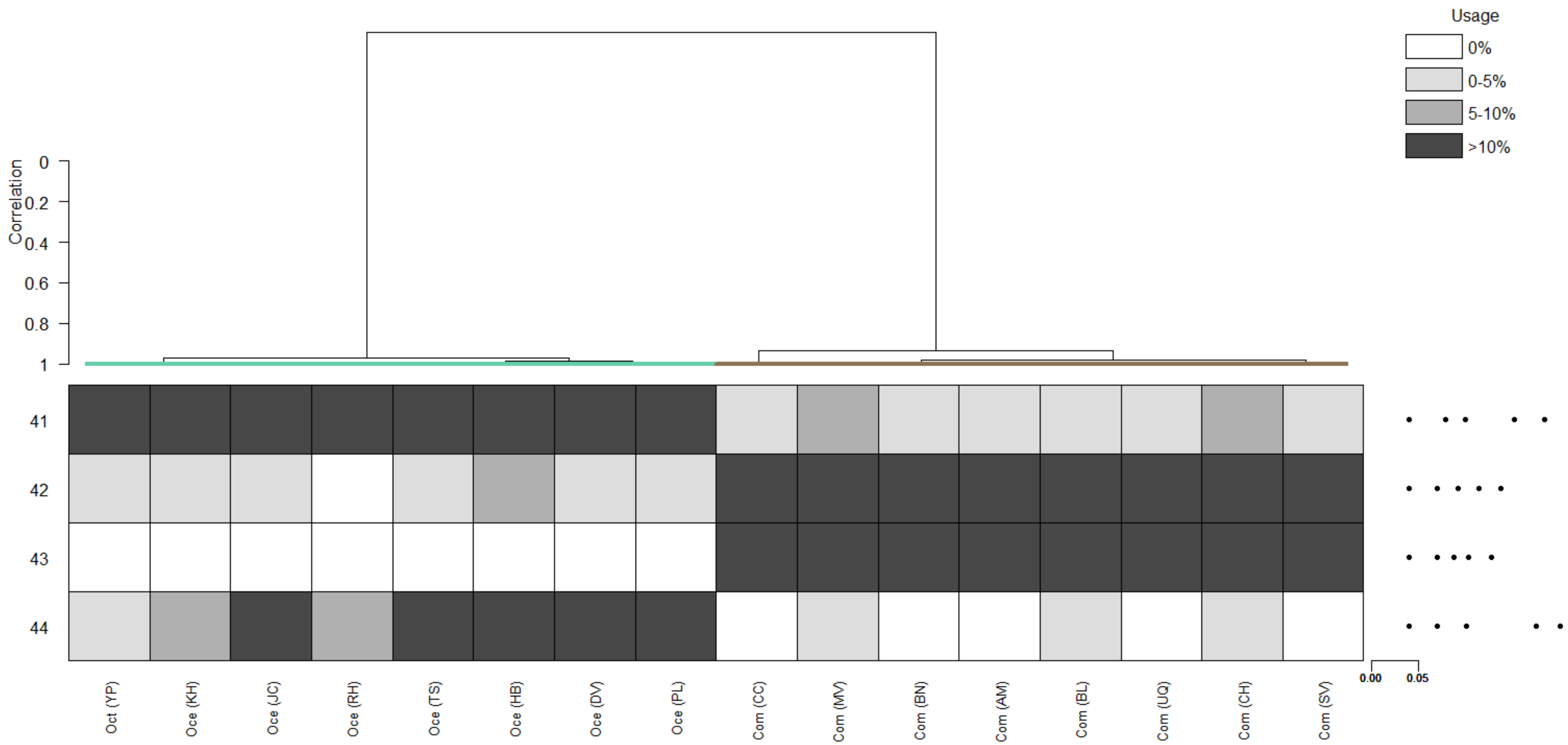

Figure S8. Dendrogram with clans and identity coda types for Atlantic/Mediterranean sperm whales using halved dataset and default values (critfact=14, minrep=6).

Average linkage hierarchical clustering dendrogram (top) depicts similarity among a randomly sampled subset of the sperm whale coda repertoires (n=41, half of the actual dataset) recorded in the Atlantic/Mediterranean (see Fig. 1 for identity clade colours). The EC2 clan (comprised of three repertoires) does not meet the *minrep* requirement and is not delineated. Heat map (bottom) depicts identity coda type usage (rows) for each repertoire (columns) (see Fig. S1 for abbreviations and Fig. 1 for detailed figure explanation). Note that the identity coda type codes here (i.e. the heat map row labels) do not match those in Fig. 1/Fig. S2/Table S12.

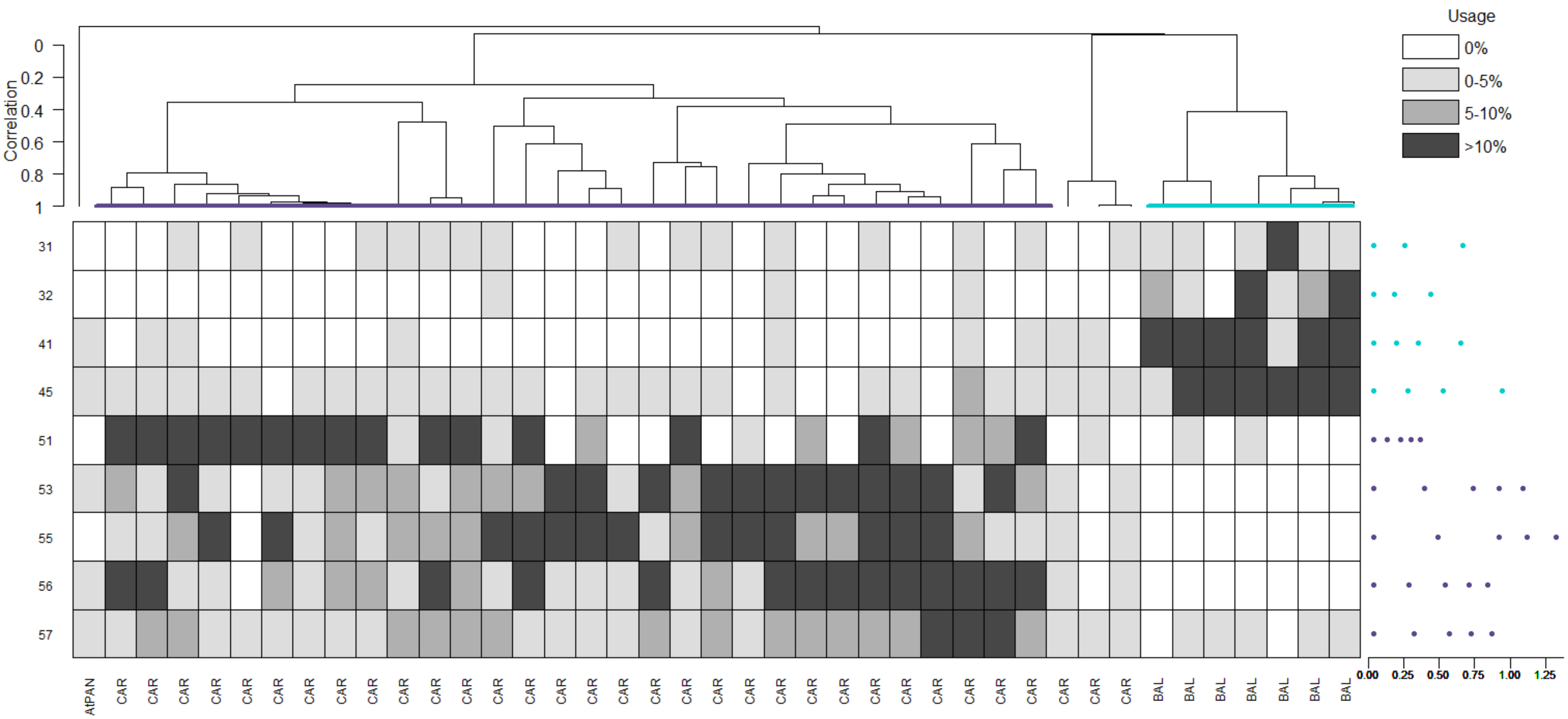

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Figure S9. Dendrogram with clans and identity coda types for Atlantic/Mediterranean sperm whales using halved dataset and modified values (critfact=14, minrep=3).

Average linkage hierarchical clustering dendrogram (top) depicts similarity among a randomly sampled subset of the sperm whale coda repertoires (n=41, half of the actual dataset) recorded in the Atlantic/Mediterranean (see Fig. 1 for identity clade colours). All three clans are delineated. Heat map (bottom) depicts identity coda type usage (rows) for each repertoire (columns) (see Fig. S1 for abbreviations and Fig. 1 for detailed figure explanation). Note that the identity coda type codes here (i.e. the heat map row labels) do not match those in Fig. 1/Fig. S2/Table S12.

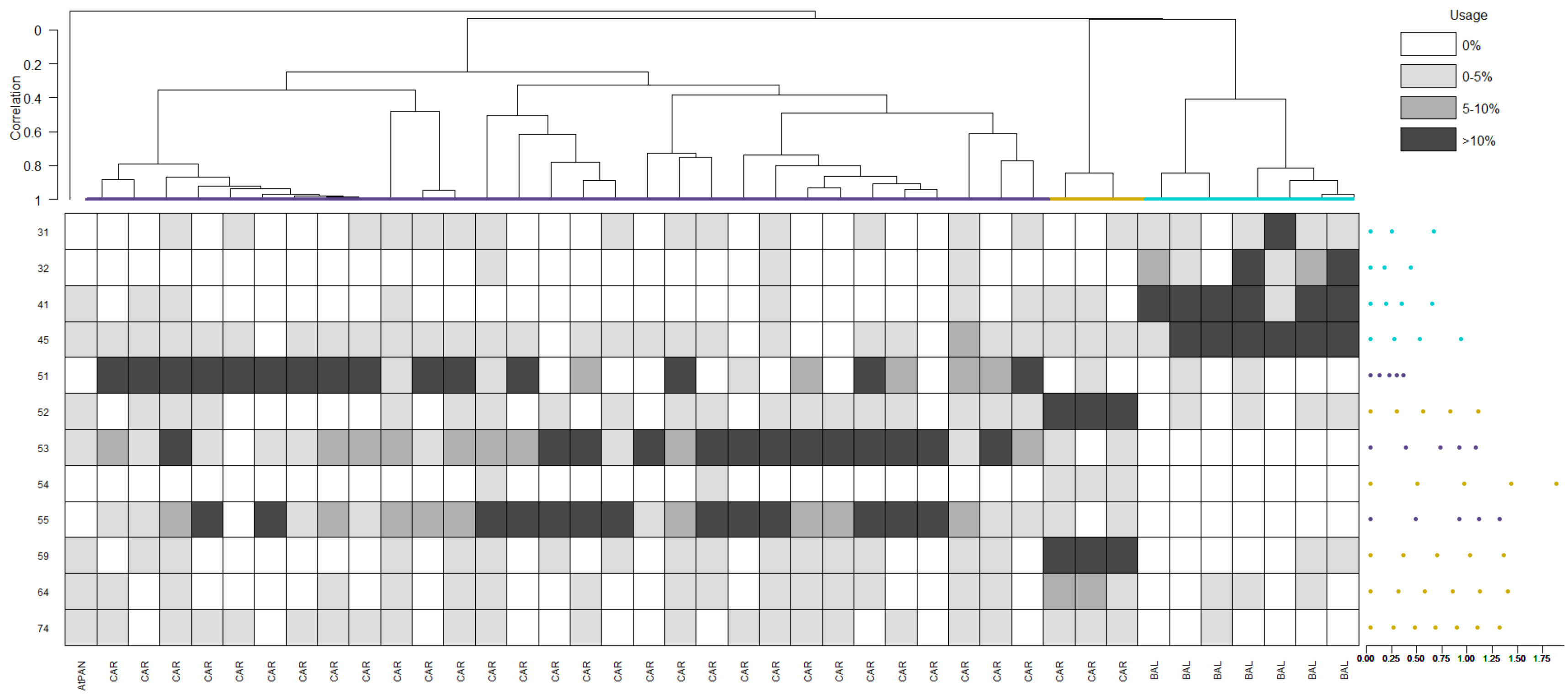
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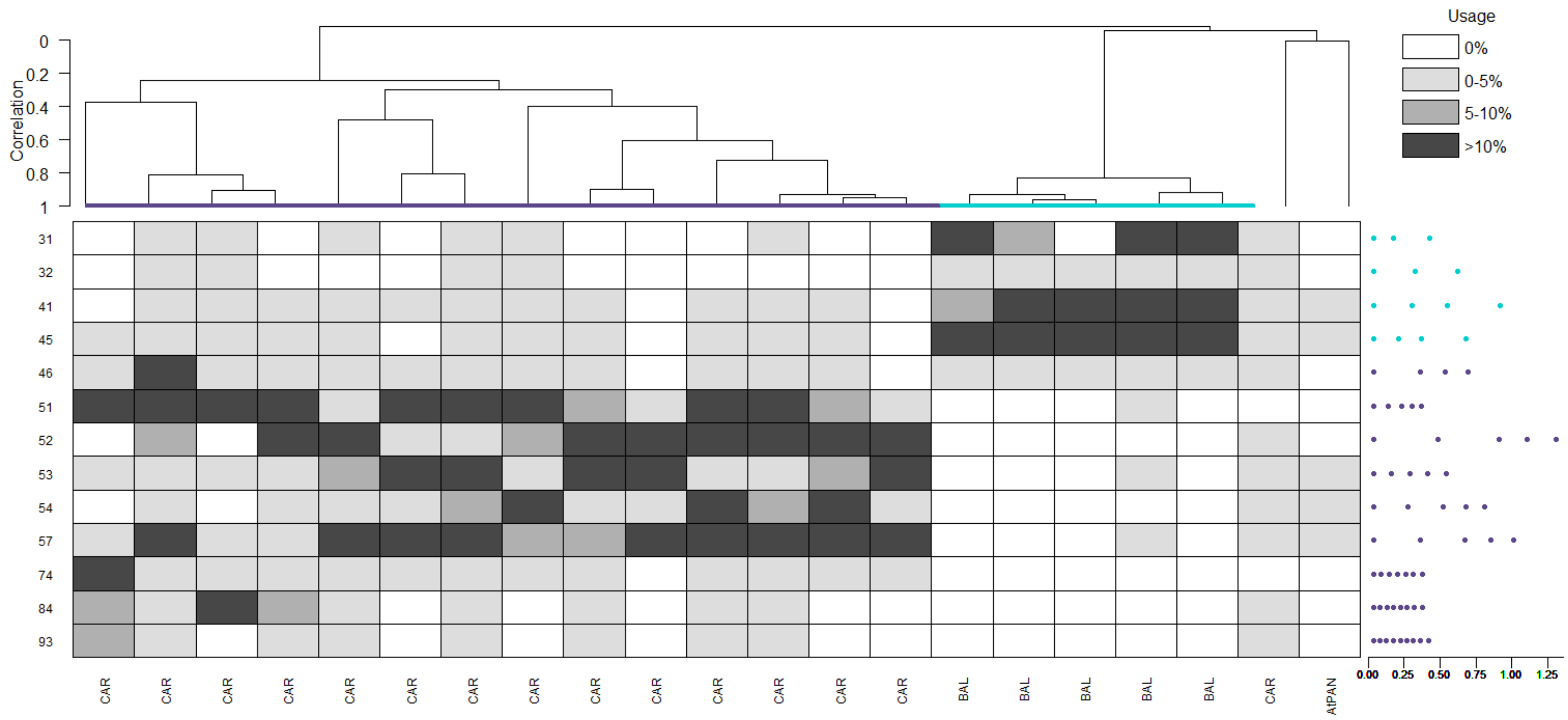
Figure S10. Dendrogram with clans and identity coda types for Atlantic/Mediterranean sperm whales using quartered dataset and modified values (critfact=14, minrep=5).

Average linkage hierarchical clustering dendrogram (top) depicts similarity among a randomly sampled subset of the sperm whale coda repertoires (n=21, approximately a quarter of the actual dataset) recorded in the Atlantic/Mediterranean (see Fig. 1 for identity clade colours). The EC2 clan (represented by just one ‘CAR’ repertoire, next to the ‘AtPan’ repertoire) does not meet the *minrep* requirement and is not delineated. Heat map (bottom) depicts identity coda type usage (rows) for each repertoire (columns) (see Fig. S1 for abbreviations and Fig. 1 for detailed figure explanation). Note that the identity coda type codes here (i.e. the heat map row labels) do not match those in Fig. 1/Fig. S2/Table S12.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Supplemental Methods

Method S1: Extracting sperm whale codas from the Watkins Marine Mammal Sound Database

The Watkins Marine Mammal Sound Database (Sayigh et al., 2016) has 308 sperm whale master tapes. One is labelled as a Galápagos Islands recording and 53 are labelled as eastern Caribbean recordings. The Galápagos recording and a majority of the eastern Caribbean recordings cannot be automatically opened due to errors that occurred when the recordings were converted from the original .KAY file format to .wav files (personal communication, Dr. Laela Sayigh, Woods Hole Oceanographic Institution). However, the files can still be opened by importing them into Audacity (version 2.3.0) as raw data and specifying the file sampling rate. The digitization parameters for each master tape .KAY file, including sampling rate, are binarily encoded in the file header. We accessed these headers using a custom MATLAB (version R2020a) script (accessible at https://osf.io/5fter/), which allowed us to correctly specify the sampling rate for each file and save it as a new, uncorrupted .wav file. All files were then audited for codas. Coda parameters (including the number of clicks and inter-click intervals; ICIs) were extracted using a custom software, called Coda Sorter, which was written by Kristian Beedholm (Marine Bioacoustics Lab, Aarhus University). Coda Sorter is implemented in LabView and run in MATLAB.

Method S2: Descriptions and processing of test datasets

Sperm whales

Due to inconsistent past marking of extremely short and long codas, and because our goal was to identify clan-specific, frequently used identity codas, only codas with three to nine clicks were analysed. Codas with more than nine clicks made up, on average, only 2.40±4.21% of each location dataset included in this study. In regions where longer codas are common (e.g. Brazil, Amorim et al., 2020), this range can be increased.

All codas recorded from a single photo-identified group of sperm whales in a year were compiled into a single repertoire for all of the Pacific locations, the Balearic Islands, the Gulf of Mexico, and the Atlantic coast of Panama (for photo-identification methods and definitions, see Cantor et al., 2016; Pirotta et al., 2011; Rendell et al., 2014; Rendell & Whitehead, 2003). For the eastern Caribbean data from Dominica, where the long-term sperm whale social structure is well-documented (Gero et al., 2014), all codas recorded from a single known social unit of sperm whales within a year were compiled into one repertoire. If the identity of recorded whales was unknown or multiple groups/units were present, all of the codas recorded on a given day were combined into a single repertoire to reduce autocorrelation and minimize the possibility of well-sampled groups/units being designated as clans due to group- or unit-specific codas (Antunes et al., 2011; Gero, Whitehead, et al., 2016; Oliveira et al., 2016). Only repertoires with at least 25 codas were analysed. For each sperm whale repertoire, it is unknown which individuals within the group produced each coda or how many individuals were vocalizing at any given time. Examples of sperm whale coda spectrograms can be viewed in Figures 1, 2, and 3 of Watkins & Schevill (1977).

Grey-breasted wood-wrens

Field and data collection methods are detailed in the original study (Halfwerk et al., 2016). The dataset contains acoustic metrics for 471 averaged song types from 66 male wrens belonging to two subspecies (*Henicorhina leucophrys hilaris* and *Henicorhina leucophrys leucophrys)*. Based on genotyping, 22 males were parental *H. l. hilaris*, 29 were parental *H. l. leucophrys*, six were first-generation (F1) hybrids, and nine were second-generation males that were backcrossed between an F1 hybrid and either *H. l. hilaris* (n=4) or *H. l. leucophrys* (n=2). Each song type has measurements for 22 temporal and spectral acoustic characteristics. Previous work found that averaged note peak frequency (F_{peak}), minimum song frequency (F_{min}), and maximum song frequency (F_{max}) differed significantly between male *H. l. hilaris* and *H. l. leucophrys* song (Dingle et al., 2008). For each song type, we created a frequency vector of F_{peak}, F_{min}, and F_{max} and logged the vectors prior to clustering to account for differences in scale of the frequency measures. Only males that had at least five song types were included in our analysis, which reduced the dataset to 396 song types and 41 males (*H. l. hilaris*=8, *H. l. leucophrys*=22, F1=5, *H. l. hilaris*/F1 backcross=5, *H. l. leucophrys*/F1 backcross=1). Each repertoire thus contains the songs from a single individual. Examples of song spectrograms can be viewed in Figure 4 of Halfwerk et al. (2016).

Australian field crickets

Field and data collection methods are detailed in the original study (Moran et al., 2020). The dataset is comprised of calling song data for male *Teleogryllus* crickets (127 *T. commodus*; 131 *T. oceanicus*). These lab-reared, first-generation crickets were derived from wild-caught individuals from 16 field sites. Each cricket's song was recorded five times and 13 song traits were extracted. The 258 songs in the dataset represent the mean of each cricket's five songs. For each song, we created an interval vector that included four of the 13 song traits: chirp pulse length, chirp interpulse interval, chirp-trill interval, and trill pulse length. We chose these traits because they were measured in the same units (seconds), had similar ranges of variation, and looked like promising candidates for species discrimination (see Figure S3 in Moran et al., 2020). Because we only had access to one averaged set of song measurements per male, we used parental field site as the repertoire grouping variable instead of individual. Each repertoire thus contains the songs from multiple individuals. Despite species sympatry at four of the 16 sites, all first-generation males derived from sympatric sites were *T. oceanicus* (Moran et al., 2020). For our purposes, eight sites were thus unambiguously *T. commodus* (sites SV, UQ, CH, BL, MV, CC, BN, AM) and the other eight were unambiguously *T. oceanicus* (KH, DV, JC, PL, YP, RH, TS, HB) (see Moran et al., 2020 for site details). Song schematics showing the various song traits can be viewed in Figure 2 of Moran et al. (2020).

Method S3: Validation of call categories

Sperm whales

A variety of approaches, each with their own assumptions and limitations, have been used to classify sperm whale codas into types and delineate repertoires into clans in the past. In the first report of sperm whale vocal clans, Rendell & Whitehead (2003) compared repertoires of codas using multivariate similarity measures and classified codas into categorical types using *k*-means clustering of standardized ICIs. Similar approaches were used to delineate sperm whale clans off Japan (Amano et al., 2014) and to look for individually distinctive features in codas recorded off Dominica (Antunes et al., 2011; Schulz et al., 2011). Due to initial emphasis on coda rhythm rather than tempo (Moore et al., 1993), many sperm whale studies standardized coda ICIs by total coda duration (e.g. Amano et al., 2014; Rendell & Whitehead, 2003; Weilgart & Whitehead, 1997). Additional research suggested that both rhythm and tempo are informative coda features (Antunes et al., 2011; Frantzis & Alexiadou, 2008; Gero, Whitehead, et al., 2016), and more recent work has used absolute coda ICIs (Amorim et al., 2020; Huijser et al., 2020; Oliveira et al., 2016).

K-means clustering as a call classification strategy has several drawbacks, a significant one being that *k*—the number of clusters—must be assigned *a priori*. Techniques exist to help with this assignation (e.g. variance ratio criterion; Caliński & Harabasz, 1974) but do not always provide consistent, unambiguous results (Huijser et al., 2020; Rendell & Whitehead, 2003a) and the choice of *k* can be arbitrary or subjective. *K*-means clustering also forces the data into spherical, equal-sized Voronoi cells; is sensitive to initial conditions and outliers; and struggles at higher dimensions (Mumtaz & Duraiswamy, 2010; Oliveira et al., 2016).

More recently, Oliveira et al. (2016) used a combination of principal components analysis and observer classification to divide codas into categorical types. A drawback of this approach is that observer classification of codas can be time consuming, subjective, and inconsistent across observers (Rendell & Whitehead, 2003a). Other studies have used the OPTICS hierarchical clustering algorithm (Ankerst et al., 1999) to divide codas into categorical types (Cantor et al., 2016; Gero, Bøttcher, et al., 2016; Huijser et al., 2020). OPTICS has several advantages over *k*-means clustering, including that points can be designated as outliers rather than forced into clusters, and the number of clusters does not need to be specified *a priori*. However, other input values still need to be set (requiring sensitivity analyses to find initialization values) and significant portions of the data are often designated as outliers (as high as 93% and 70% for certain click lengths in Cantor et al., 2016 and Huijser et al., 2020, respectively), which makes OPTICS extremely conservative and limits the scope and types of questions that can be addressed.

Our ability to delineate clans in an "automated, objective, and meaningful way independent of the dataset" remains a challenging but essential research objective, especially as datasets scale up in size (Frantzis & Alexiadou, 2008). IDcall addresses some of these challenges by focusing on common and distinctive coda types. It takes advantage of two well-supported assumptions: different sperm whale clans make different types of codas (Gero et al., 2016; Rendell & Whitehead, 2003b) and certain coda types are favored by each clan (Amano et al., 2014; Gero et al., 2016; Huijser et al., 2019; Rendell & Whitehead, 2003b). There is theoretical support for this approach (e.g. repetition and redundancy of call types to improve signal-to-noise ratios) if the production of identity codas is more salient to the whales than general repertoire similarity (Wiley, 2013). This is especially the case if some coda types function as 'passwords' or badges of clan membership (e.g. Tyack, 2008), as has been hypothesized by several researchers (Cantor & Whitehead, 2013; Gero et al., 2016a; Whitehead & Rendell, 2014).

One noteworthy deviation of IDcall compared to previously used methods is that call classification is initially probabilistic rather than categorical (i.e. each call is assigned a probability of belonging to each call type rather than being given a single call type designation). The user can eventually choose to categorically assign calls to types (whereby each call is assigned to the type for which is has the highest probability) but does not have to. The user can also choose whether to exclude calls designated as outliers from future analyses.

We compared the coda types detected by IDcall to those detected in previous Atlantic/Mediterranean (Table S12) and Pacific (Table S13) analyses. Despite variability and inconsistency in coda naming practices across studies, many coda types found in previous Atlantic/Mediterranean and Pacific sperm whale research have similar rhythmic patterns to coda types found using IDcall. We did not attempt to look for 1:1 type matches between IDcall and previous methods given that many past studies determined coda types after standardizing codas for duration (e.g. Rendell & Whitehead, 2003b) or did not provide detailed durational information for each type (e.g. Pavan et al., 2000). Some types were unique to IDcall, which may reflect the previously discussed differences in call classification methodology (many of these unique types are rare and were likely excluded as outliers in past studies) and the increase in sample size in both datasets from previous studies (3,091 new Atlantic/Mediterranean codas; 2,765 new Pacific codas). Additionally, a small number of types were present in past work but not in the present work, which could again relate to sample size increases or variability in naming practices. Importantly, many of the coda types previously recognized as frequent and important in the other studies we consulted were also found by IDcall.

Grey-breasted wood-wrens

Male *H. l. leucophrys* song is generally slower, contains fewer notes, and is higher and broader in frequency than male *H. l. hilaris* song (Dingle et al., 2008; Halfwerk et al., 2016). Accordingly, the centroid averaged note peak frequency, minimum song frequency, and maximum song frequency were all lower in the *H. l. hilaris* identity song compared to the *H. l. leucophrys* identity songs, matching past work.

Australian field crickets

T. commodus songs generally have a lower carrier frequency, fewer trills, and a shorter chirp-trill interval than *T. oceanicus* songs (Moran et al., 2020). Accordingly, the centroid chirp-trill interval in the *T. commodus* identity songs was shorter than in the *T. oceanicus* identity songs, matching past work.

Supplemental Methods References

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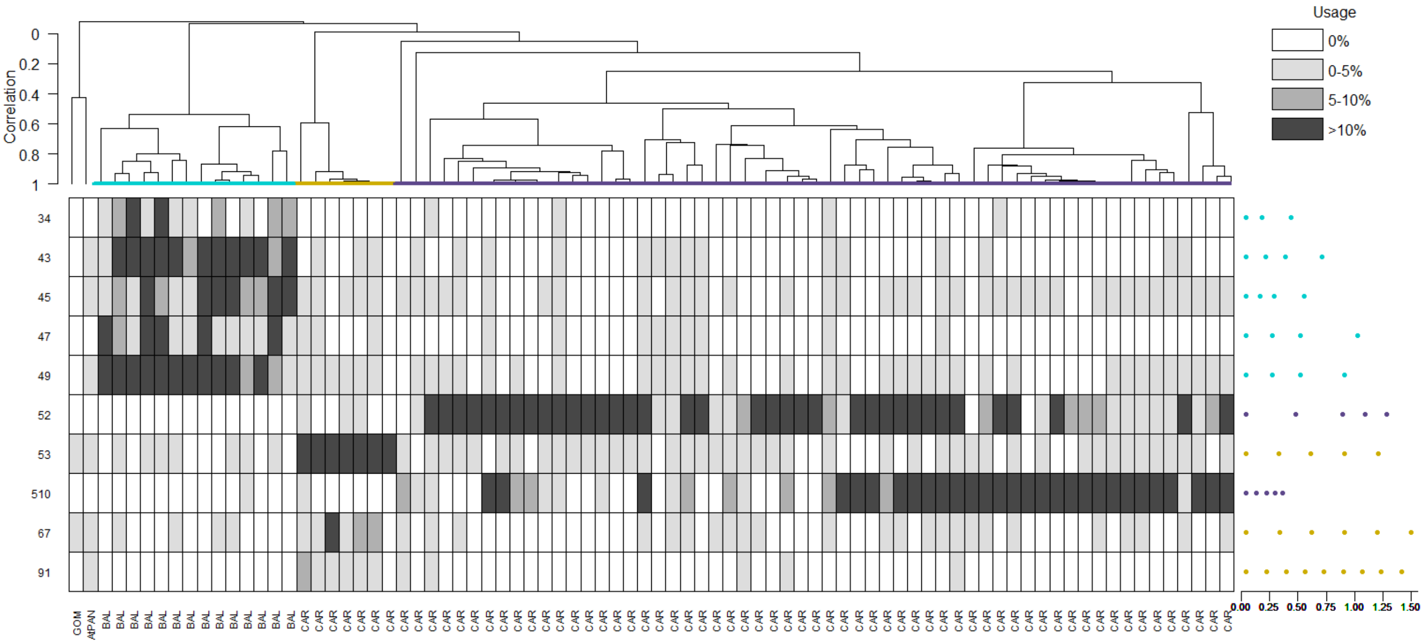
Supplemental Data

Option/parameter trial dendrograms

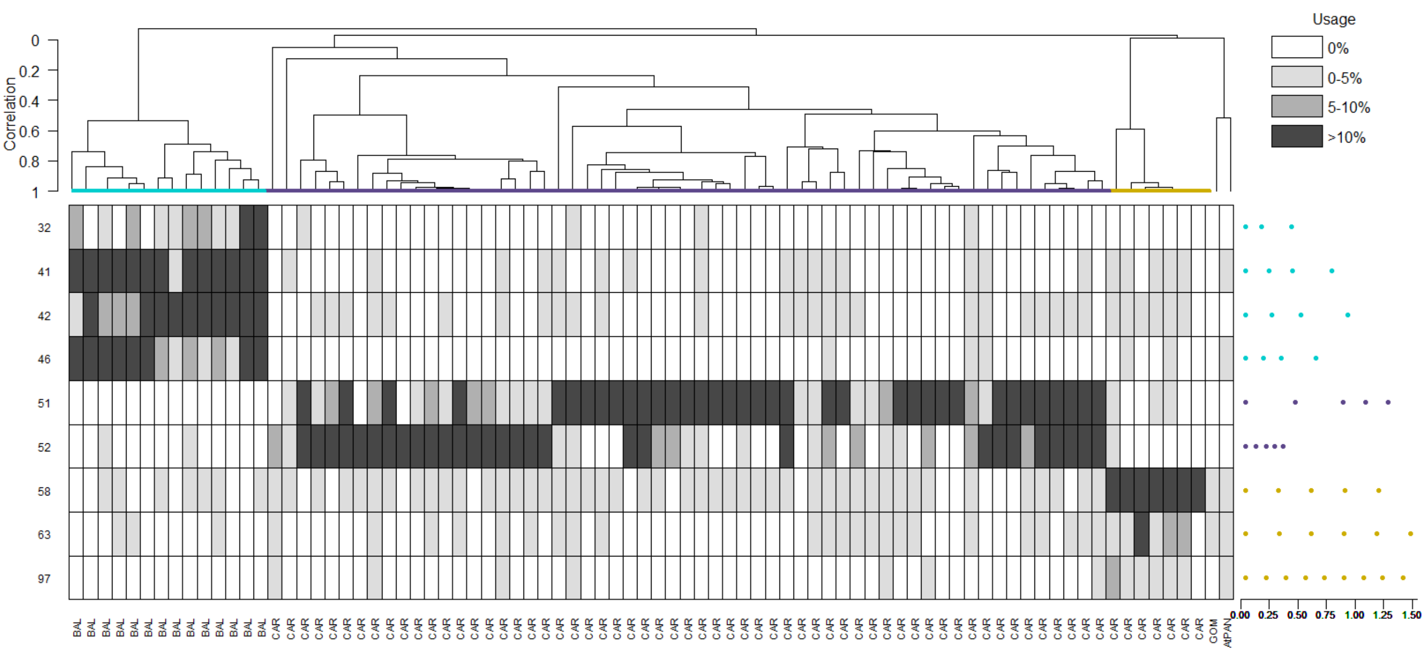
For each dataset, 19 trials were conducted. The resultant dendrogram for each trial is presented below, with trial numbers corresponding to those in Tables S4 (Atlantic/Mediterranean sperm whales), S5 (Pacific sperm whales), S6 (wrens), and S7 (crickets). The varied option/parameter is listed after each trial number. All other options/parameters were kept at the default values (Table 2). For detailed, dataset-specific figure descriptions, see Figs. 1–4. Acoustic clade colours match those used in Figs. 1–4, as well as those used in Figs. S4 and S5 for Pacific sperm whales. Numeric call type codes (i.e. heat map row labels) are consistent for the baseline dendrogram and the trials varying the hierarchical clustering linkage strategy, *critfact*, and *minrep*, but not necessarily for the trials varying the ECM algorithm initialization strategy, the information criterion used during call classification, or the default value reruns.

Atlantic/Mediterranean sperm whales

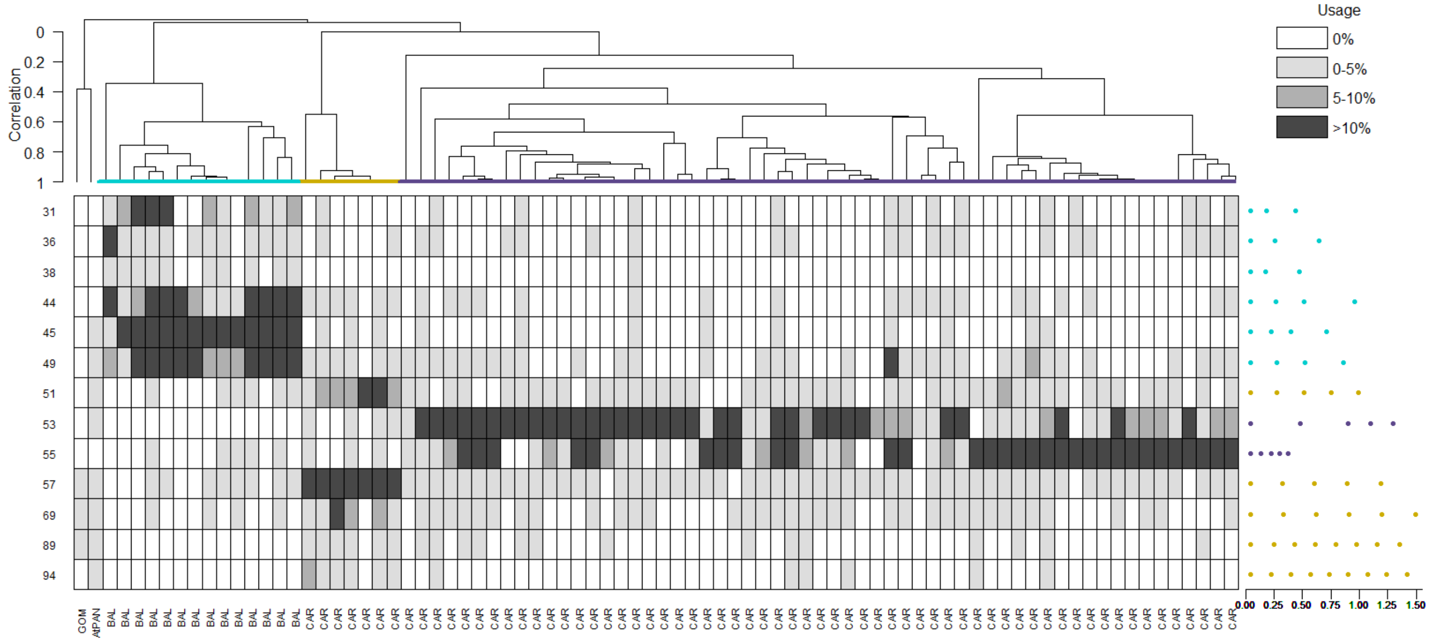
Trial 1 (initialization=random.post)



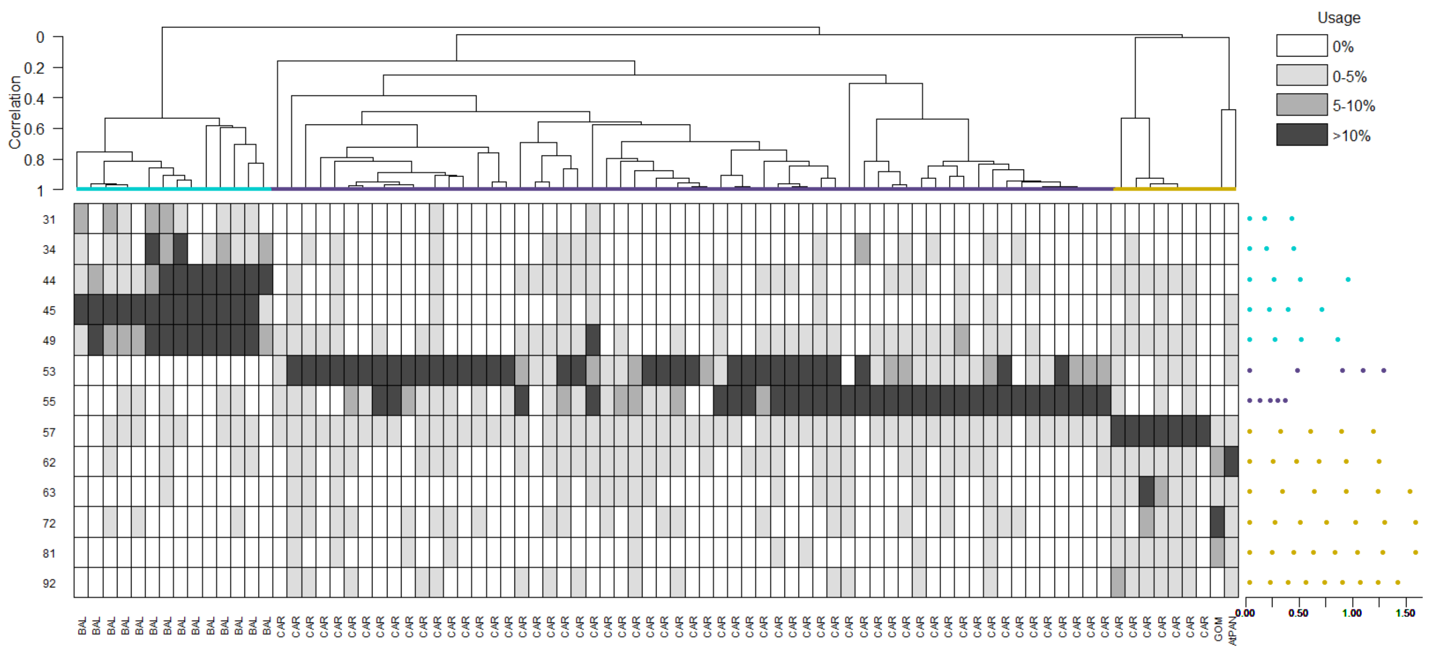
Trial 2 (initialization=random.clas)



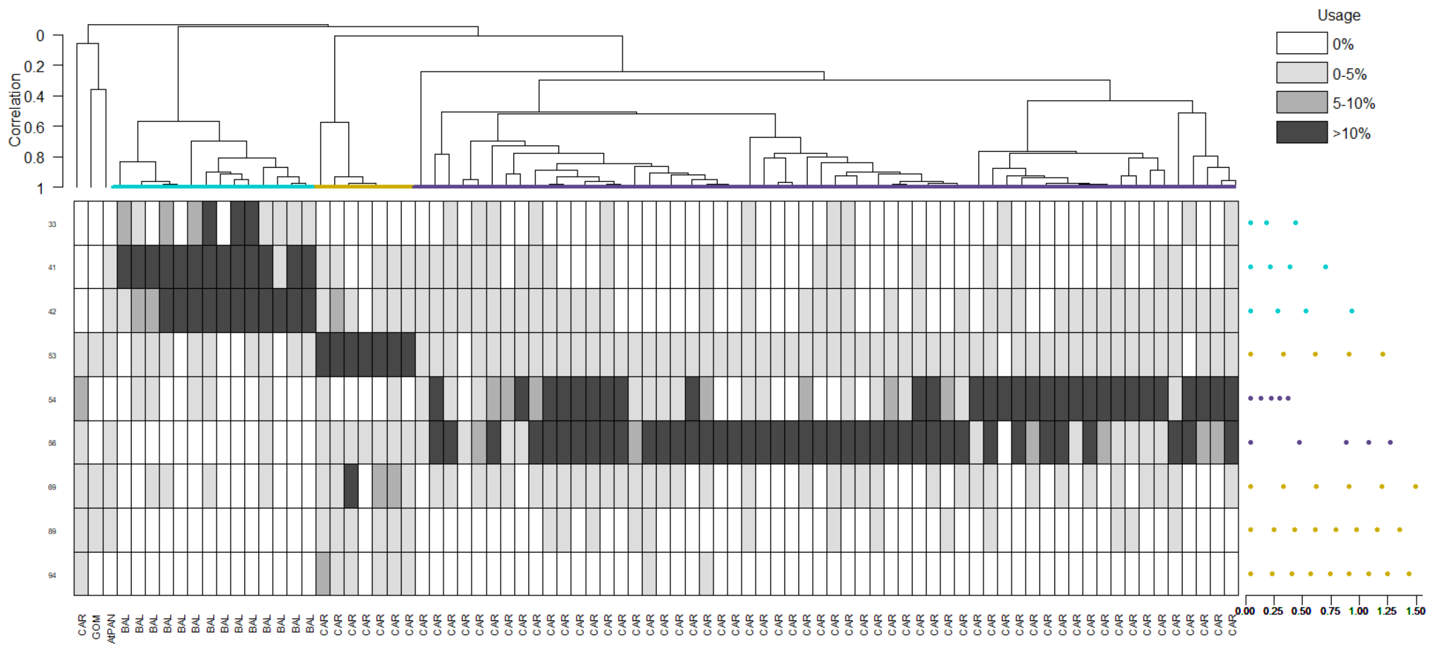
Trial 3 (information criterion=AICc)



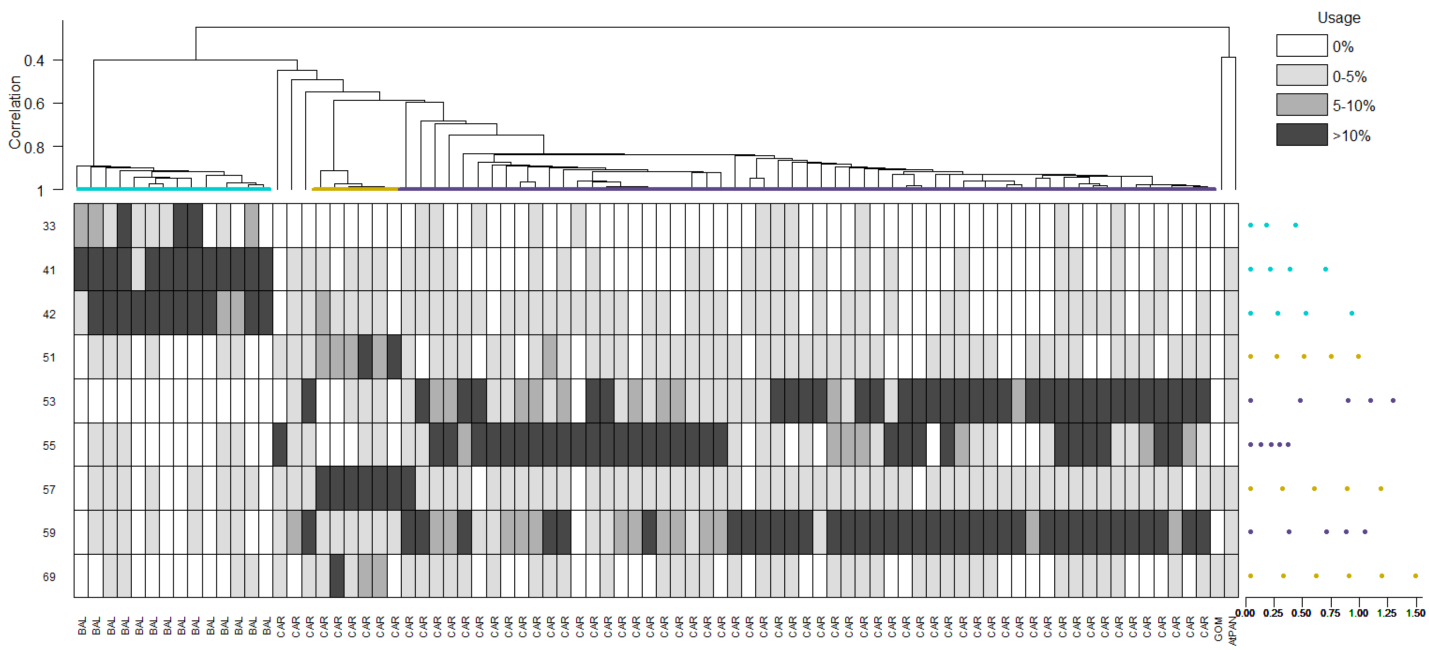
Trial 4 (information criterion=AIC)



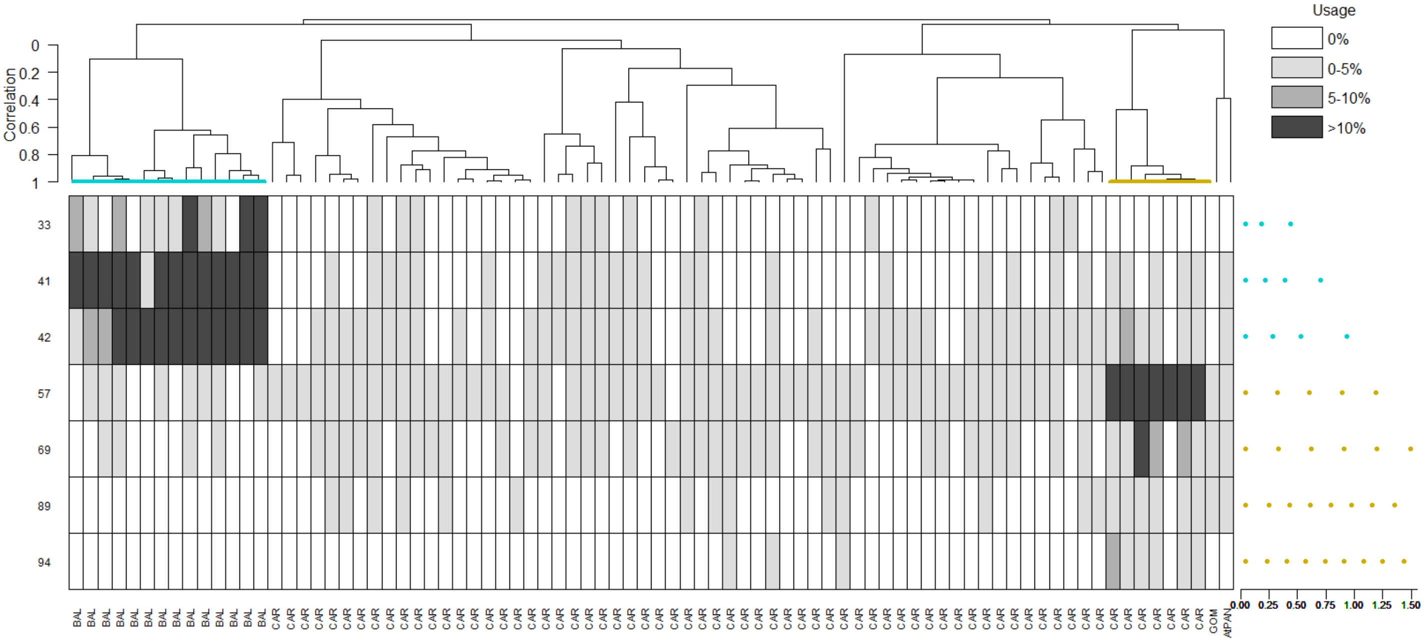
Trial 5 (information criterion=ICL)



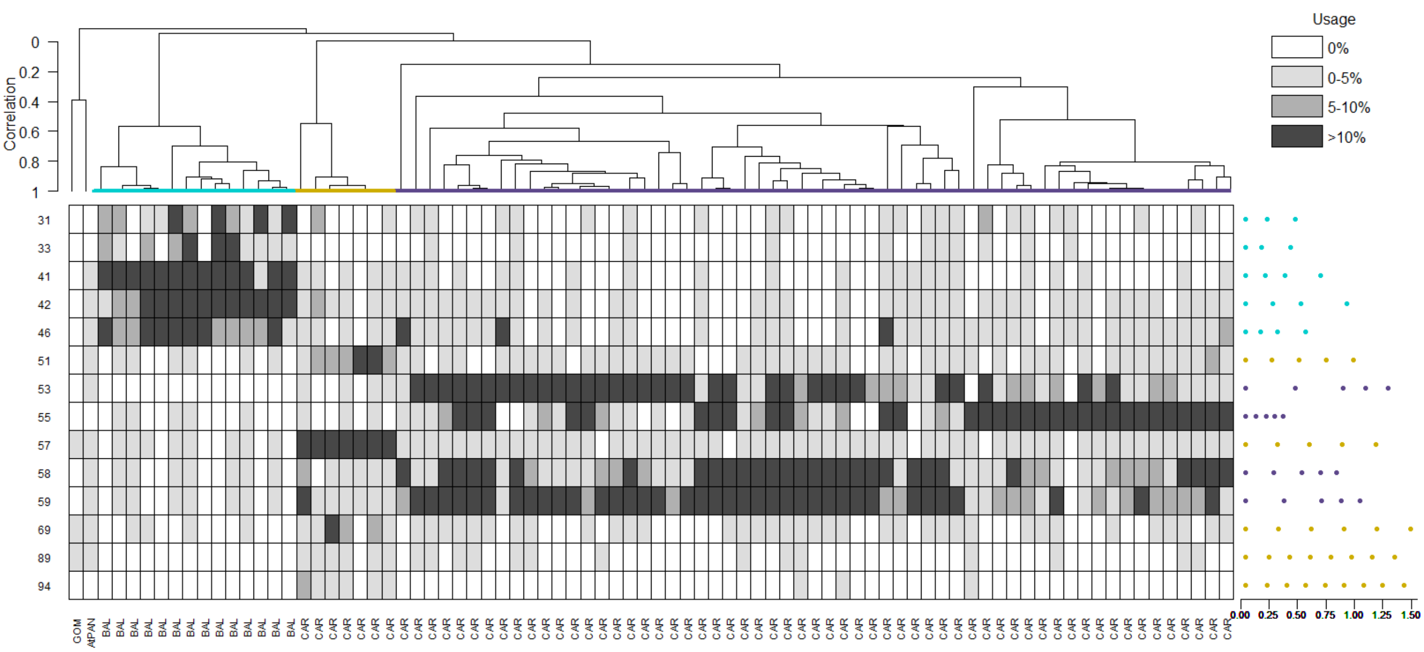
Trial 6 (linkage=single)



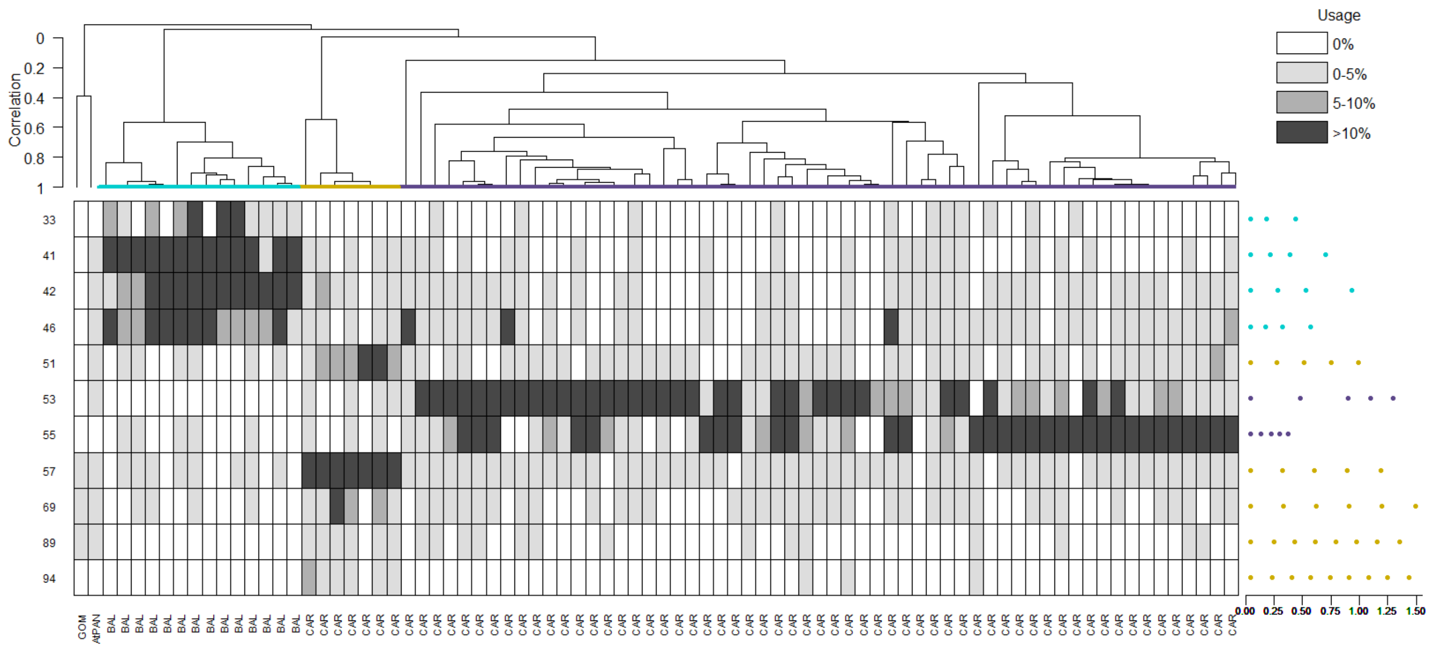
Trial 7 (linkage=complete)



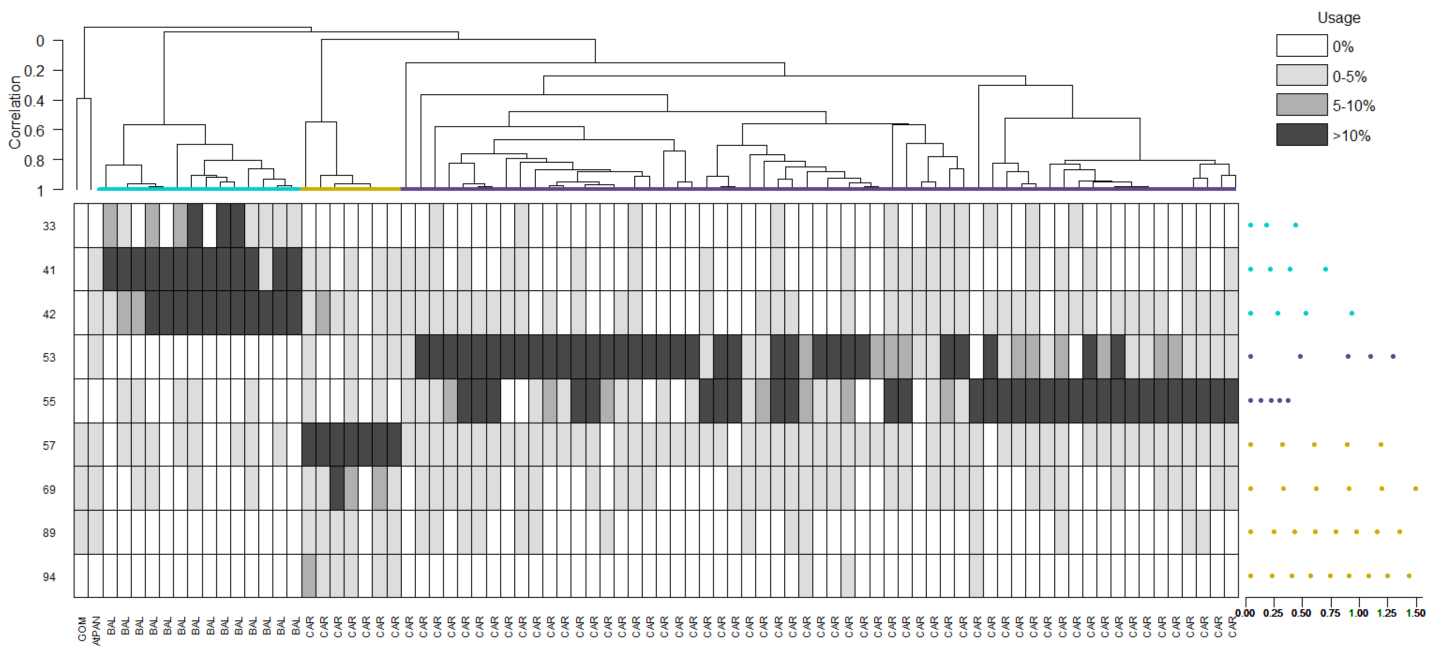
Trial 8 (critfact=6)



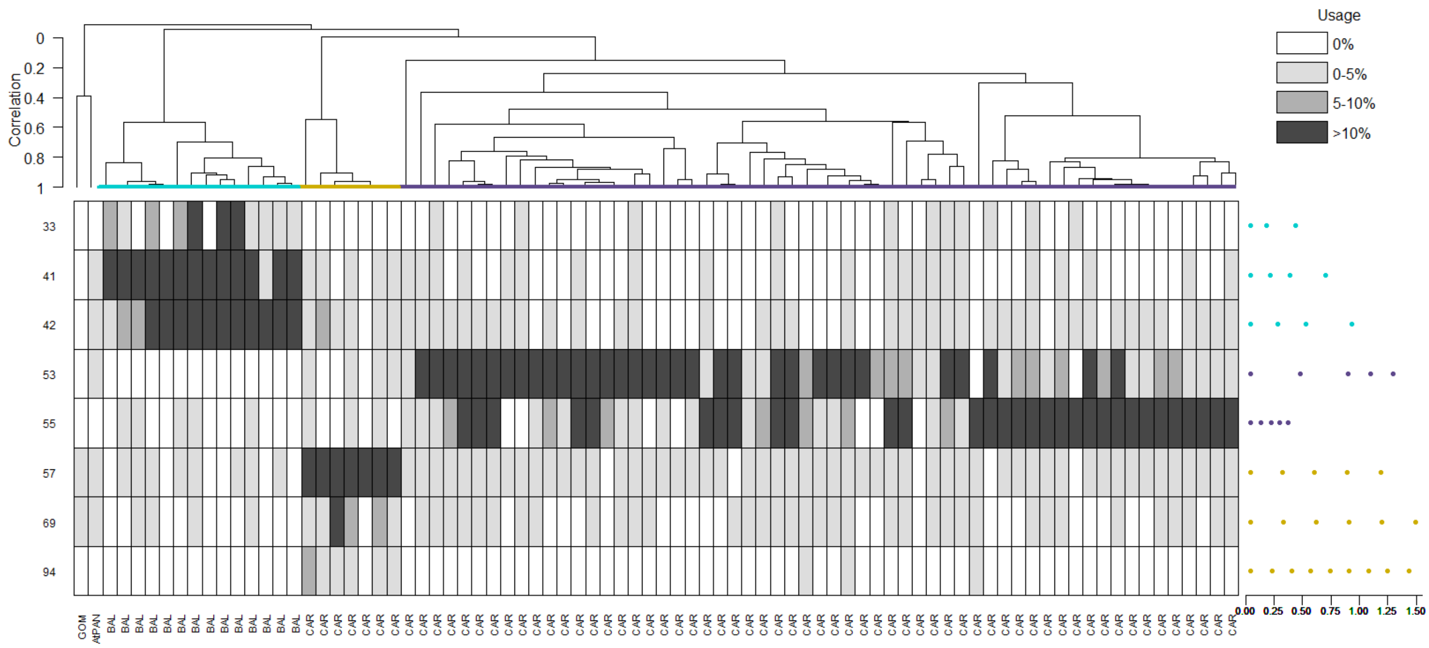
Trial 9 (critfact=10)



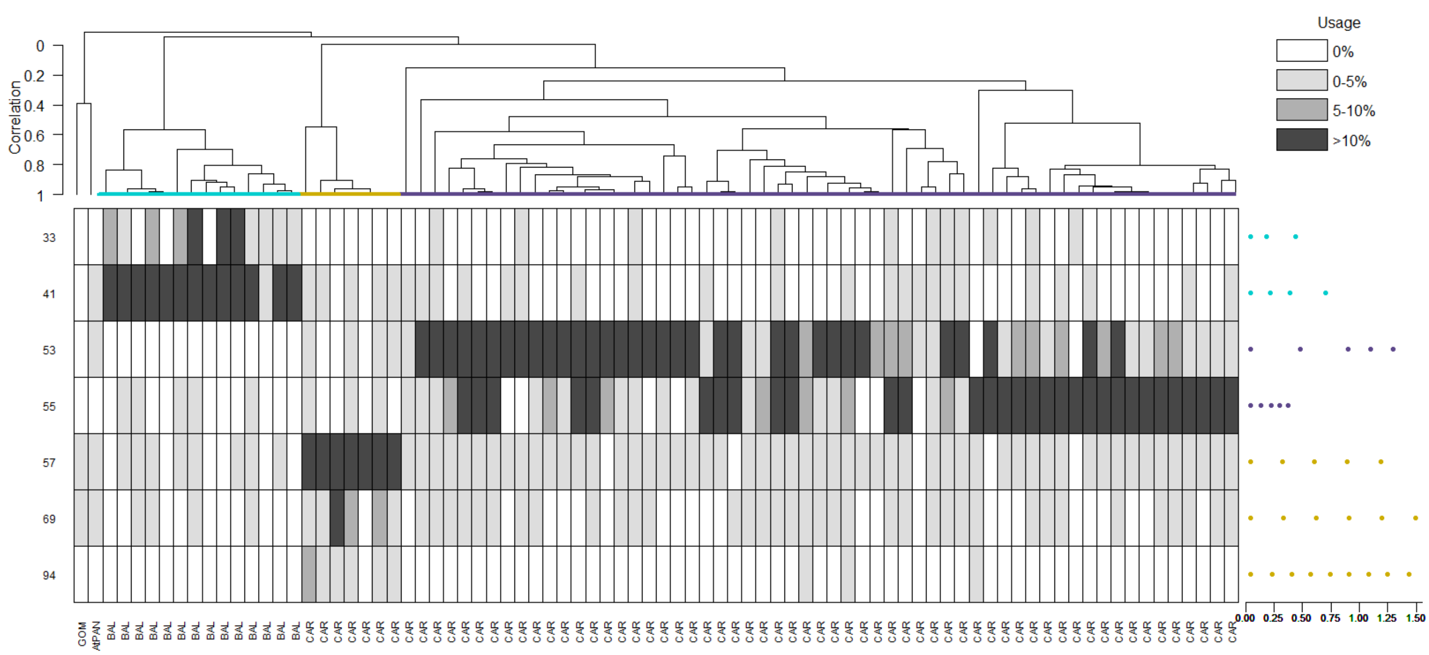
Trial 10 (critfact=18)



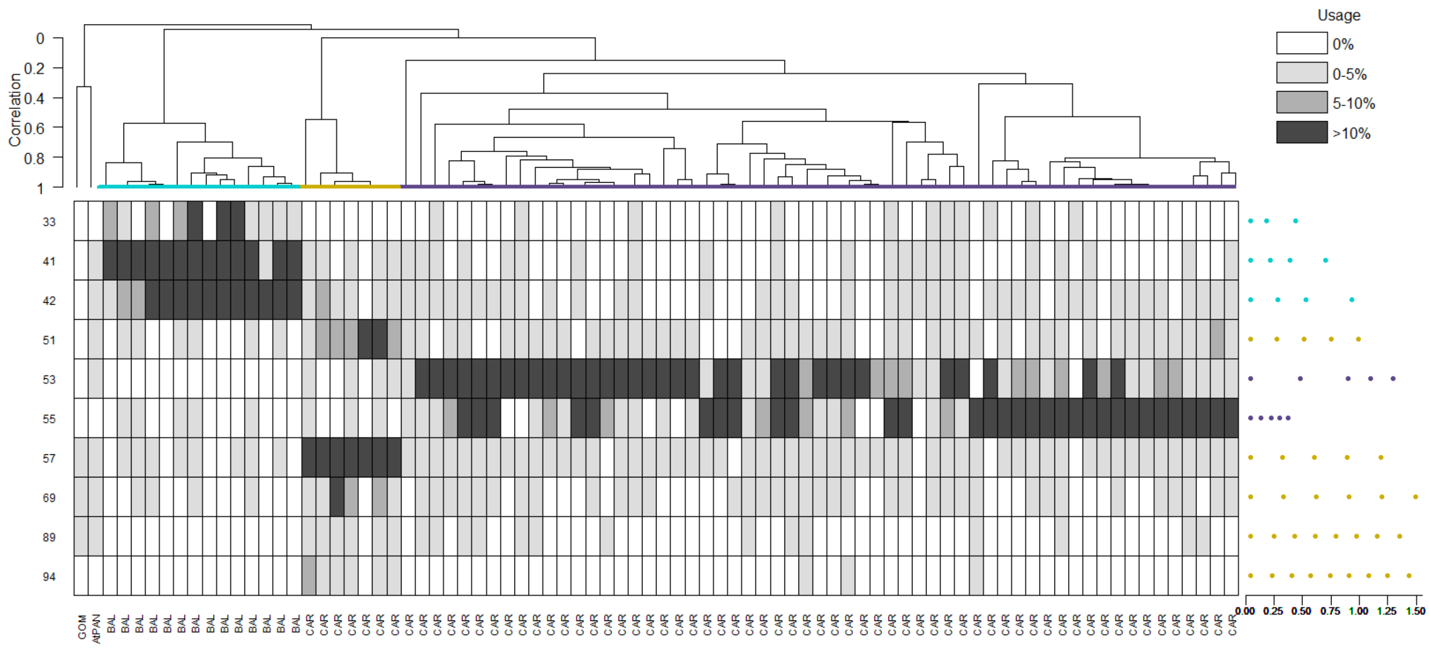
Trial 11 (critfact=22)



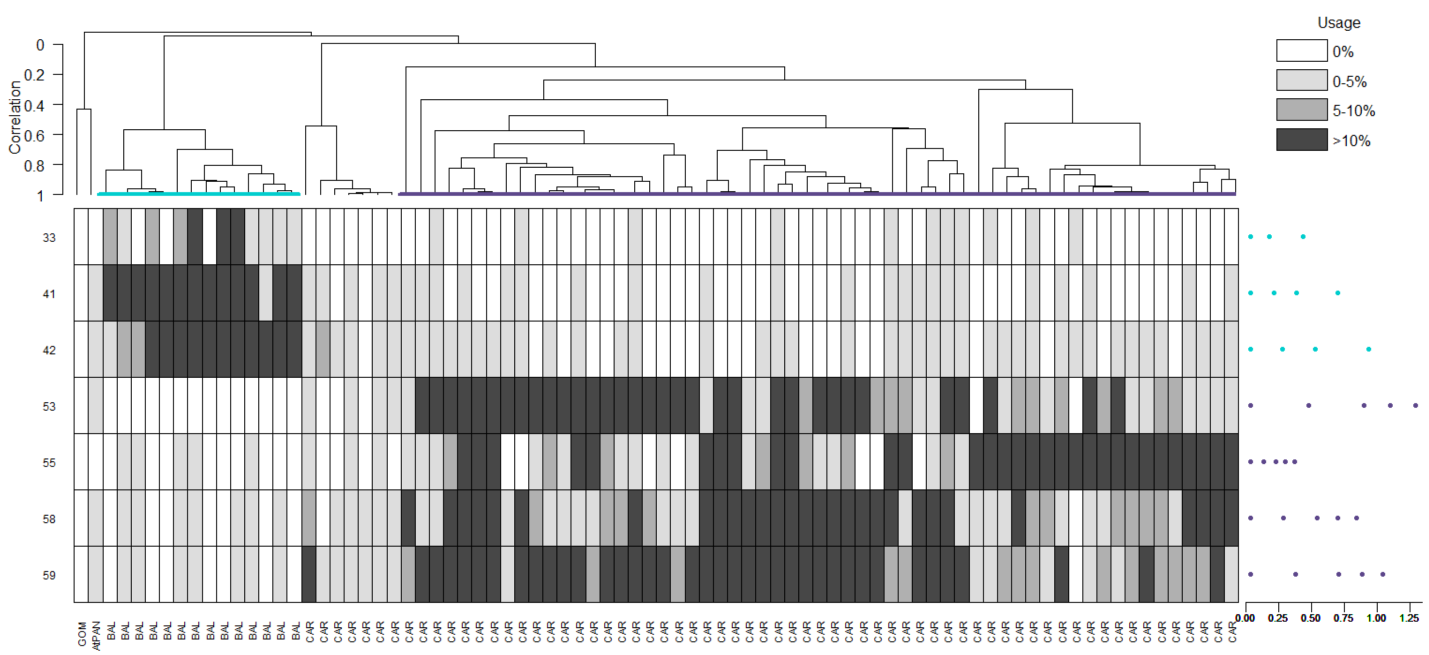
Trial 12 (critfact=26)



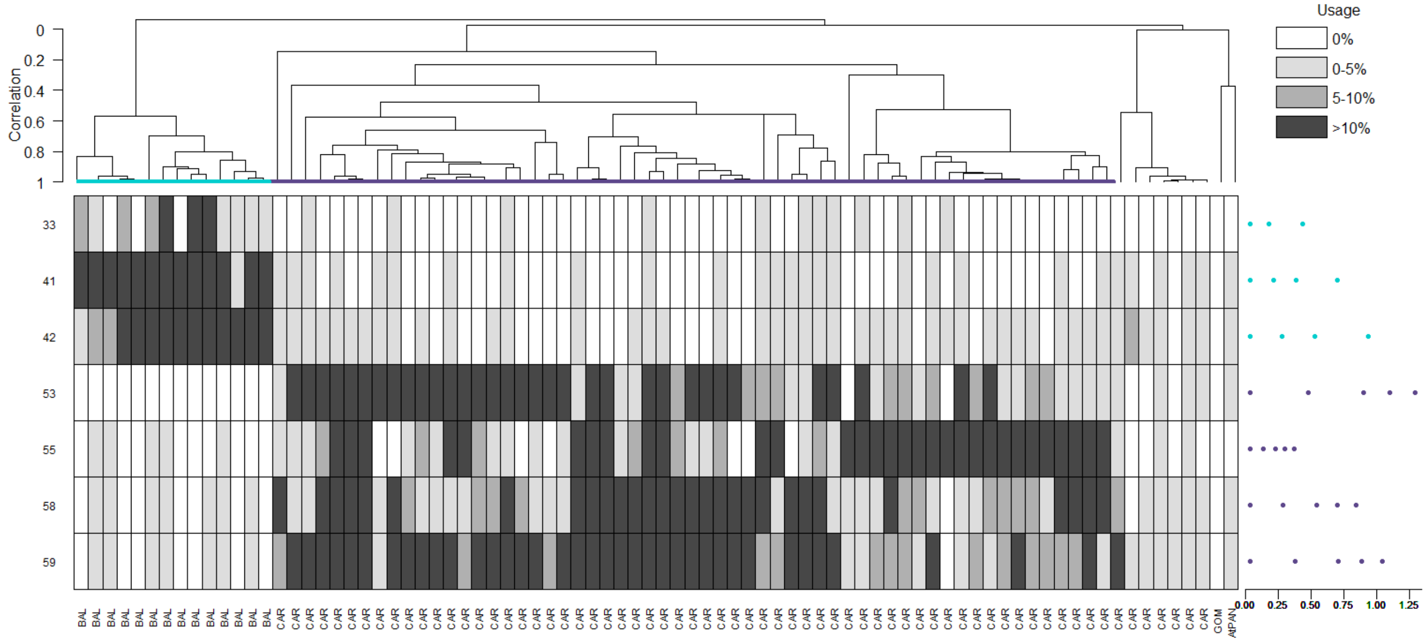
Trial 13 (minrep=4)



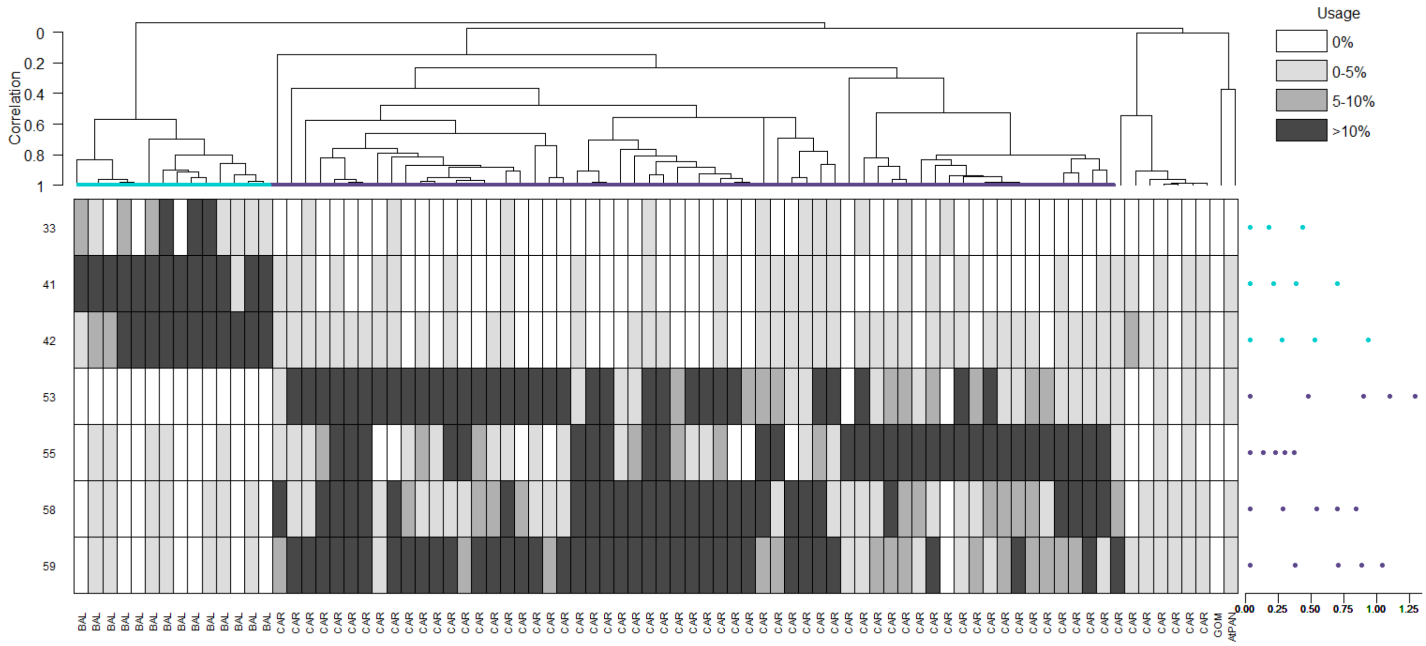
Trial 14 (minrep=8)



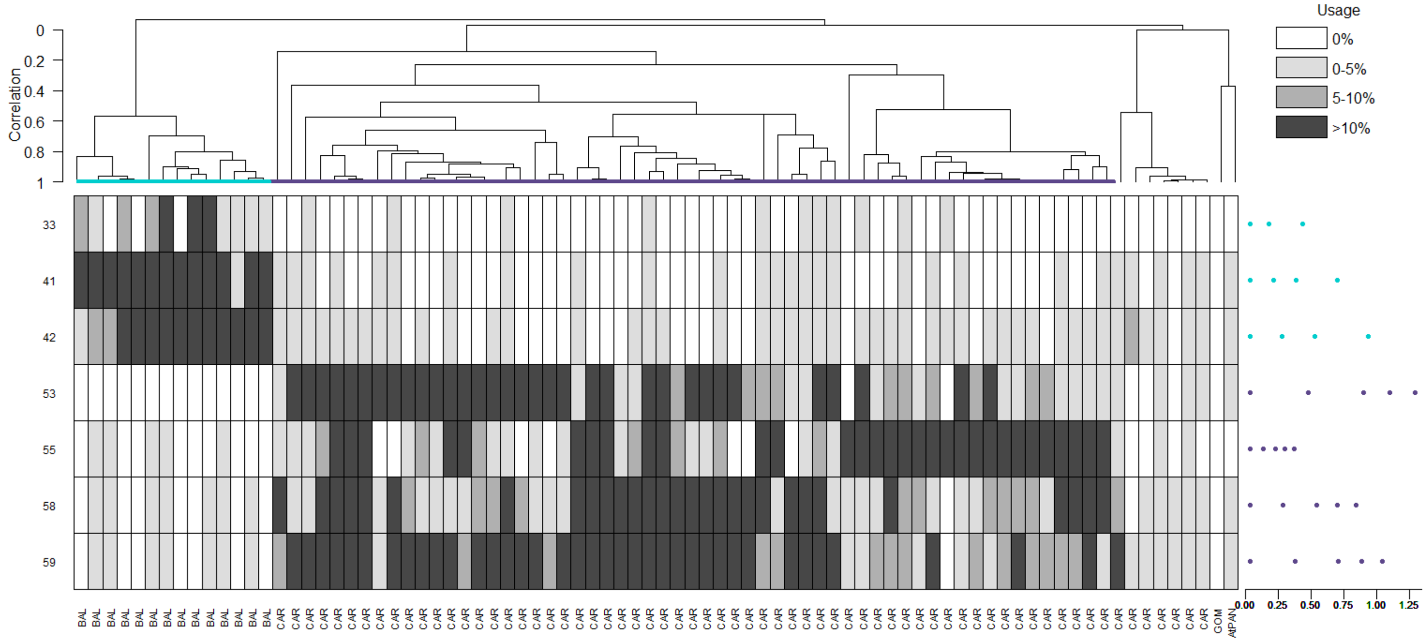
Trial 15 (minrep=10)



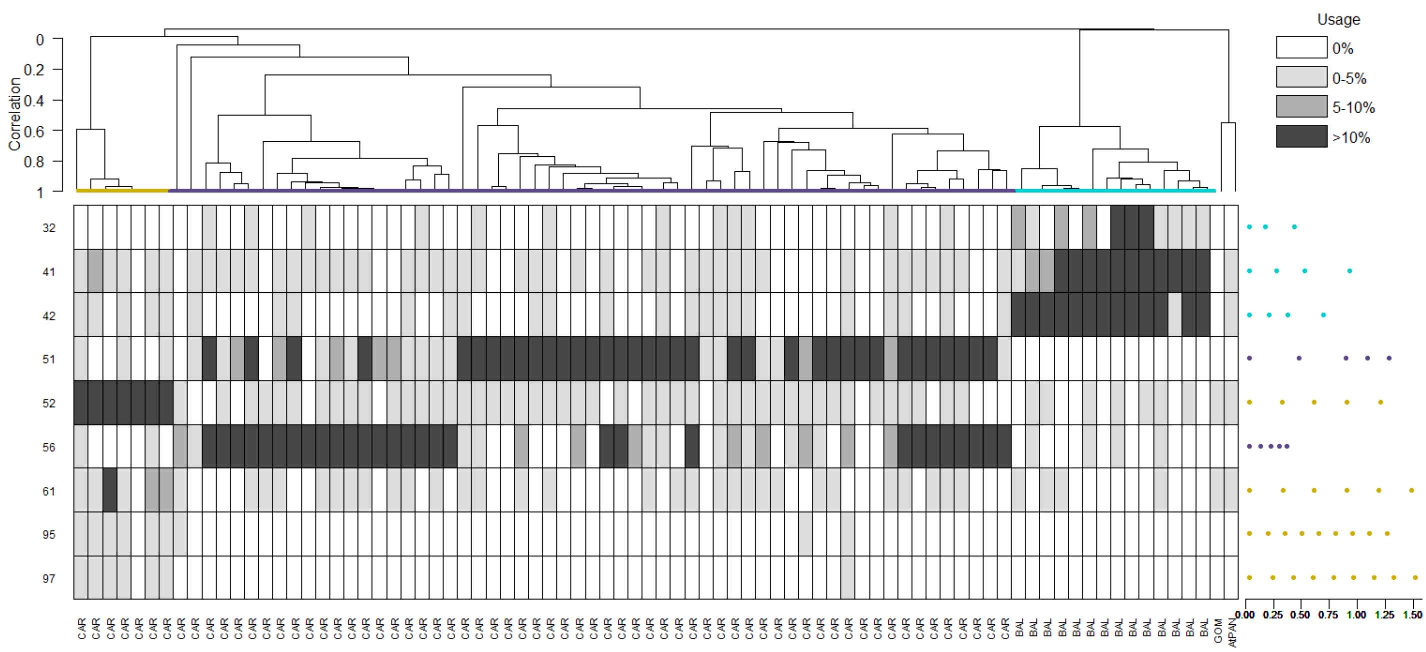
Trial 16 (minrep=12)



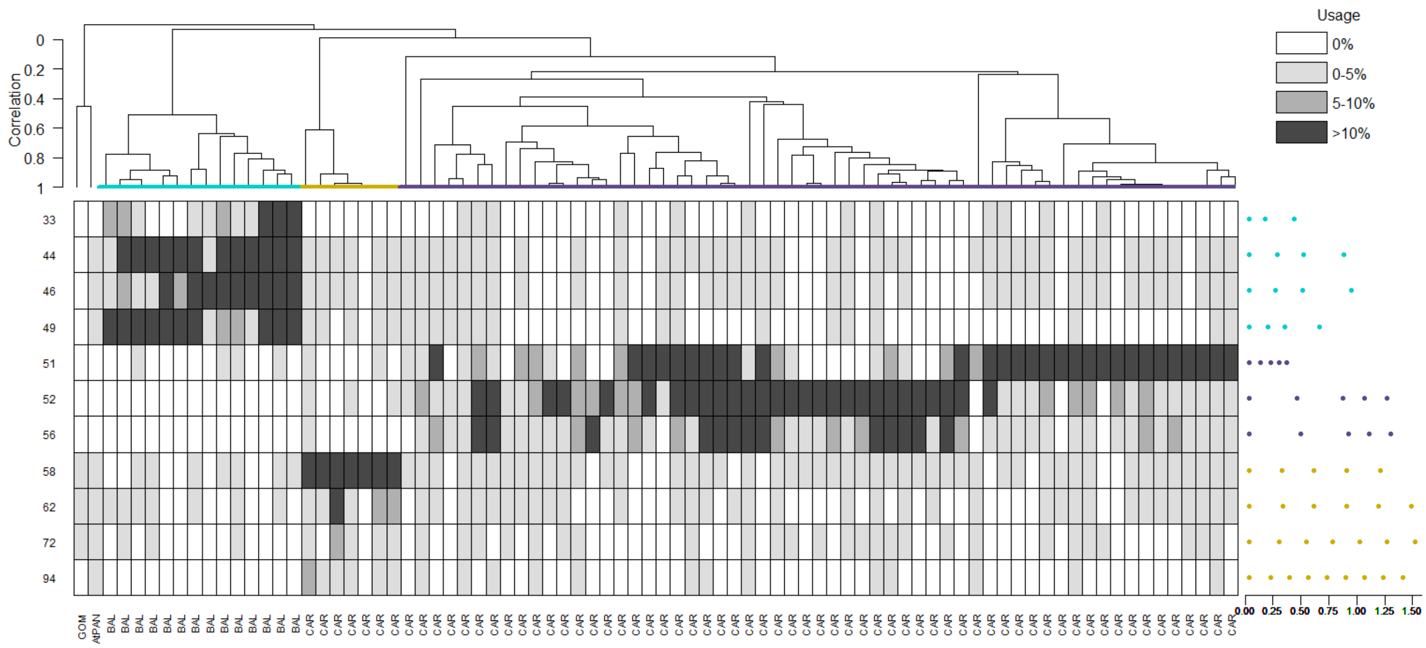
Trial 17 (minrep=14)



Trial 18 (baseline run 2)

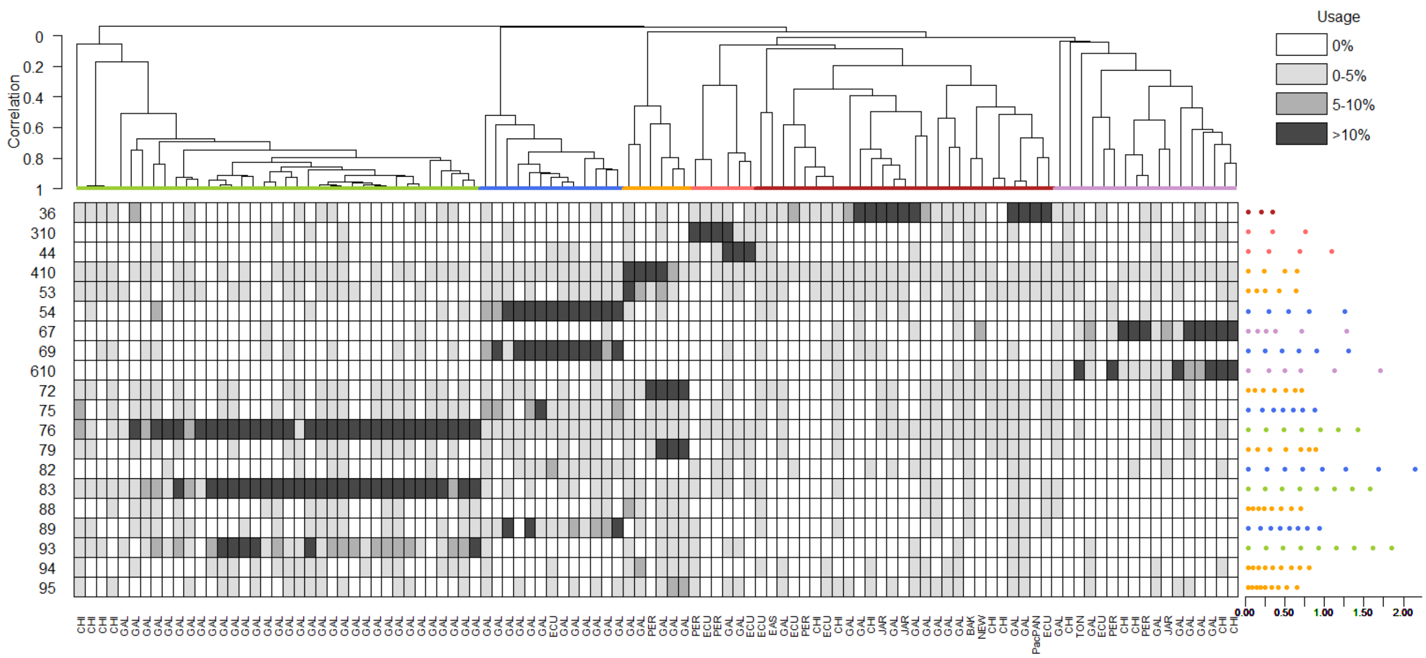


Trial 19 (baseline run 3)

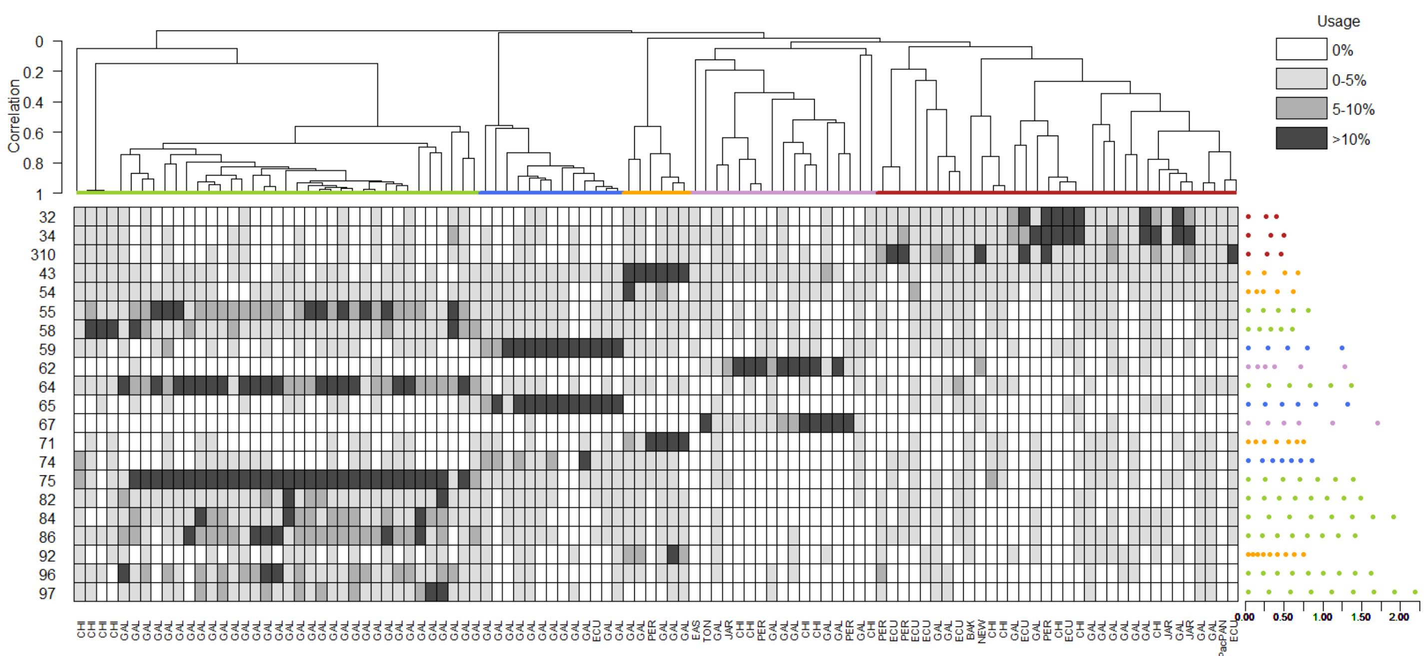


Pacific sperm whales

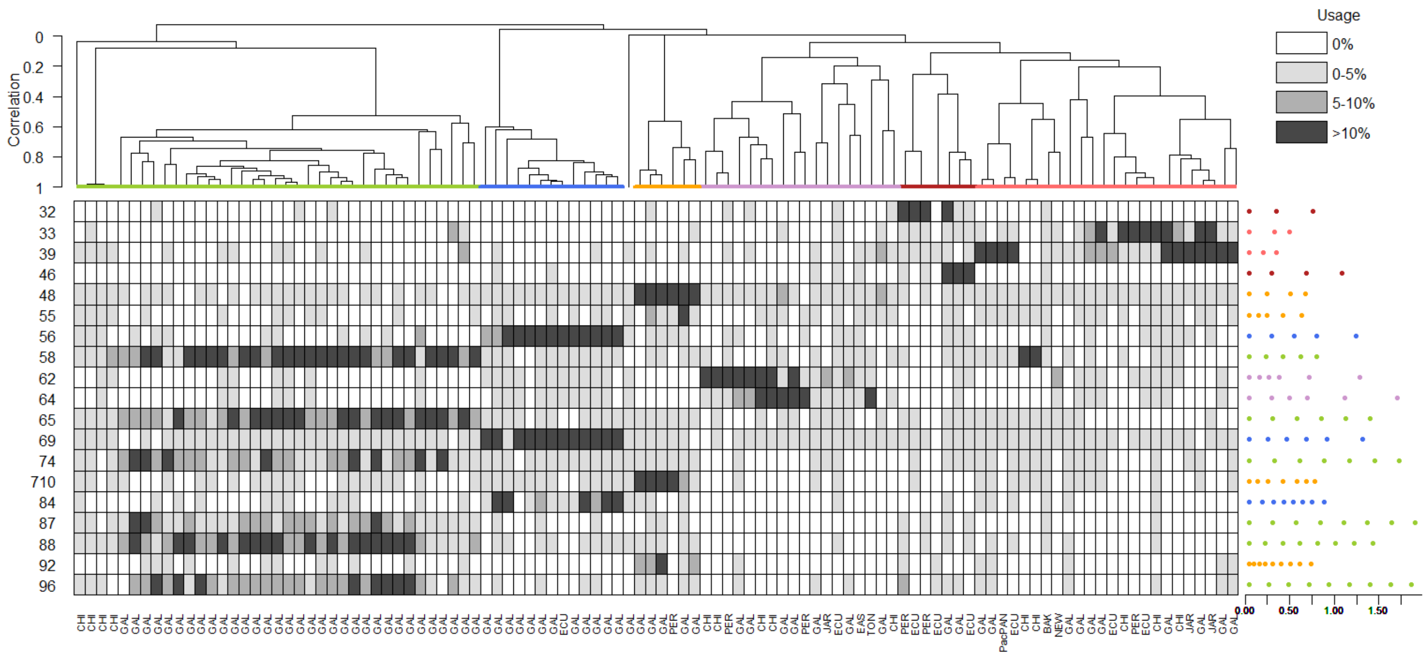
Trial 1 (initialization=random.post)



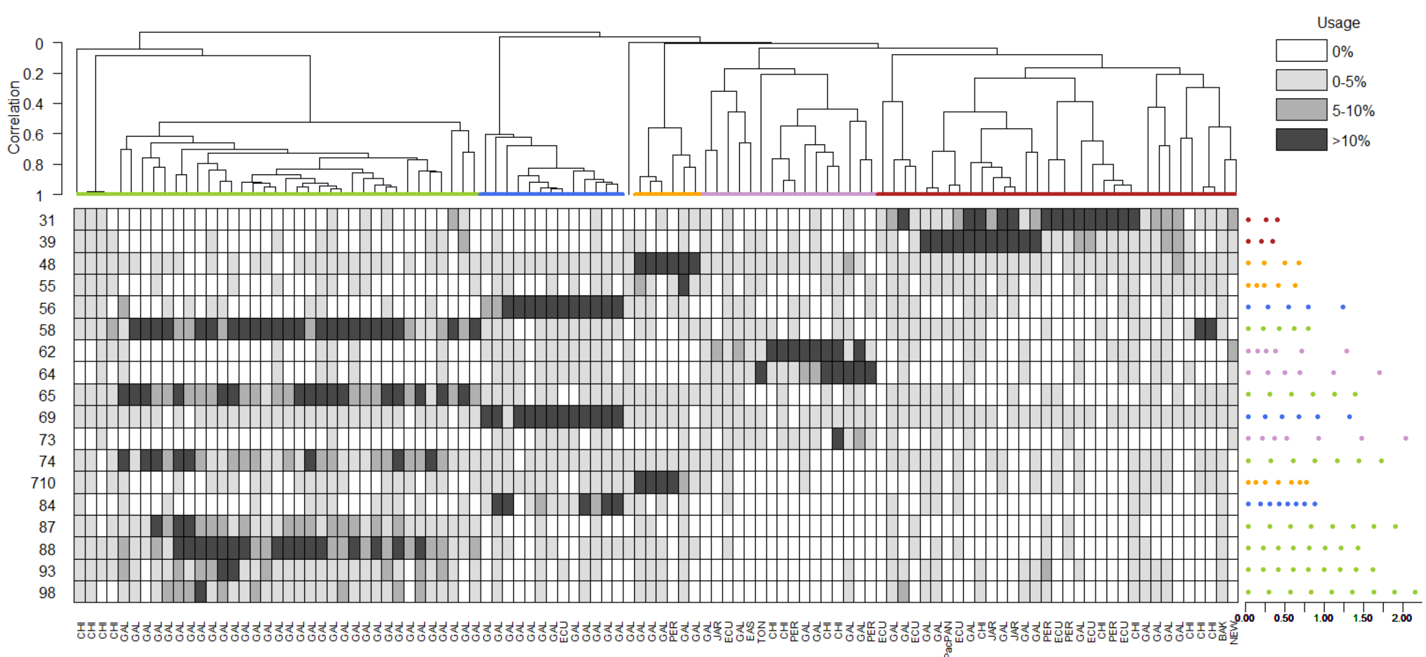
Trial 2 (initialization=random.clas)



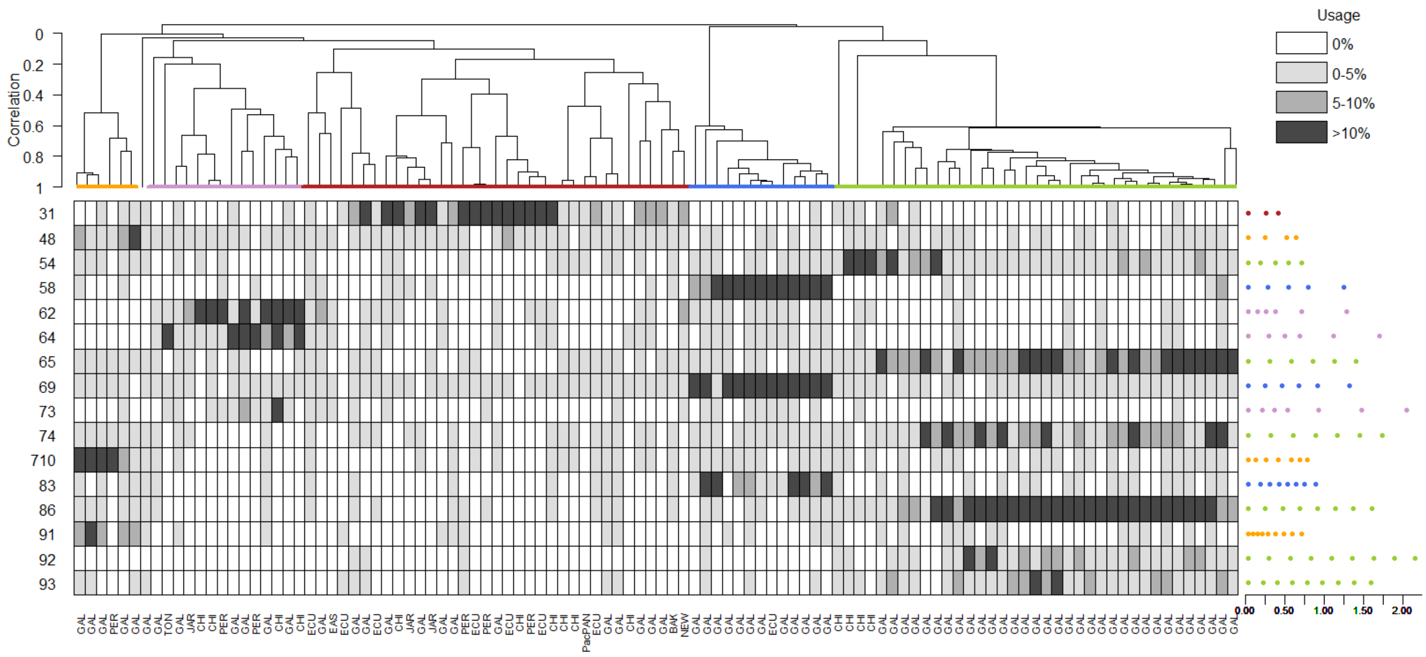
Trial 3 (information criterion=AICc)



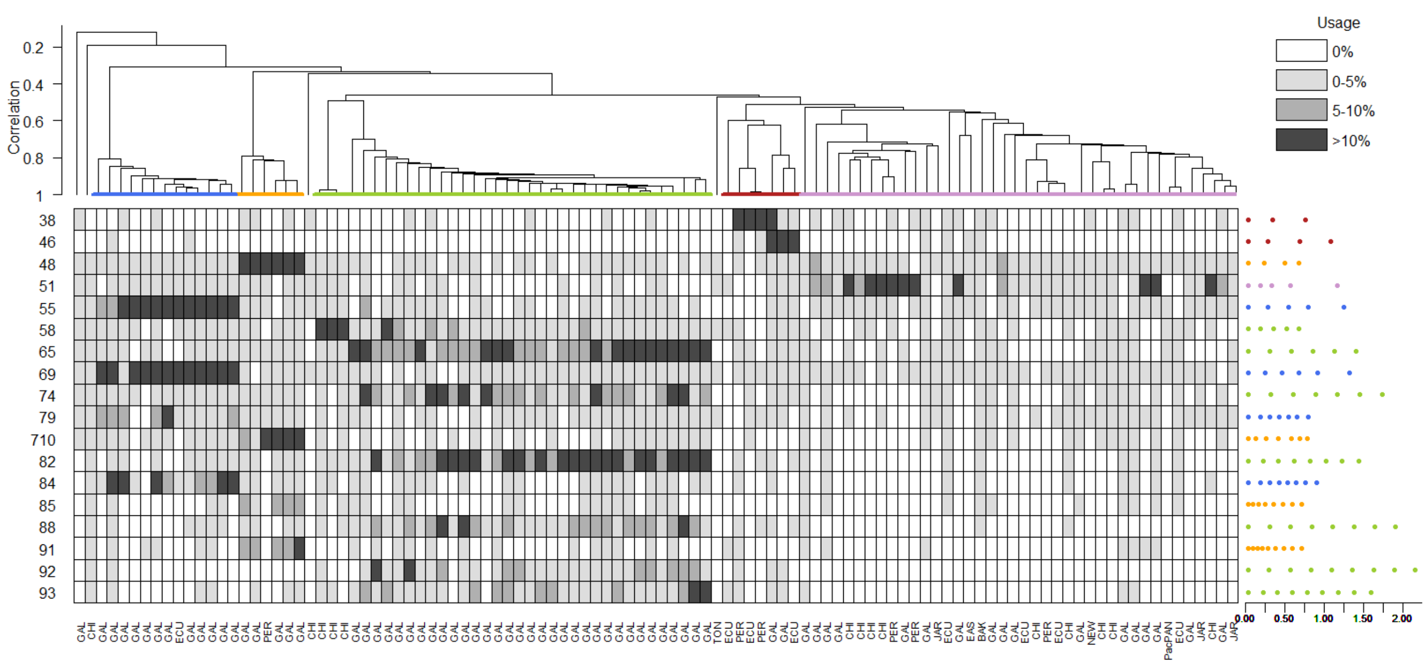
Trial 4 (information criterion=AIC)



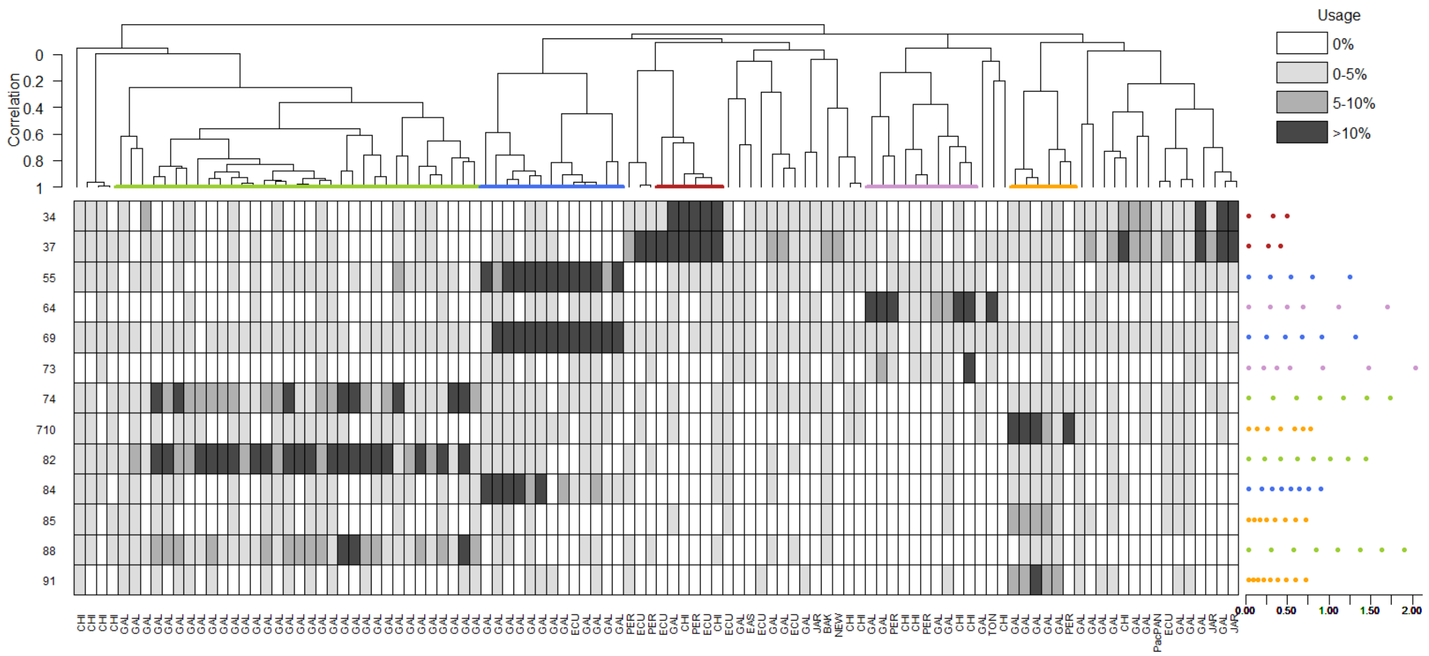
Trial 5 (information criterion=ICL)



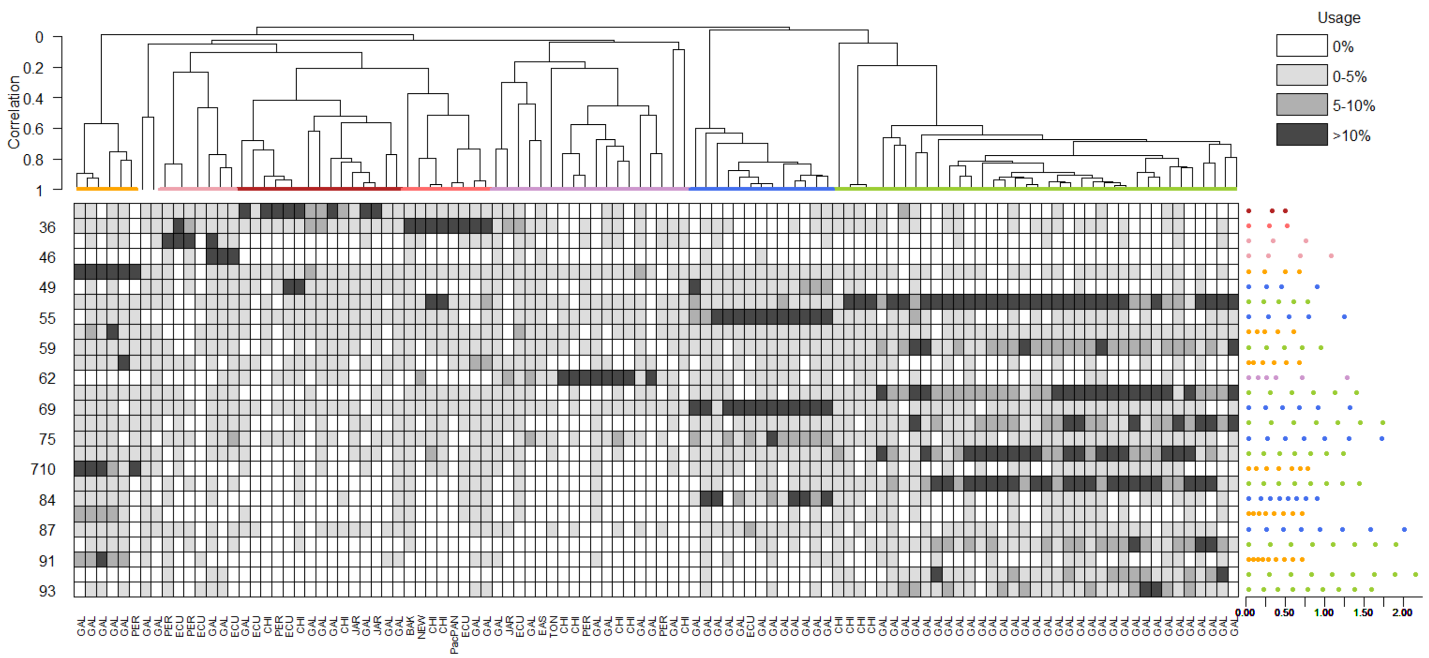
Trial 6 (linkage=single)



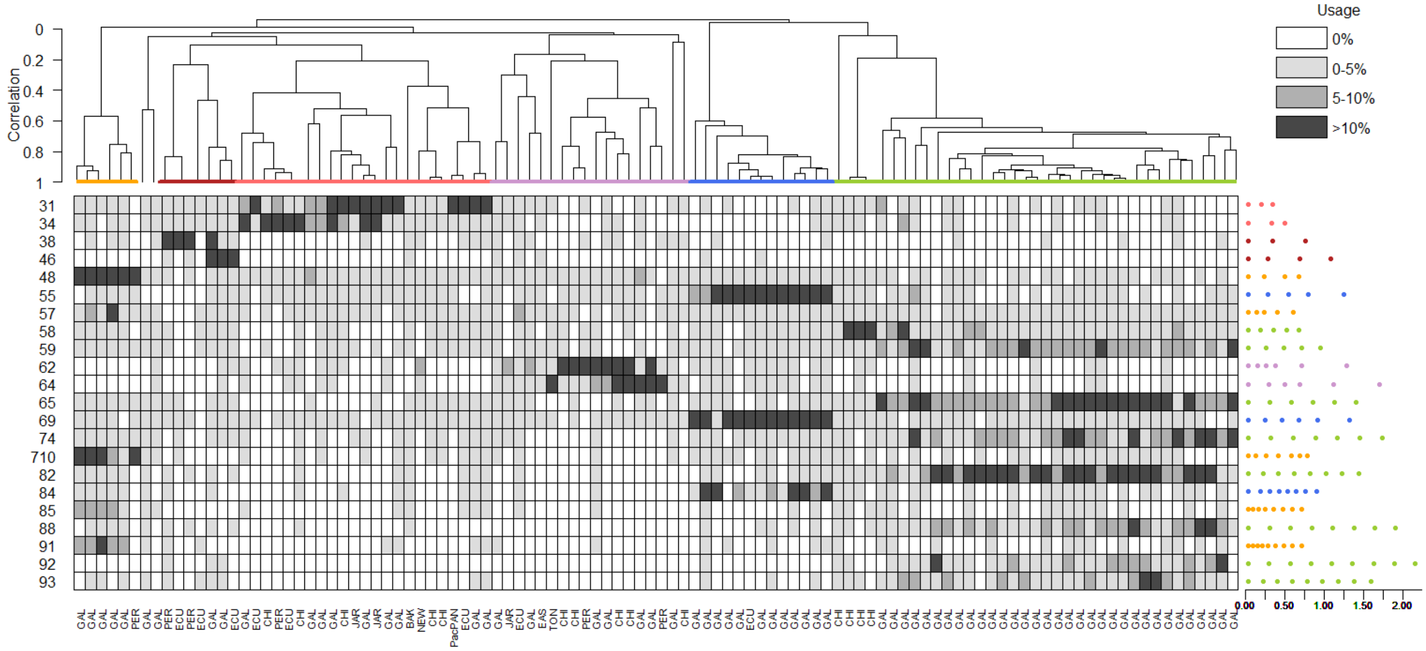
Trial 7 (linkage=complete)



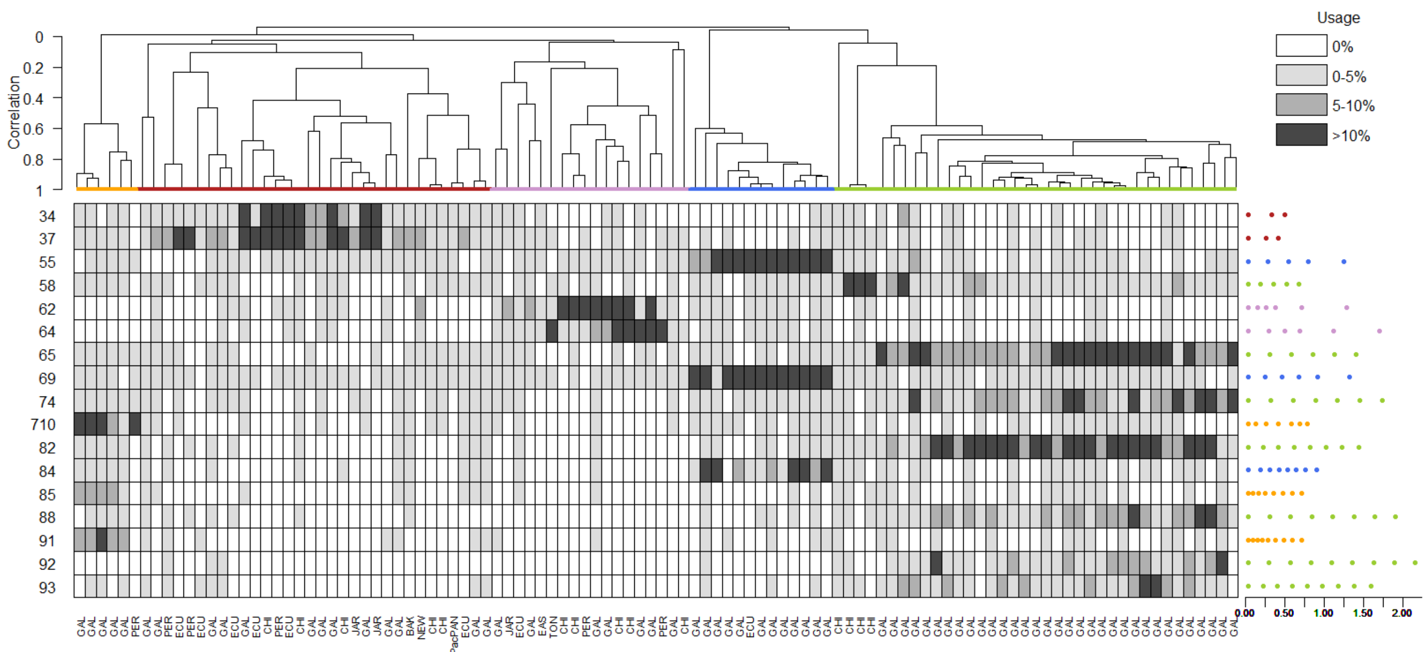
Trial 8 (critfact=6)



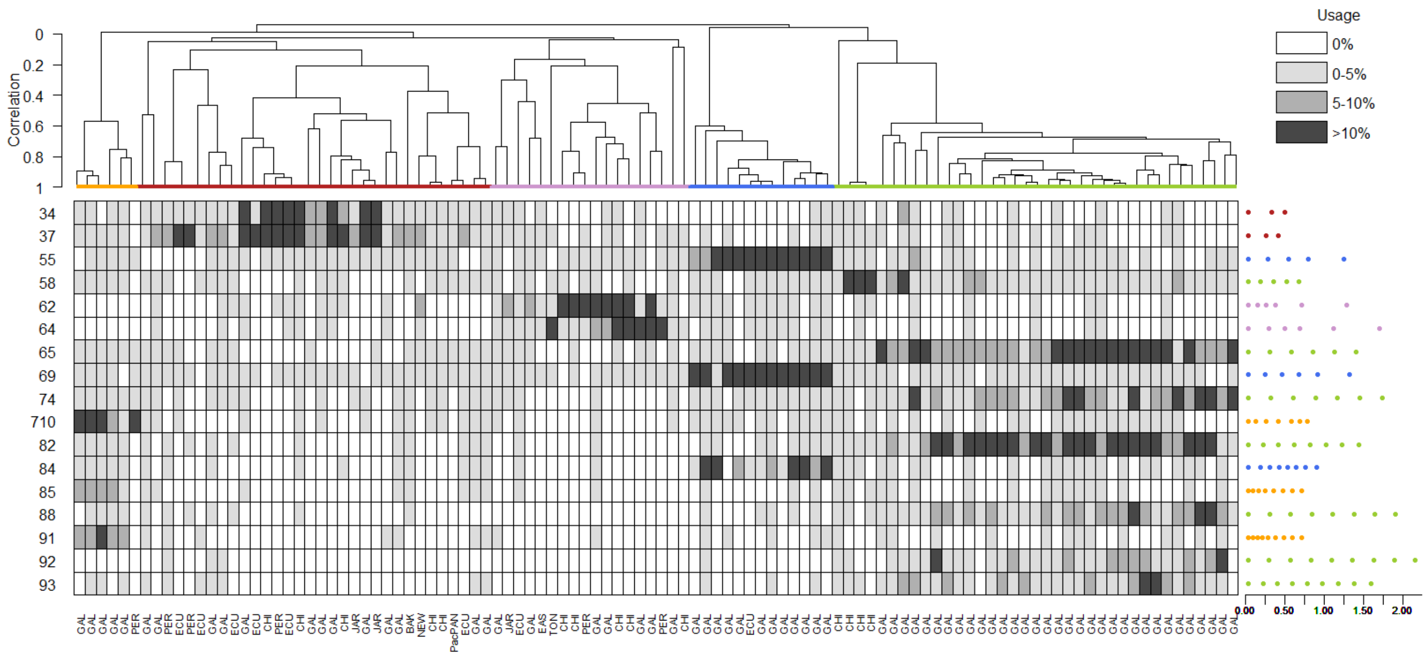
Trial 9 (critfact=10)



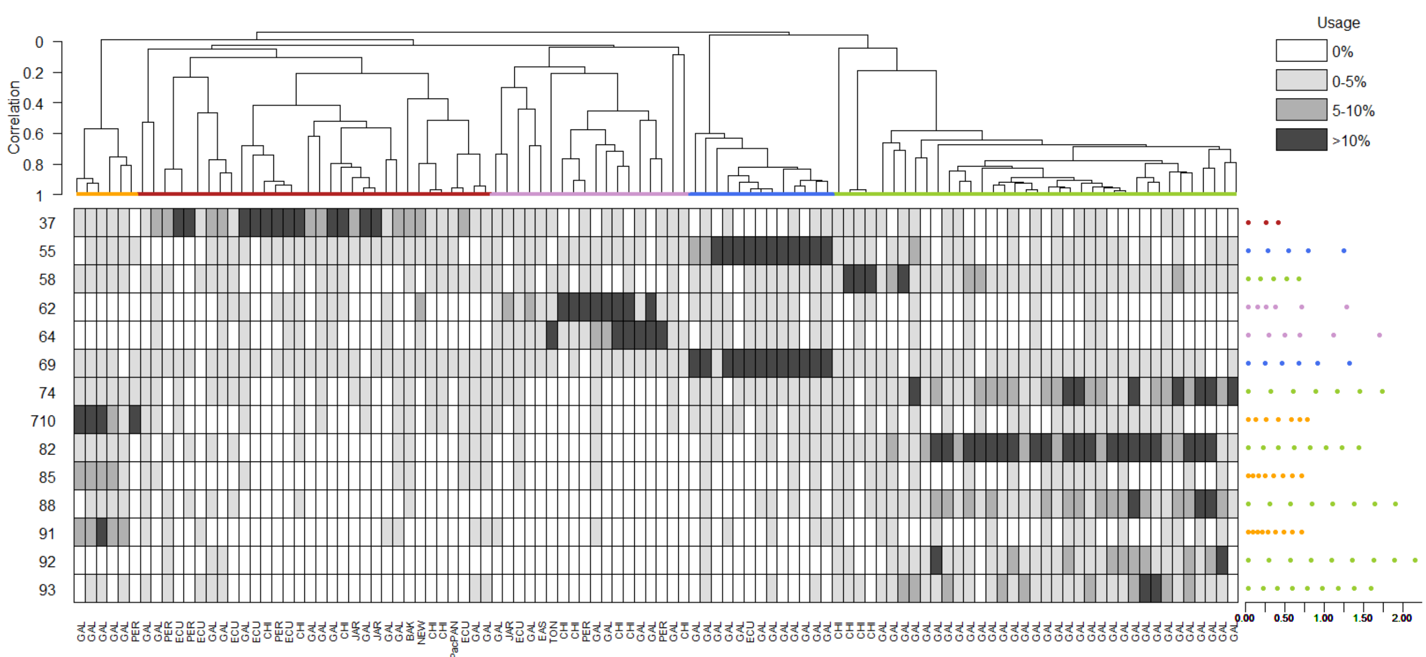
Trial 10 (critfact=18)



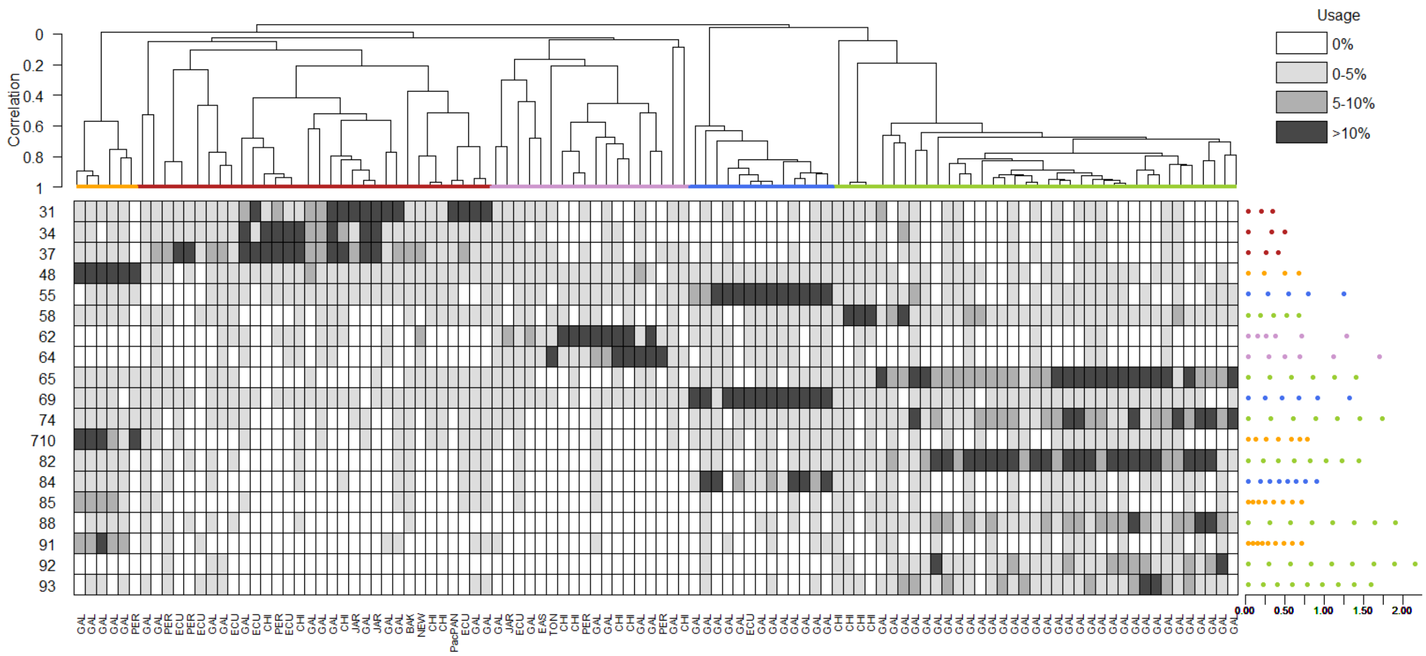
Trial 11 (critfact=22)



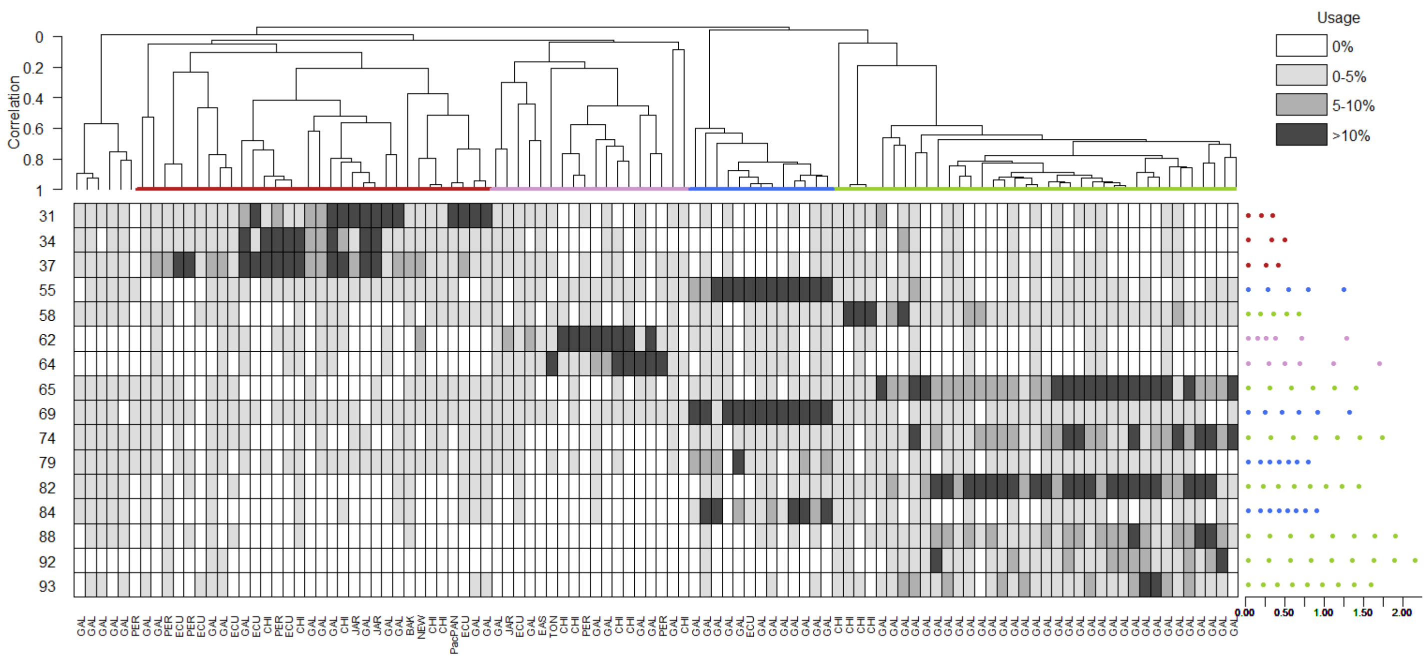
Trial 12 (critfact=26)



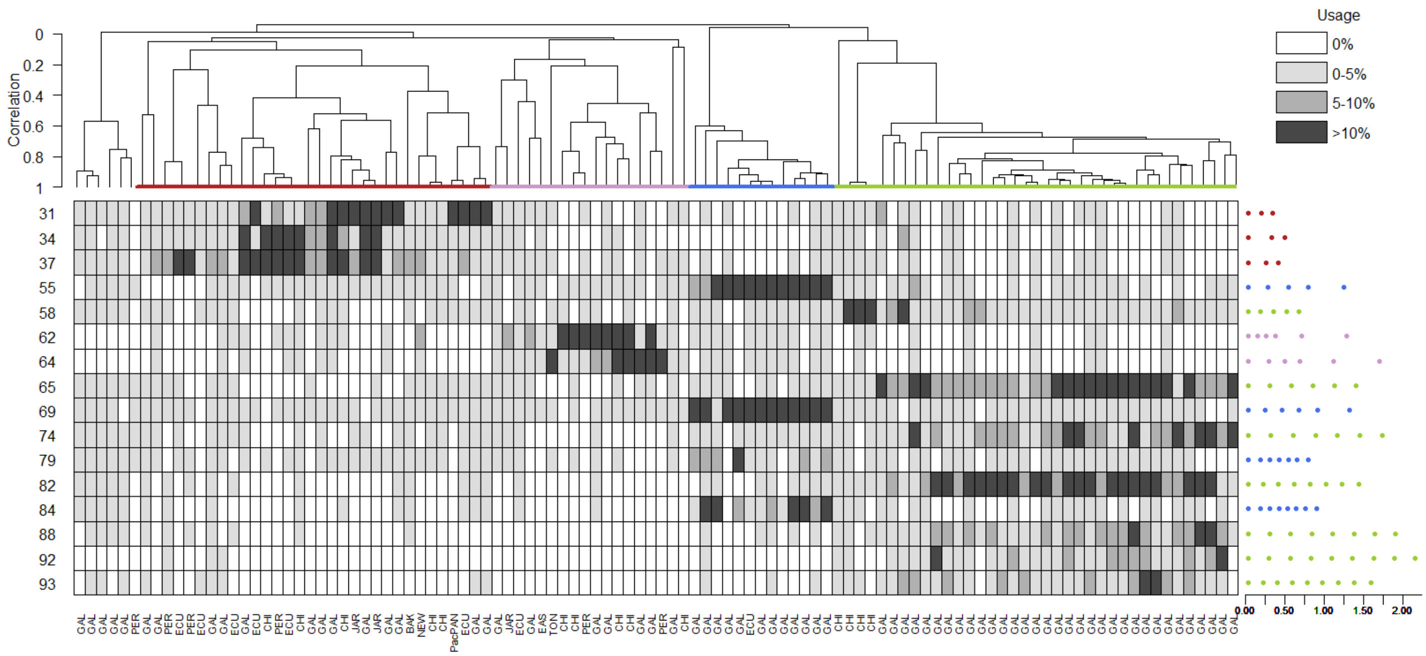
Trial 13 (minrep=4)



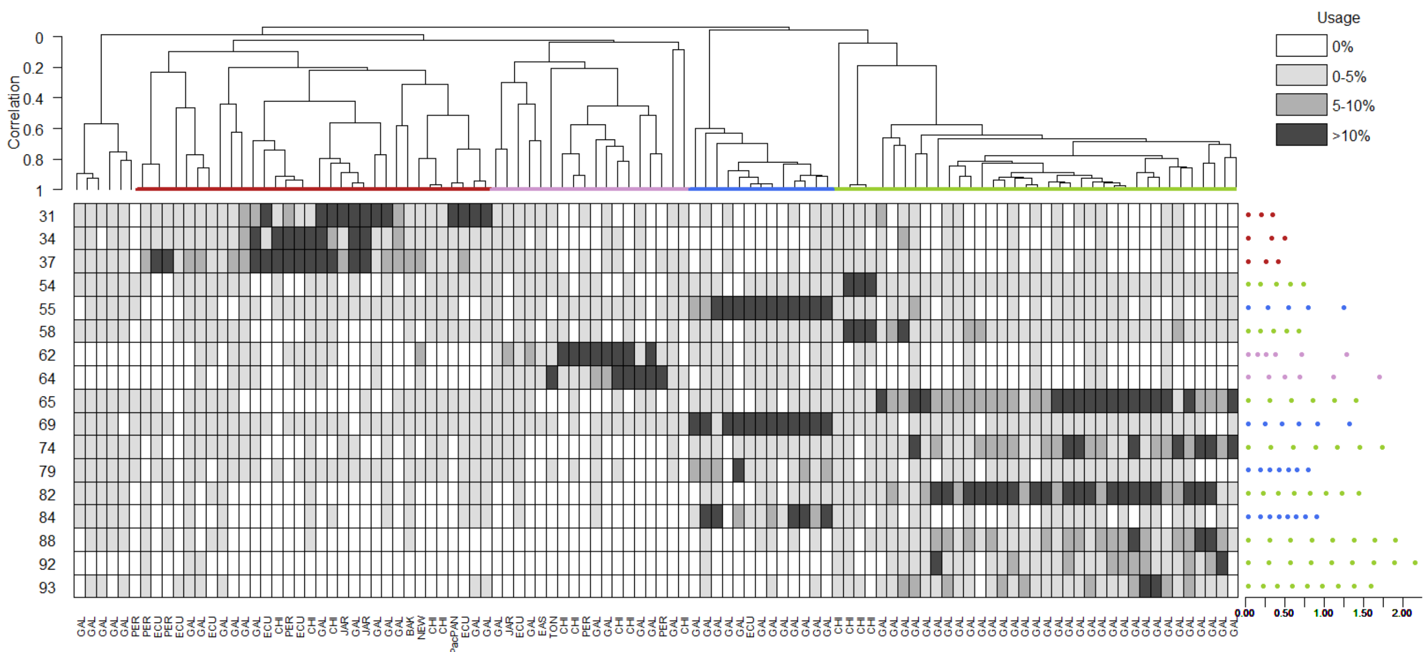
Trial 14 (minrep=8)



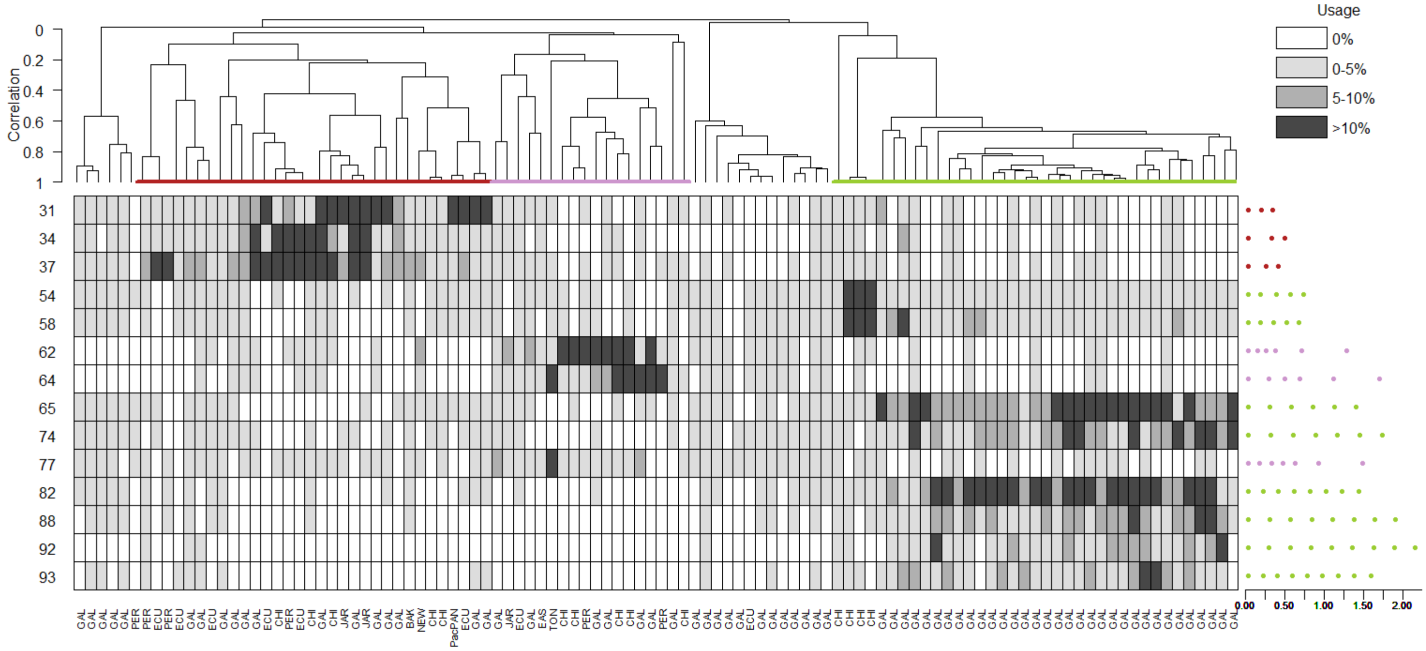
Trial 15 (minrep=10)



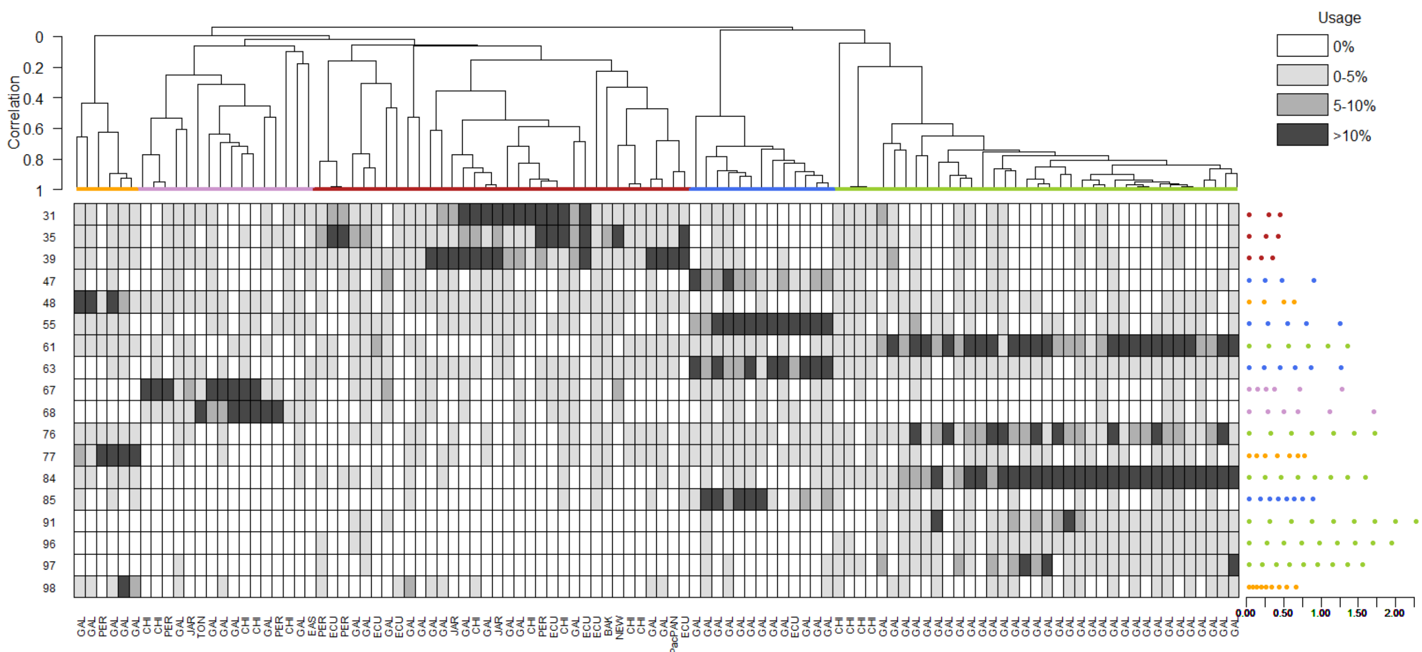
Trial 16 (minrep=12)



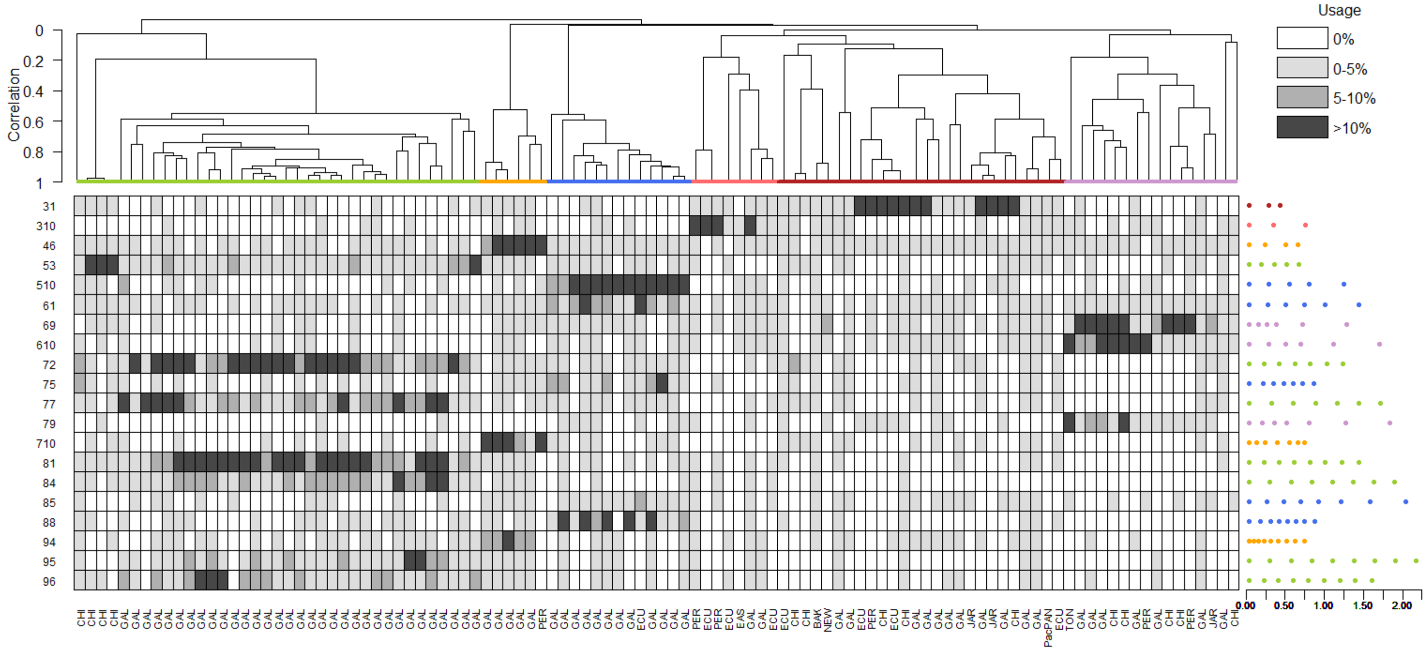
Trial 17 (minrep=14)



Trial 18 (baseline run 2)

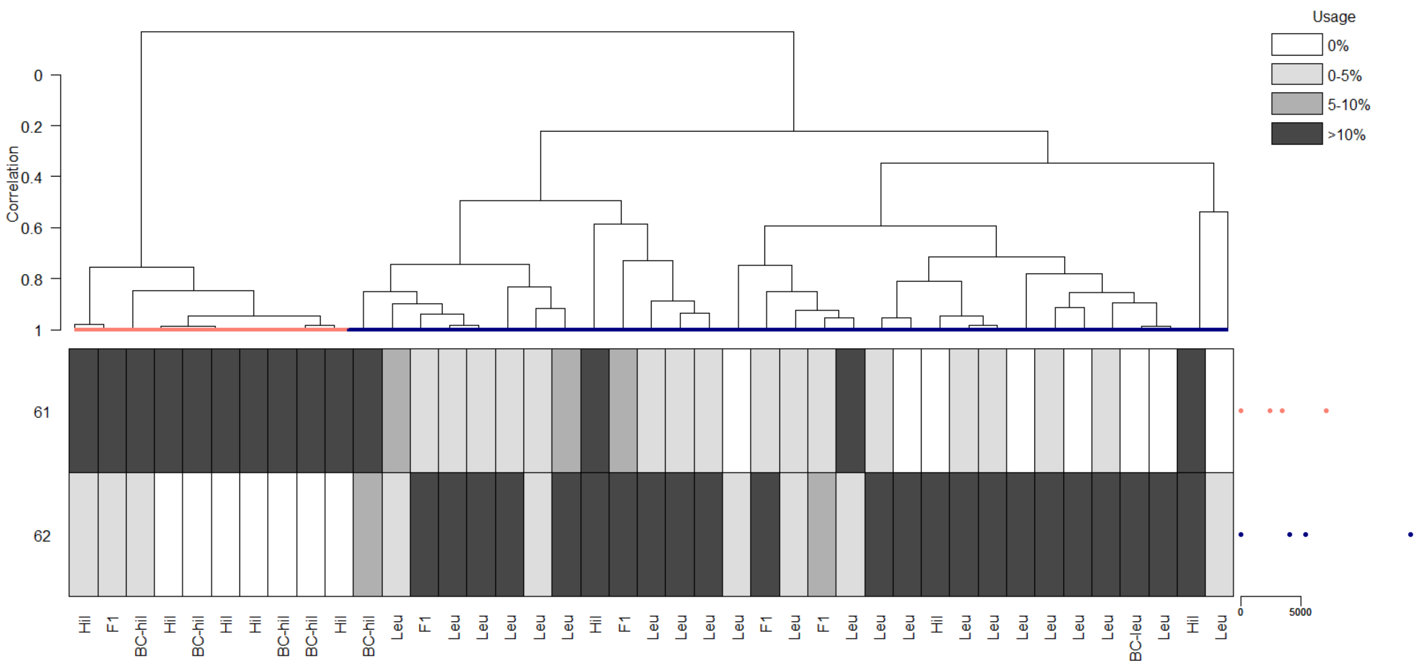


Trial 19 (baseline run 3)



Grey-breasted wood-wrens

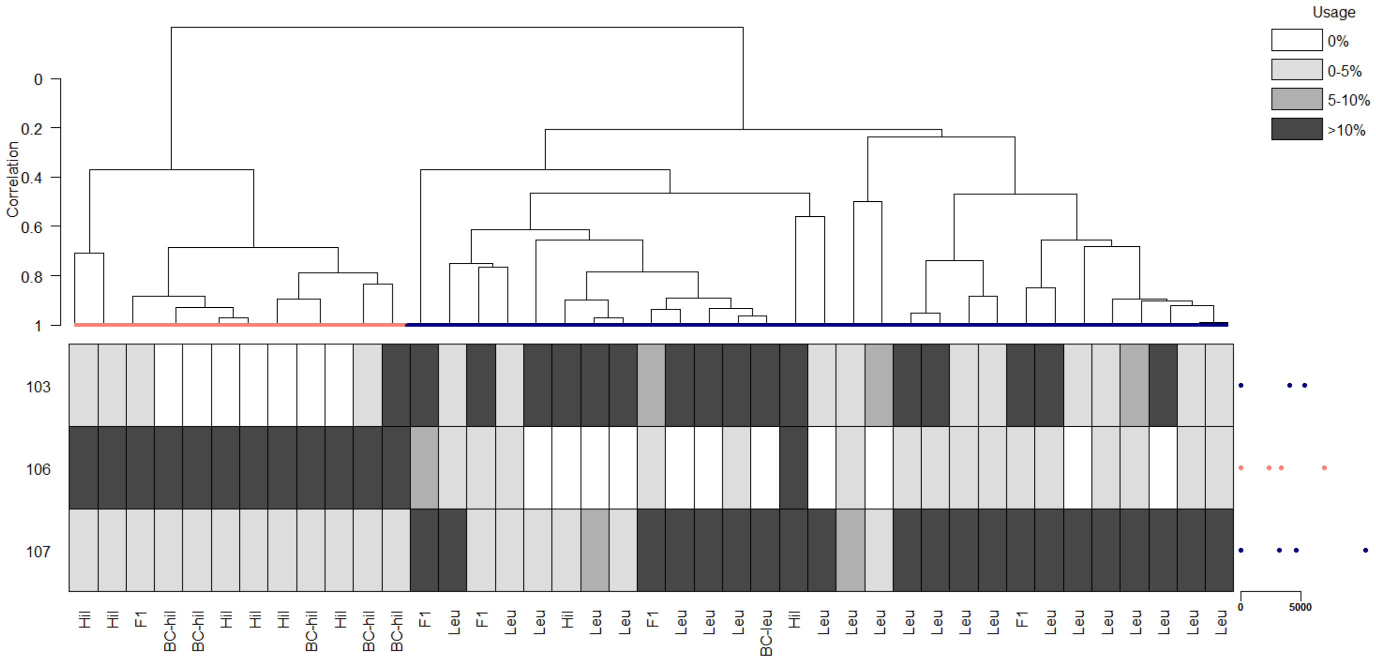
Trial 1 (initialization=random.post)



Trial 2 (initialization=random.clas)

- No identity calls or clades delineated

Trial 3 (information criterion=AIC)



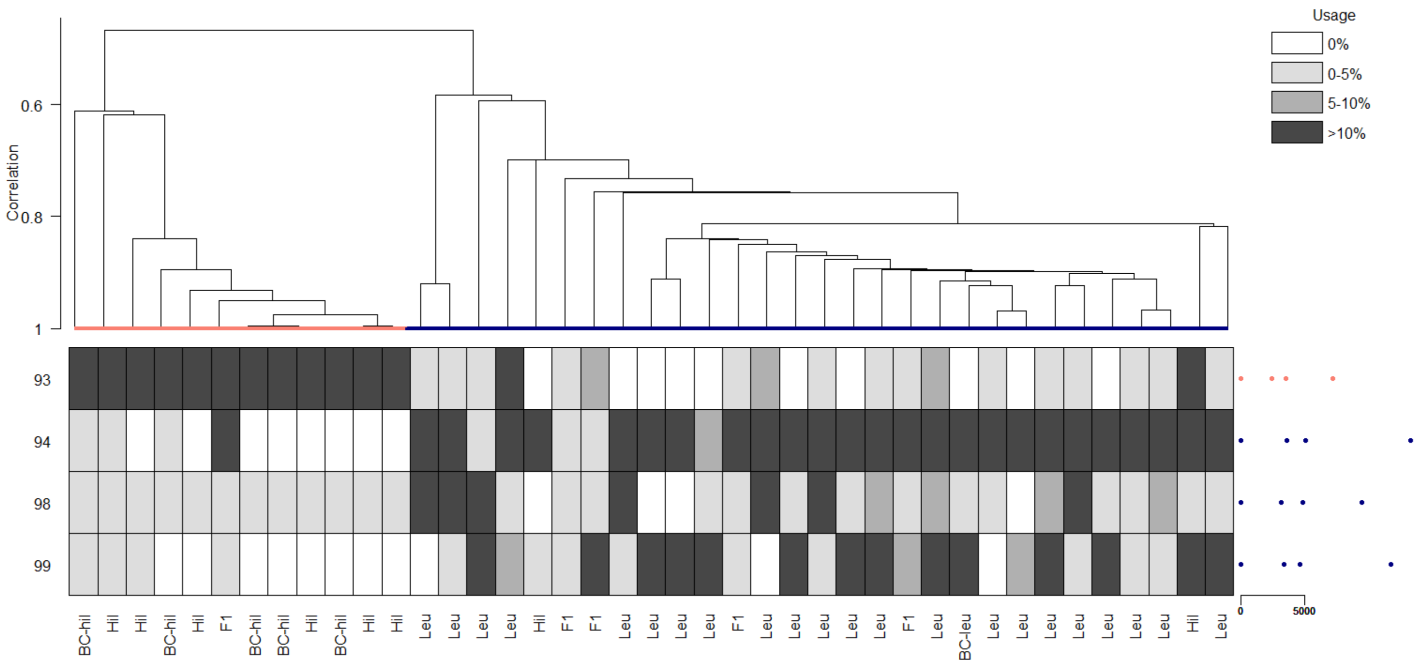
Trial 4 (information criterion=BIC)

- No identity calls or clades delineated

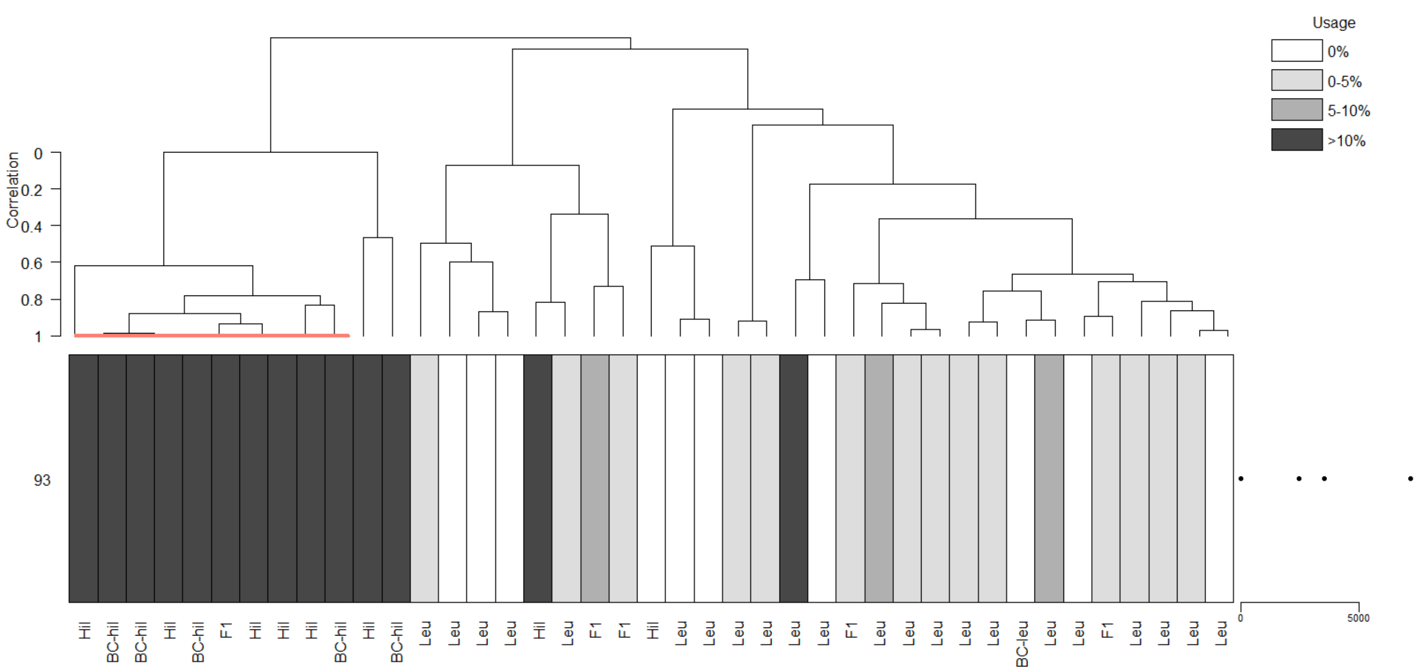
Trial 5 (information criterion=ICL)

- No identity calls or clades delineated

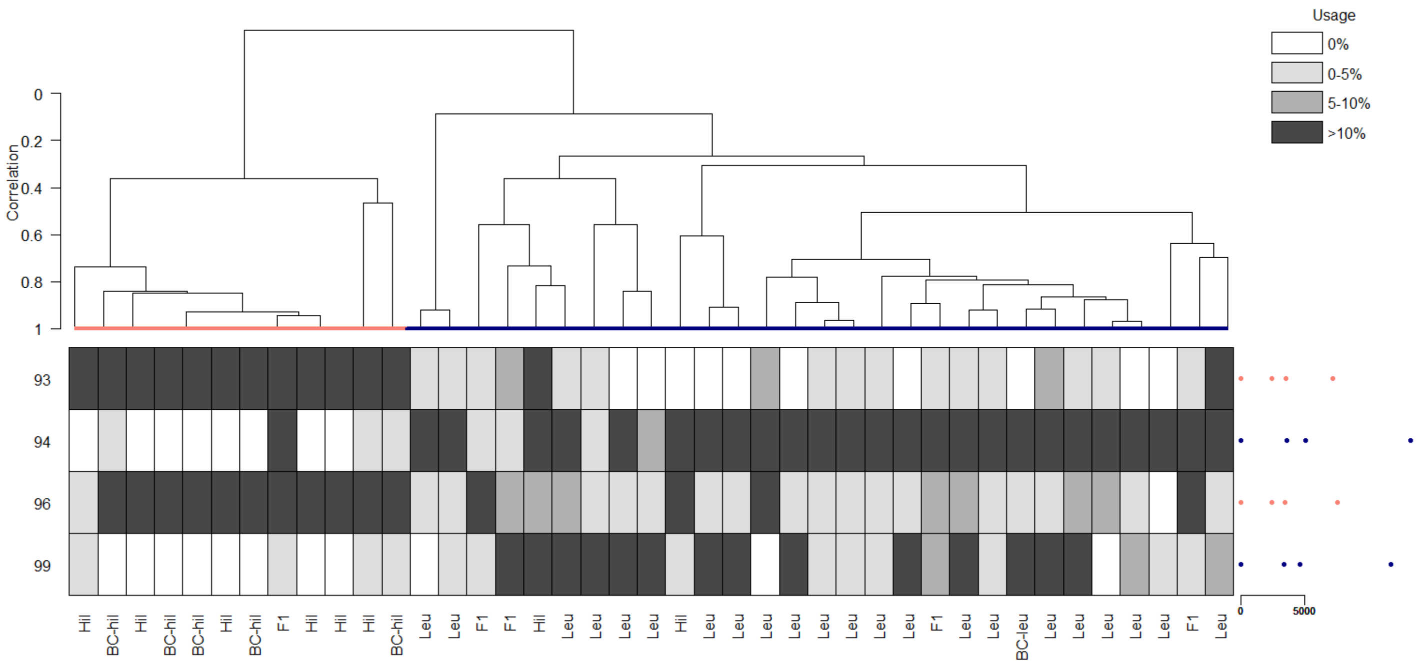
Trial 6 (linkage=single)



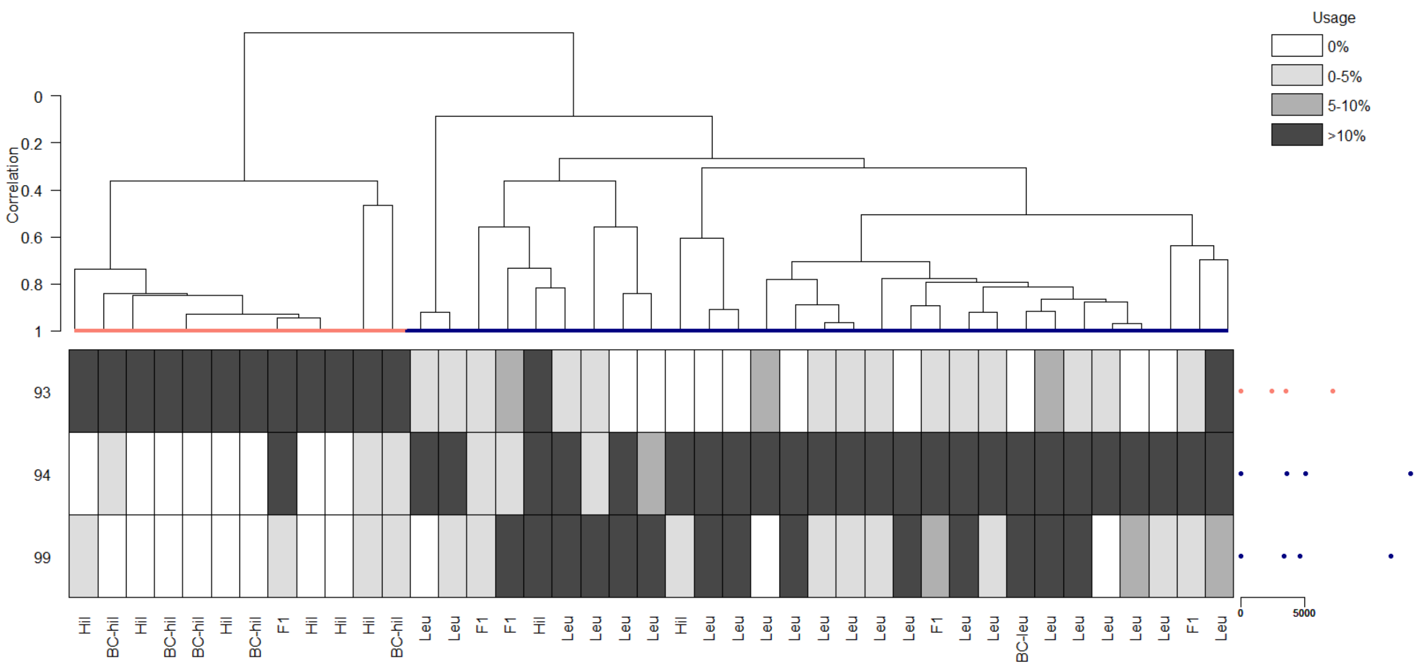
Trial 7 (linkage=complete)



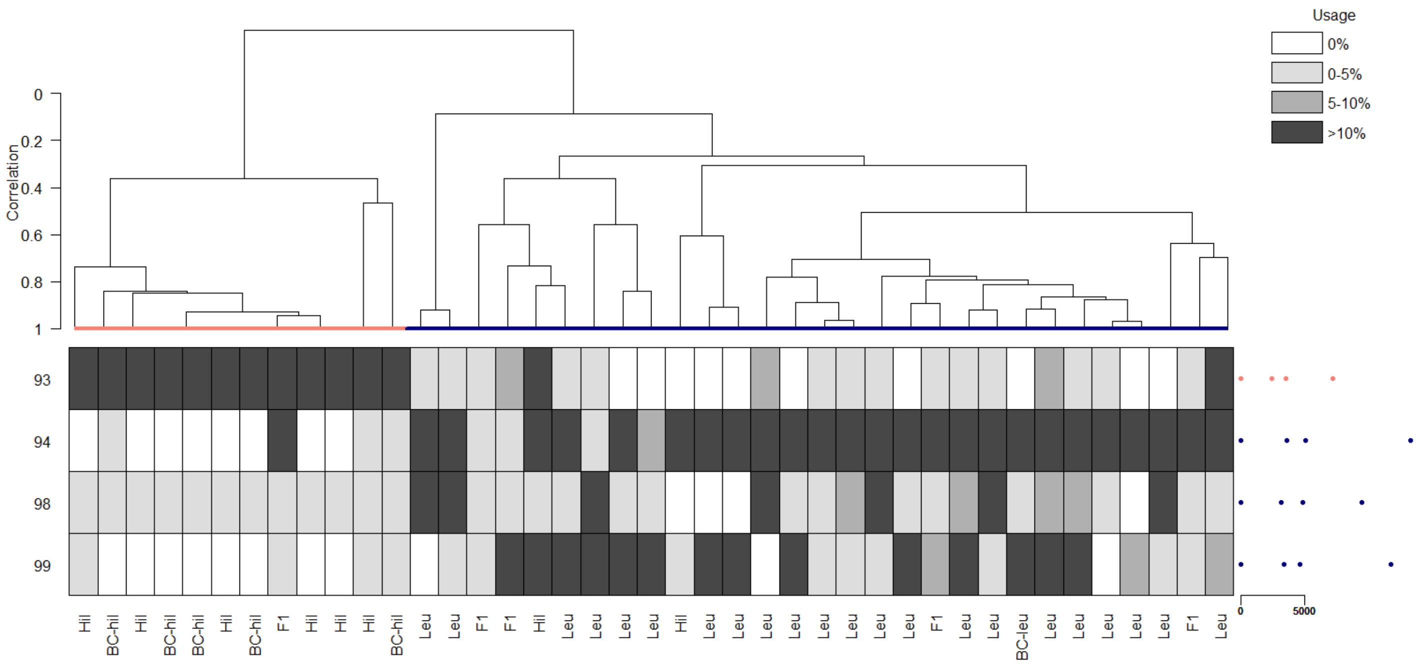
Trial 8 (critfact=6)



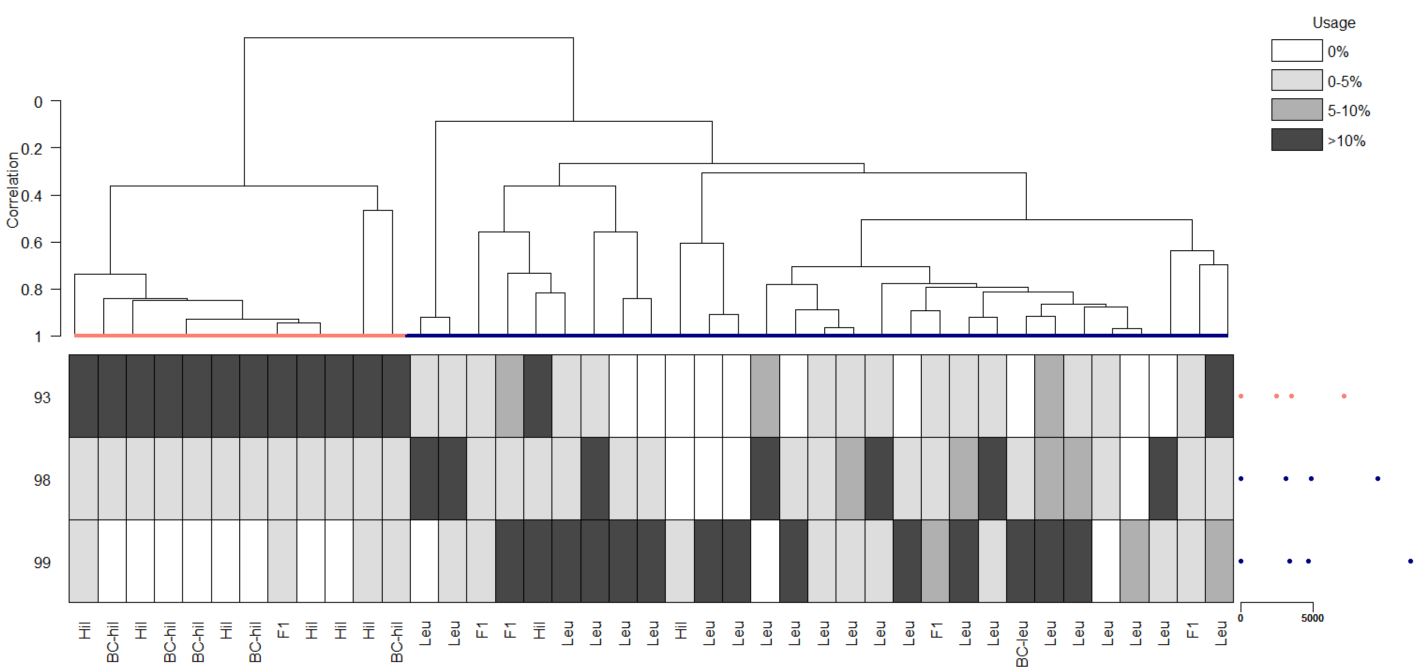
Trial 9 (critfact=10)



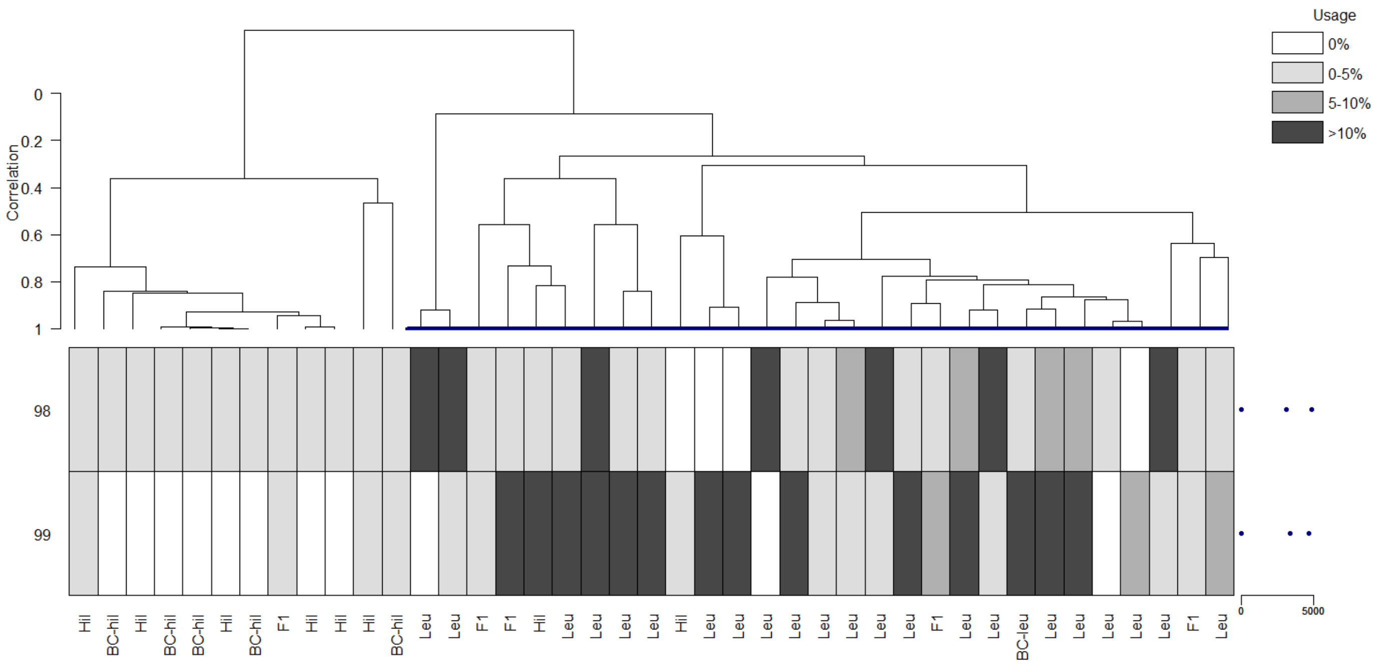
Trial 10 (critfact=18)



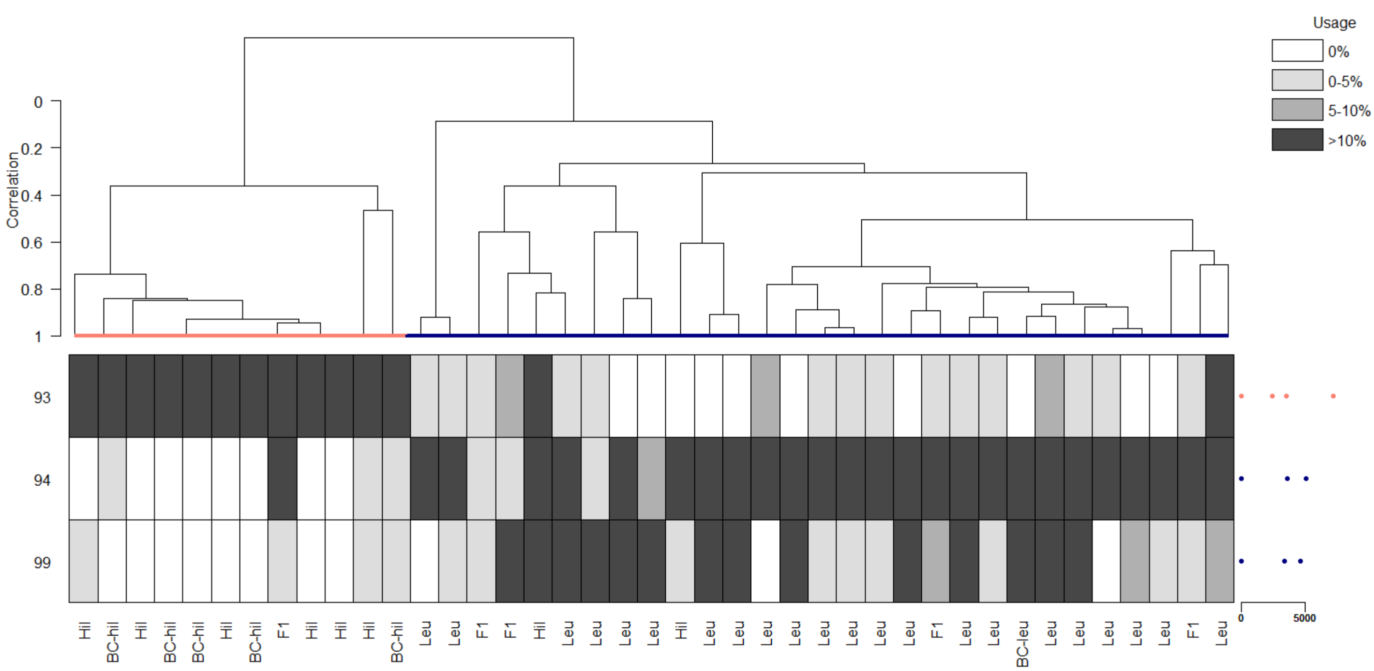
Trial 11 (critfact=22)



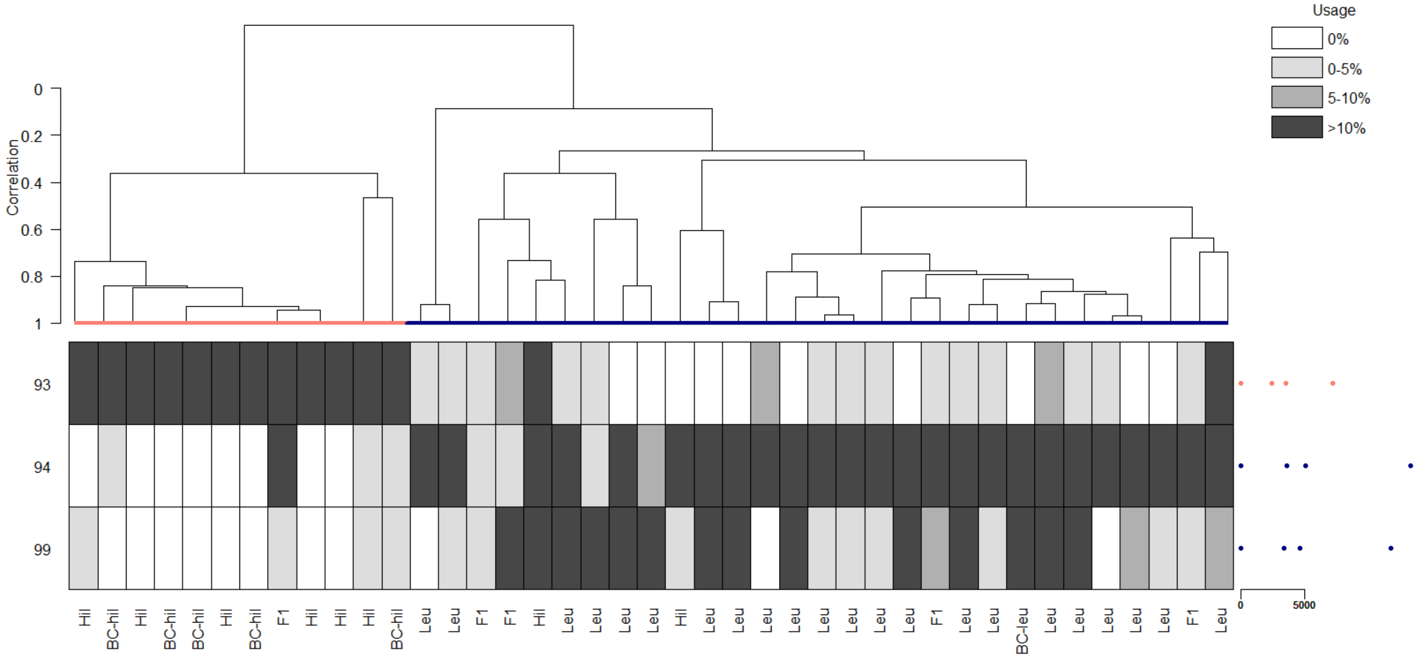
Trial 12 (critfact=26)



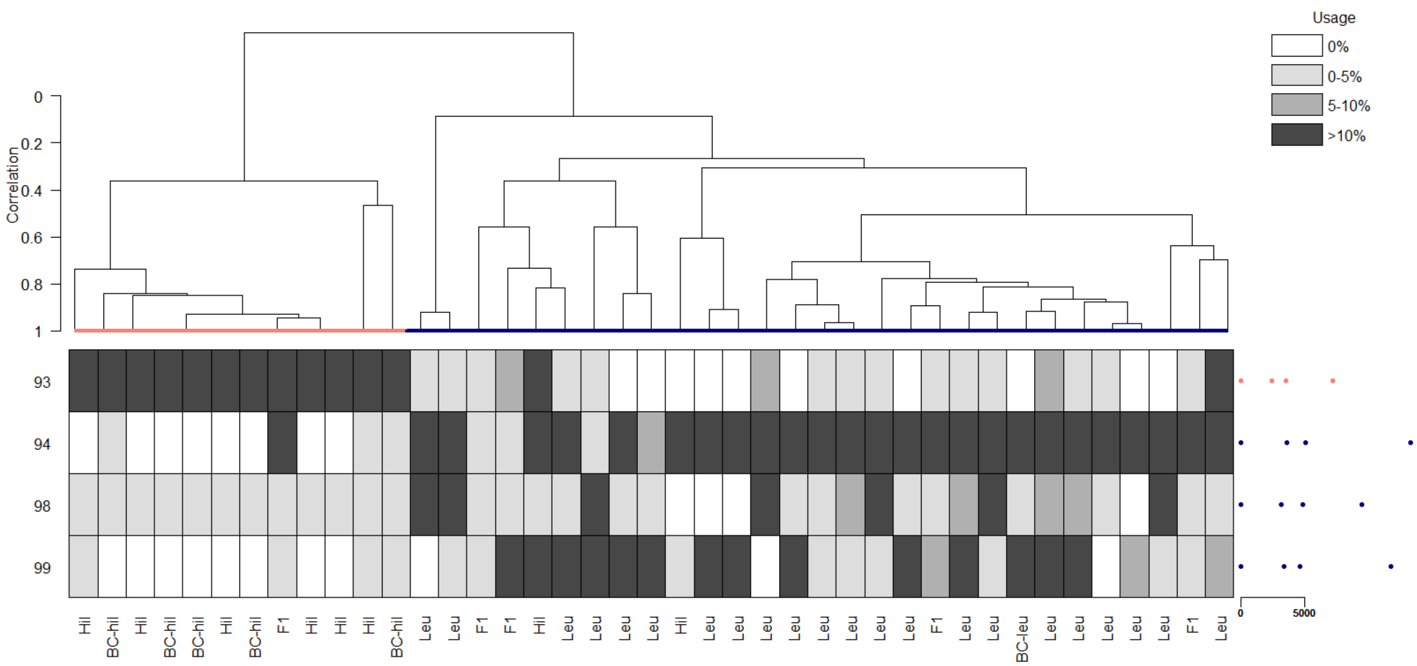
Trial 13 (minrep=3)



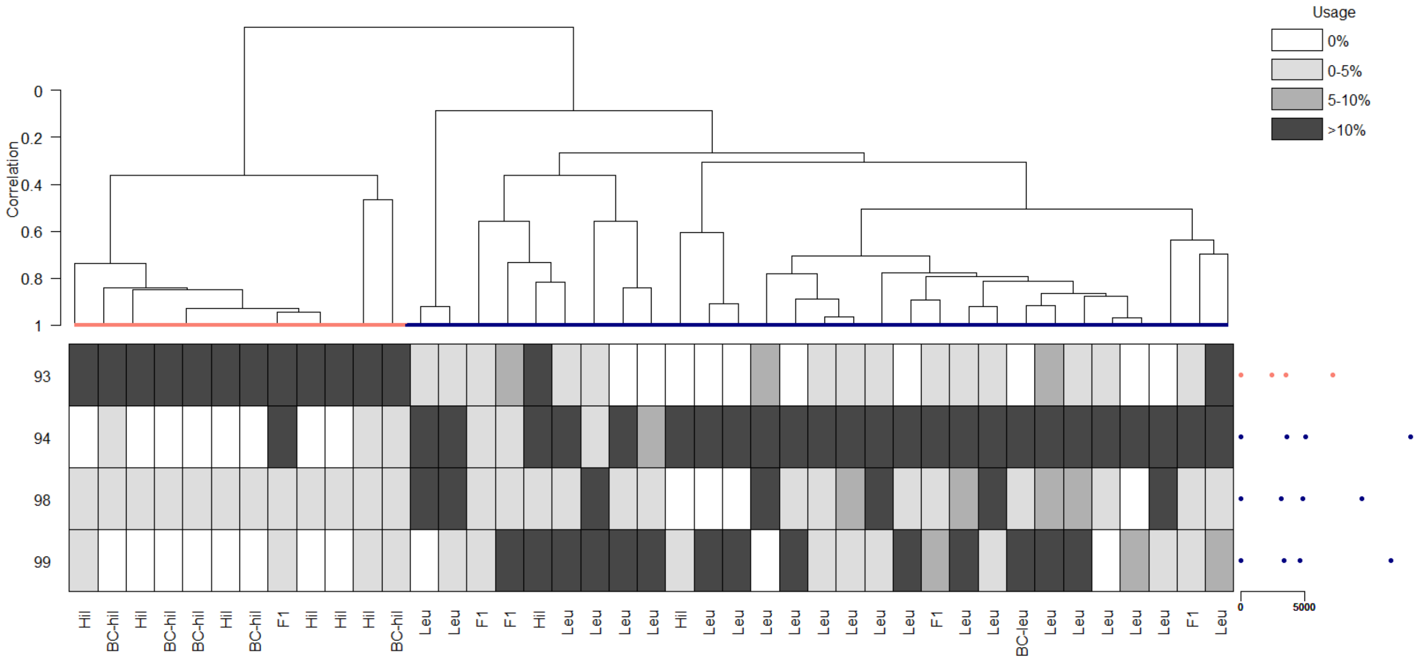
Trial 14 (minrep=5)



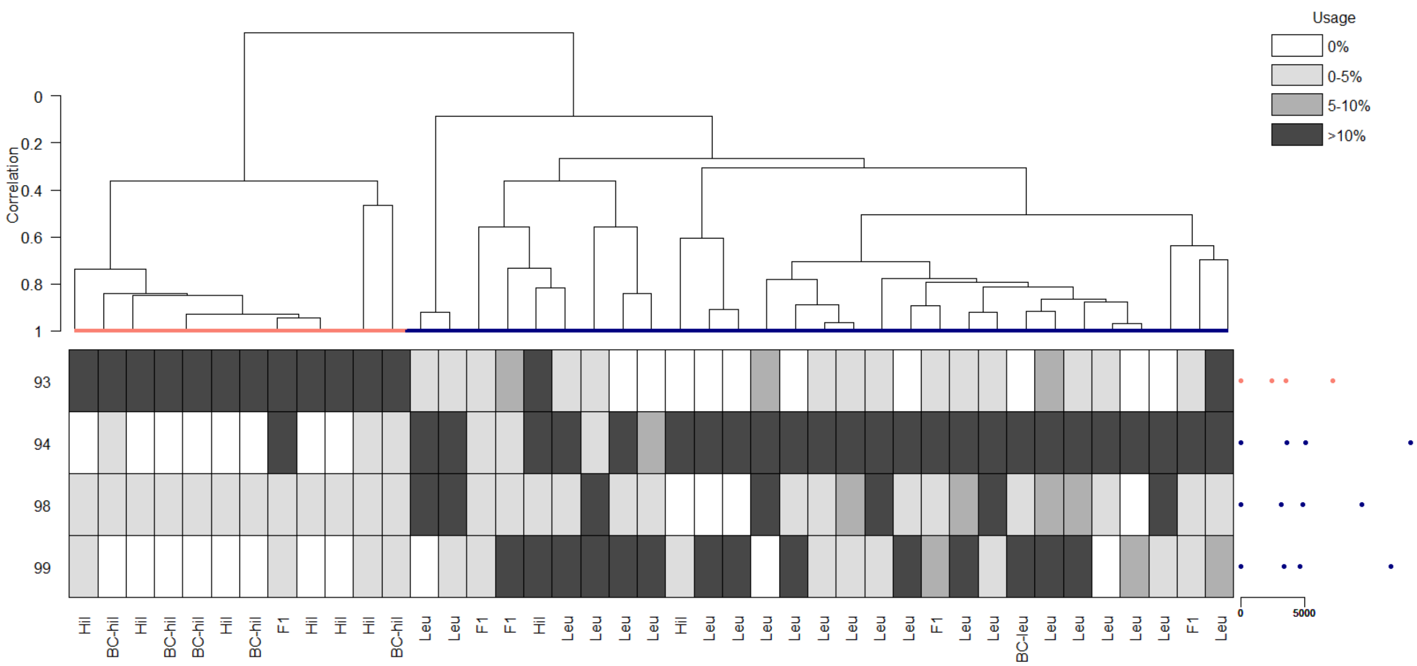
Trial 15 (minrep=7)



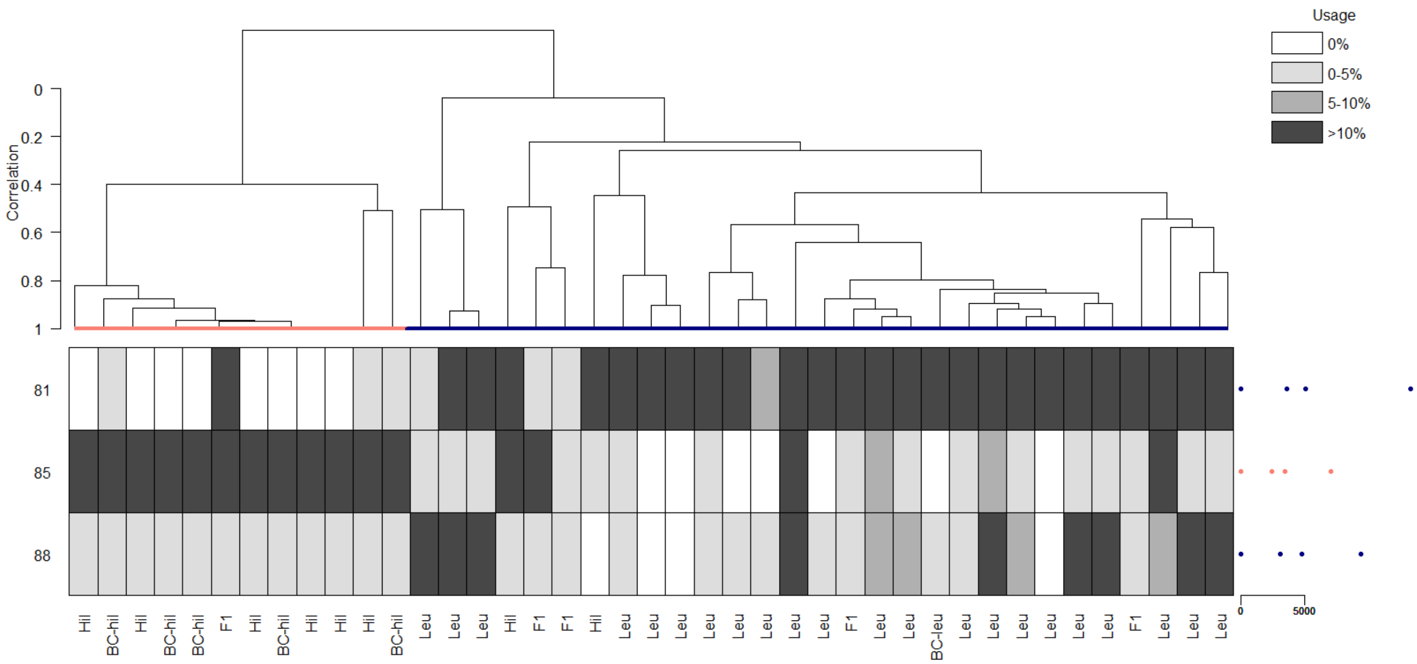
Trial 16 (minrep=9)



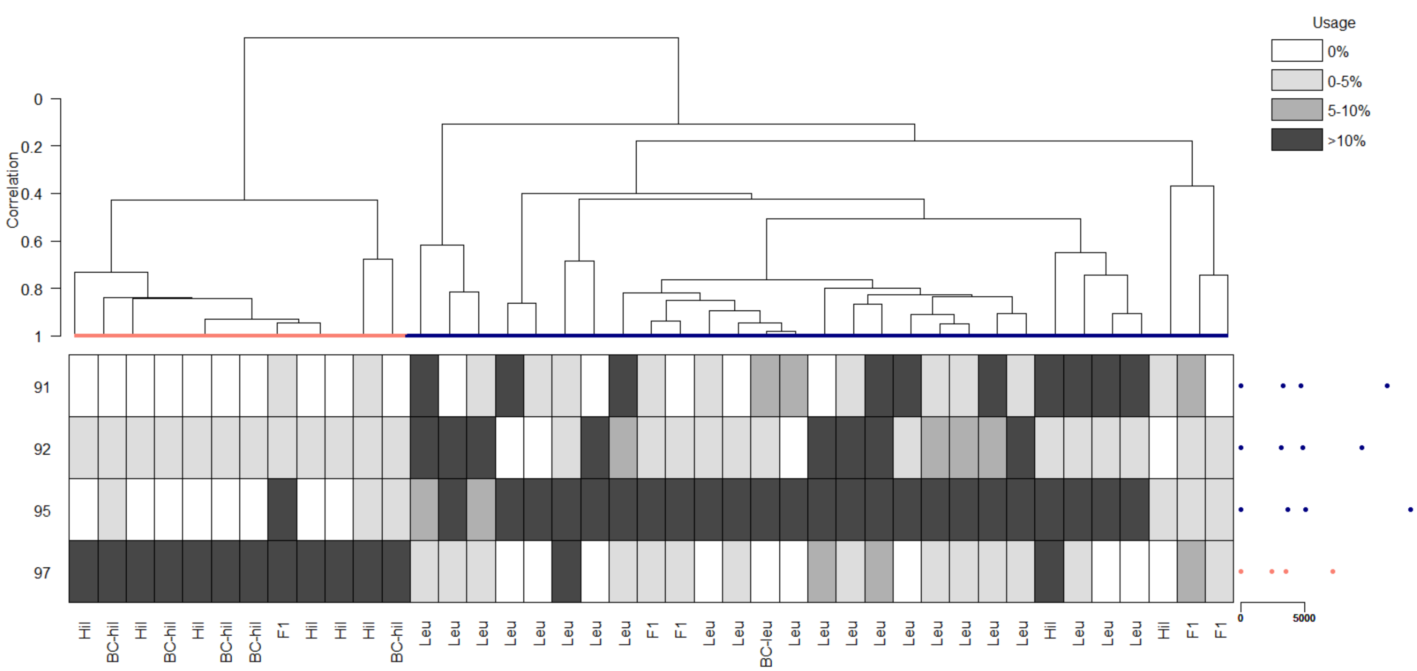
Trial 17 (minrep=11)



Trial 18 (baseline run 2)

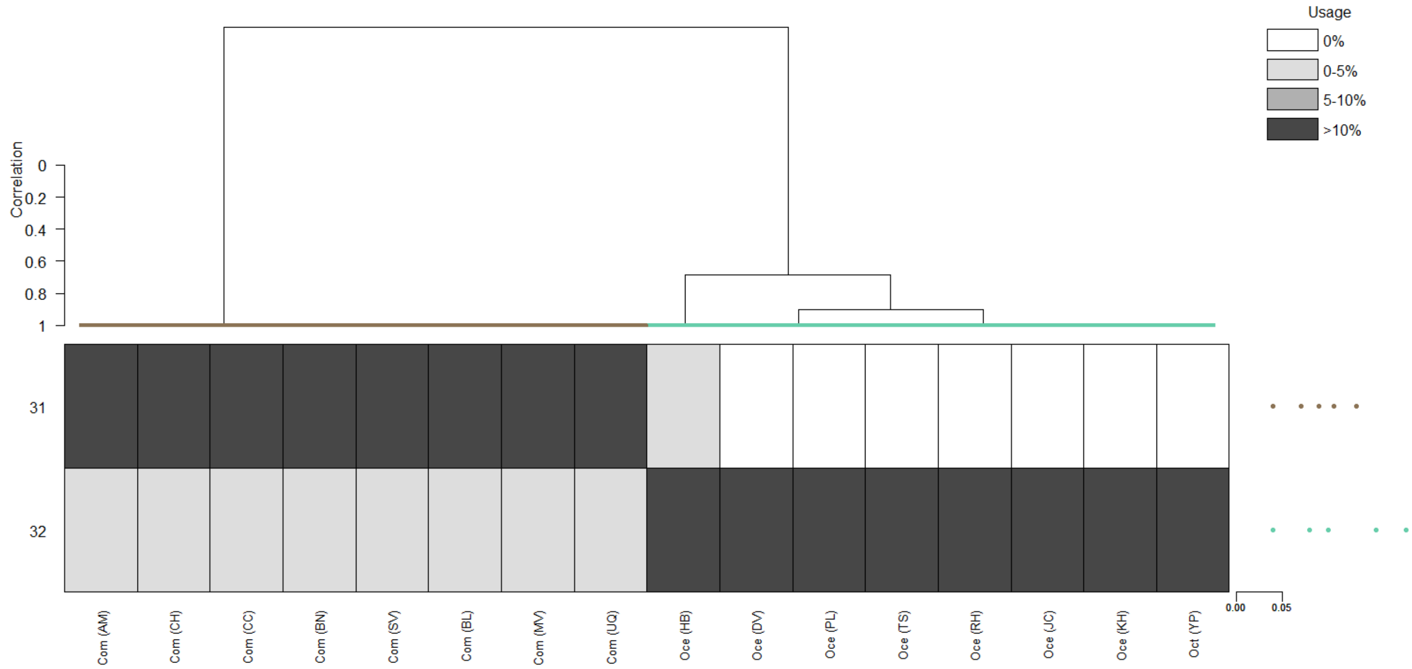


Trial 19 (baseline run 3)

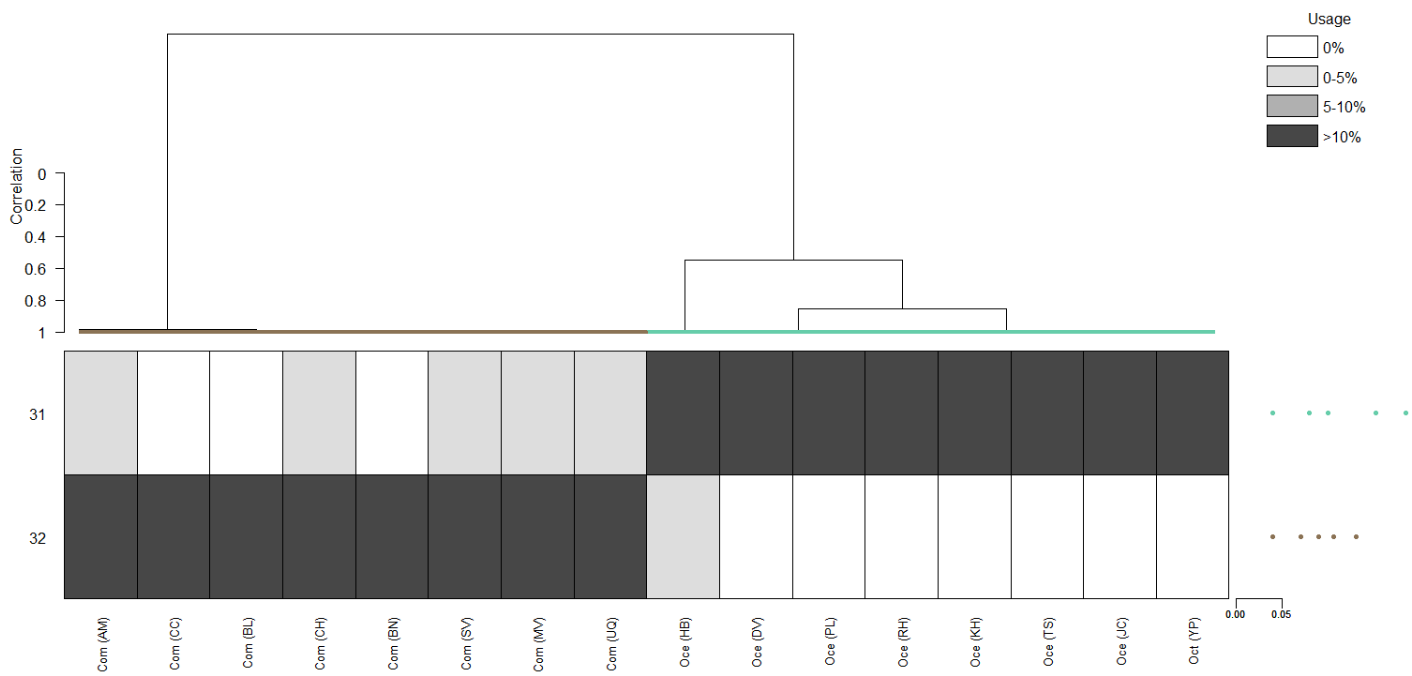


Australian field crickets

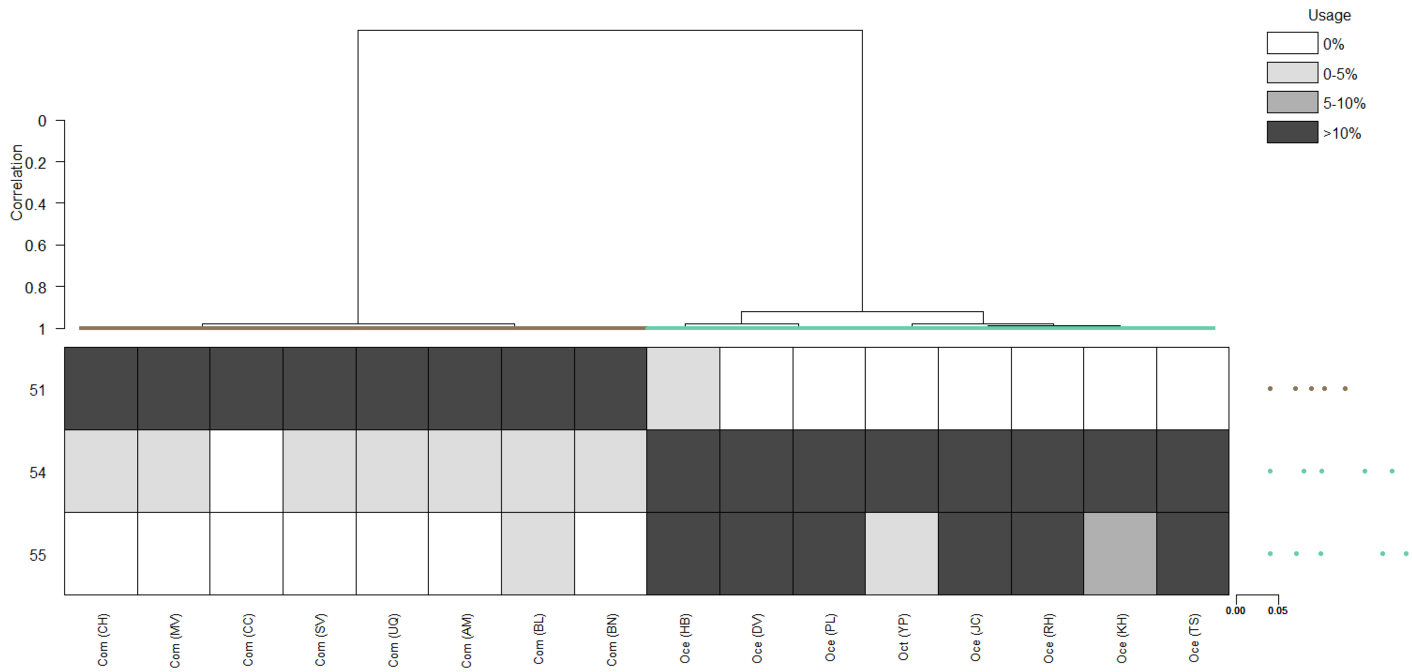
Trial 1 (initialization=random.post)



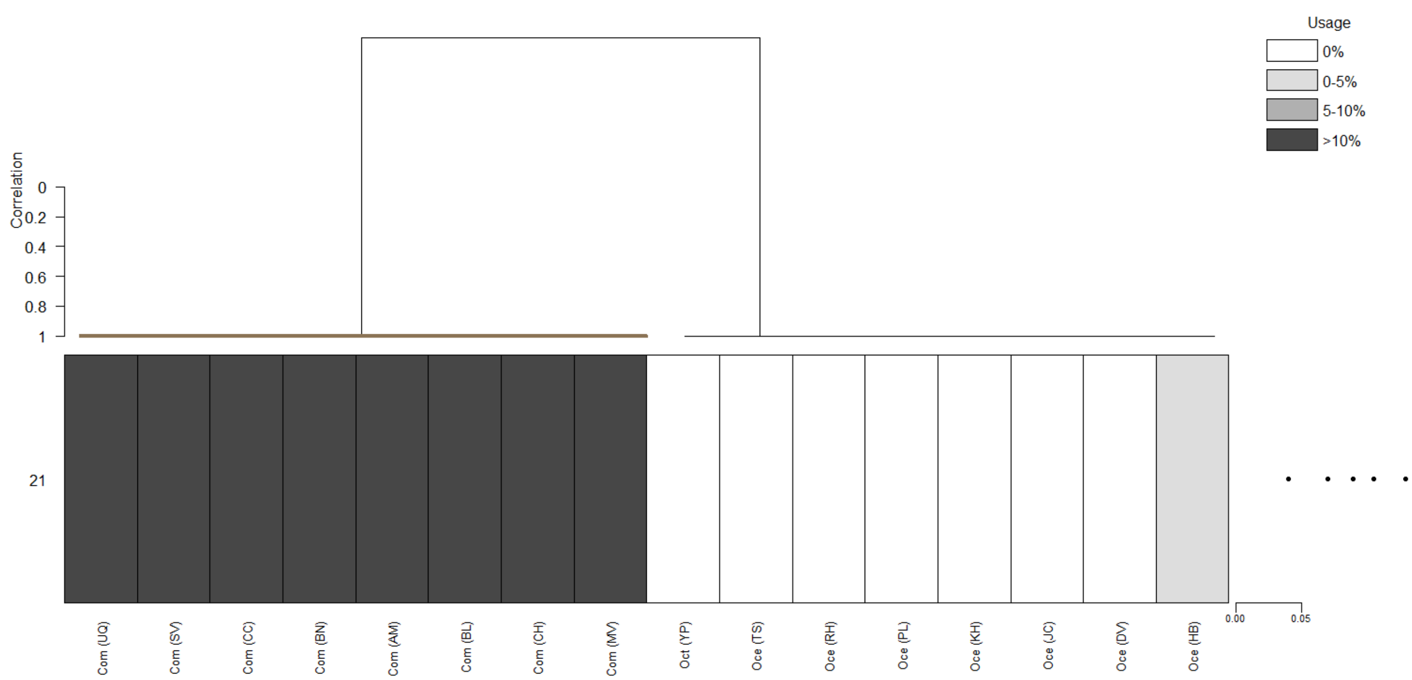
Trial 2 (initialization=random.clas)



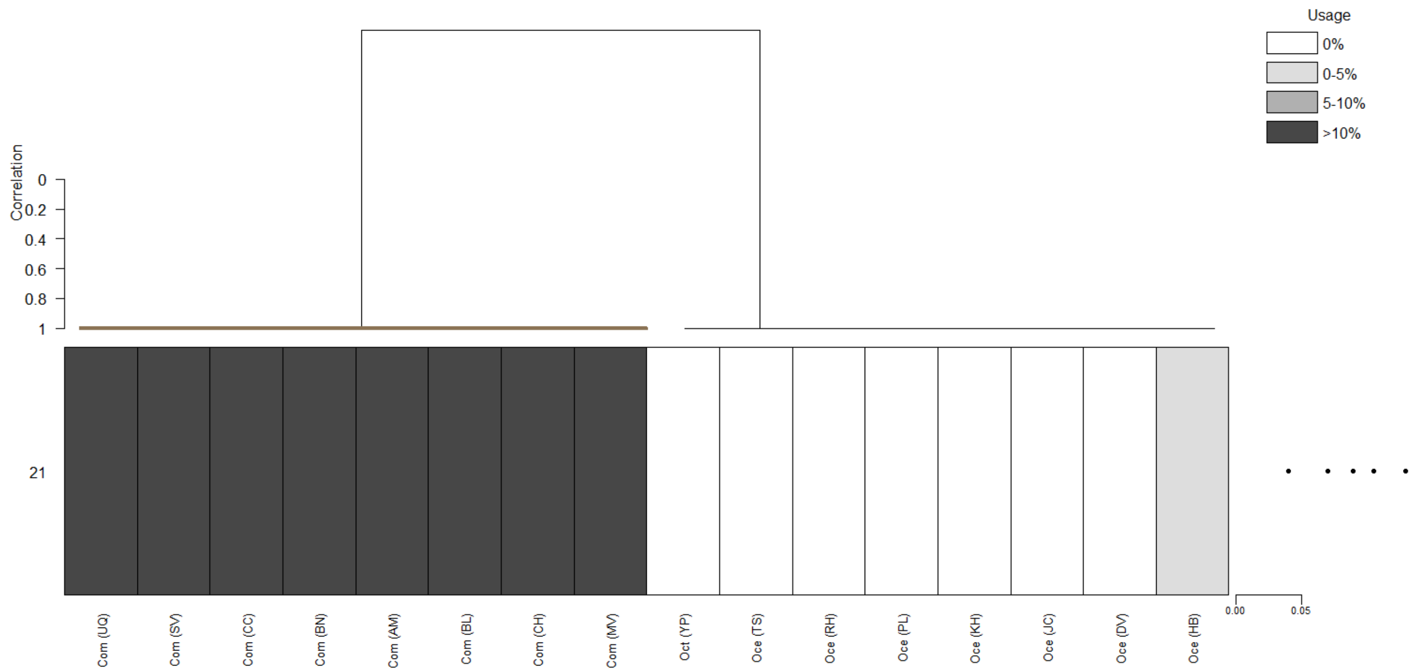
Trial 3 (information criterion=AIC)



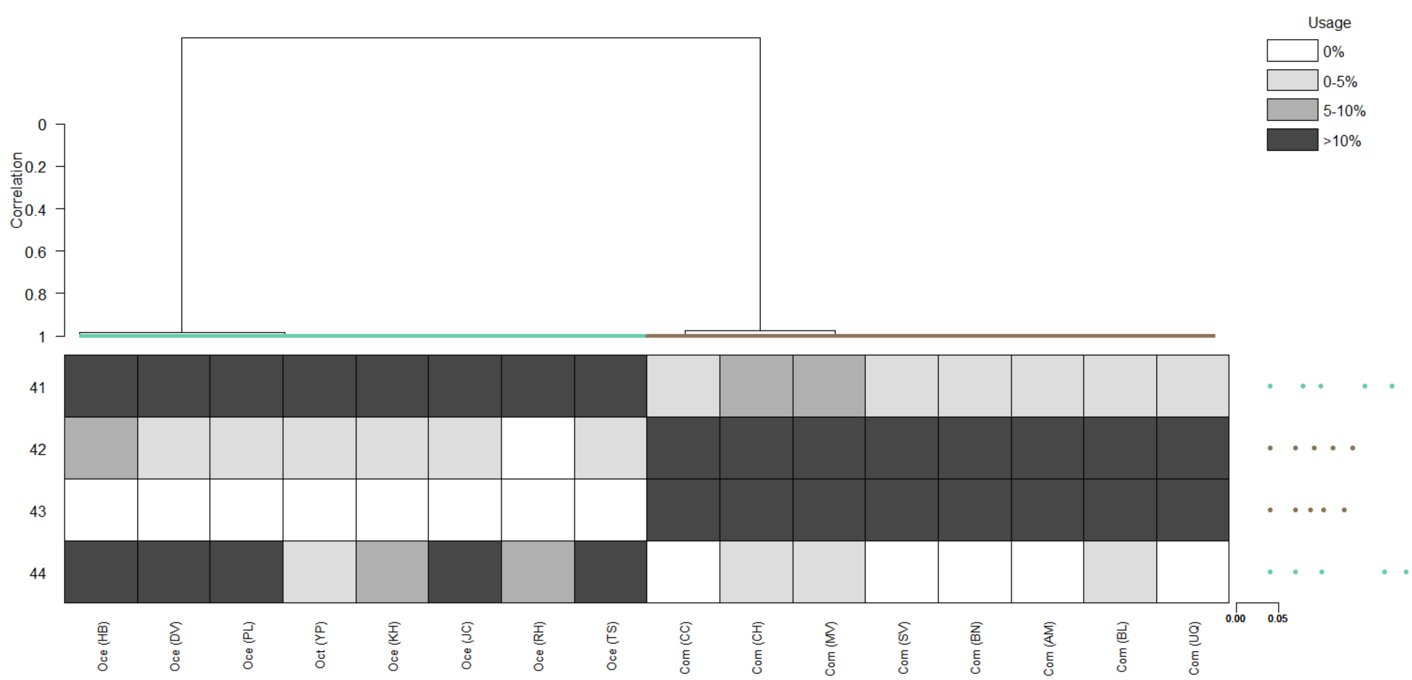
Trial 4 (information criterion=BIC)



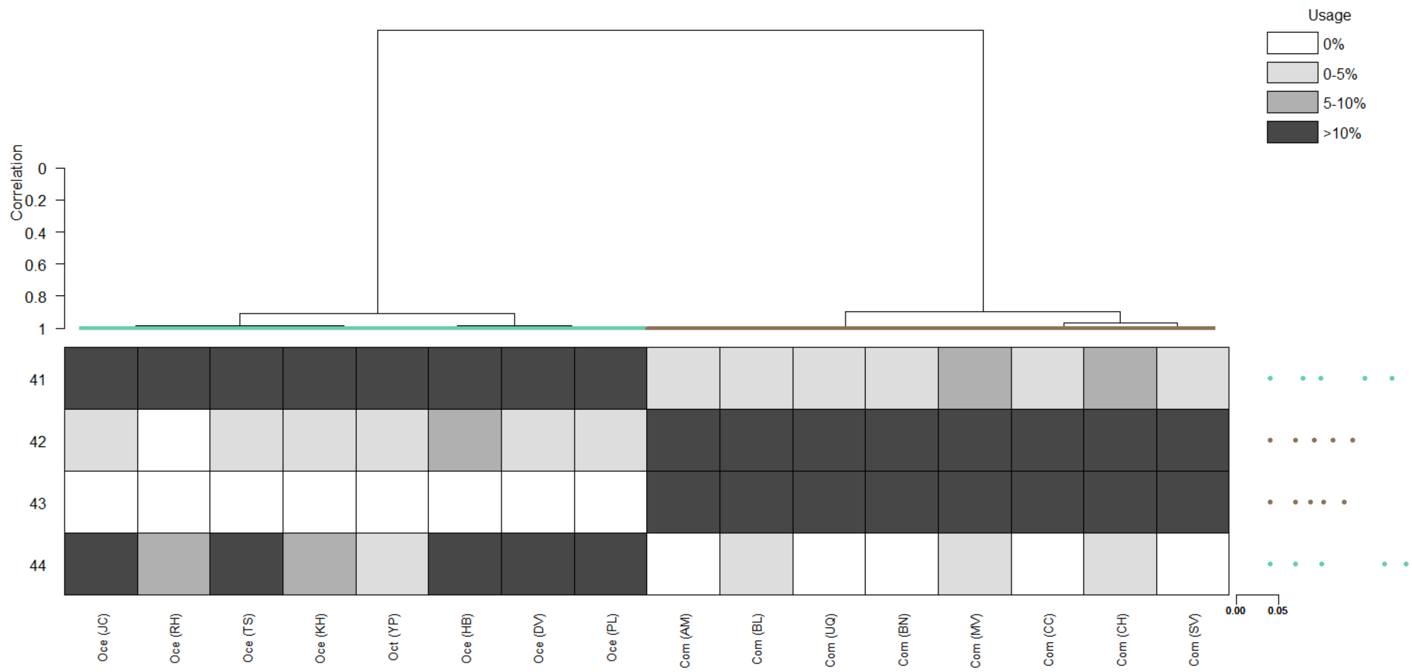
Trial 5 (information criterion=ICL)



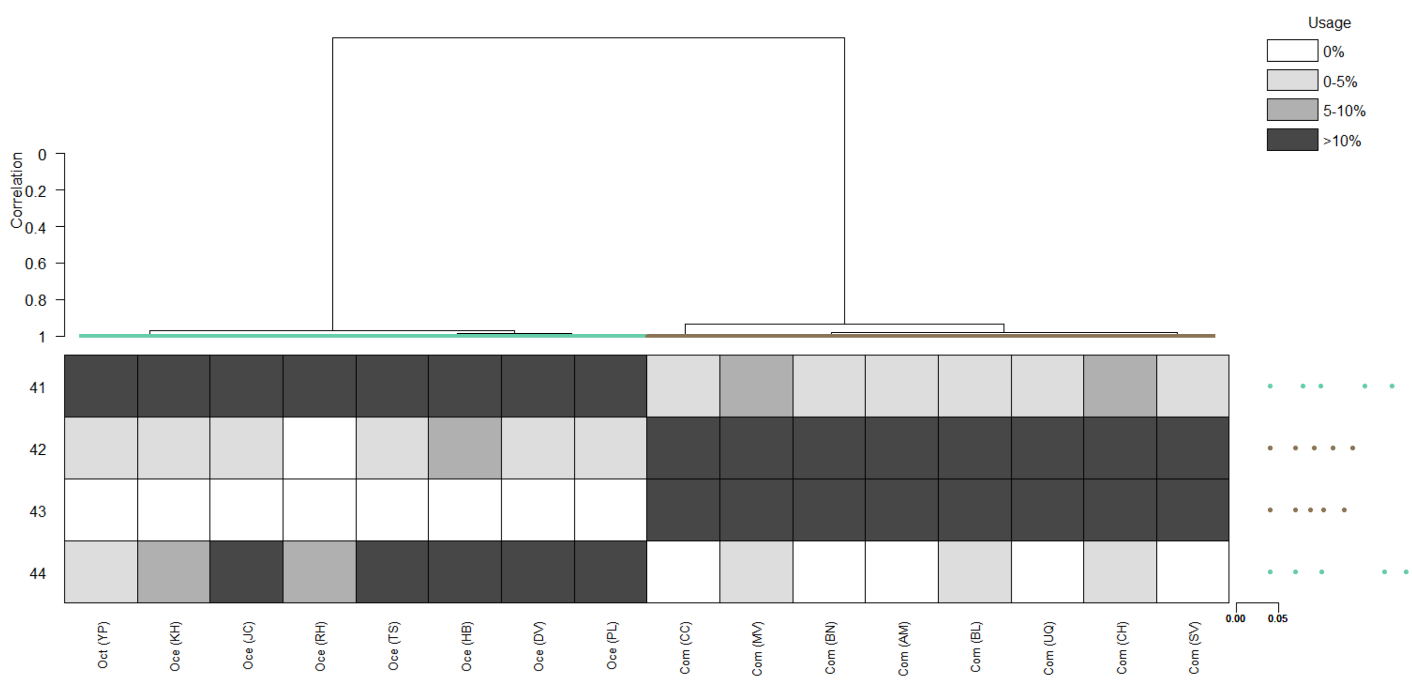
Trial 6 (linkage=single)



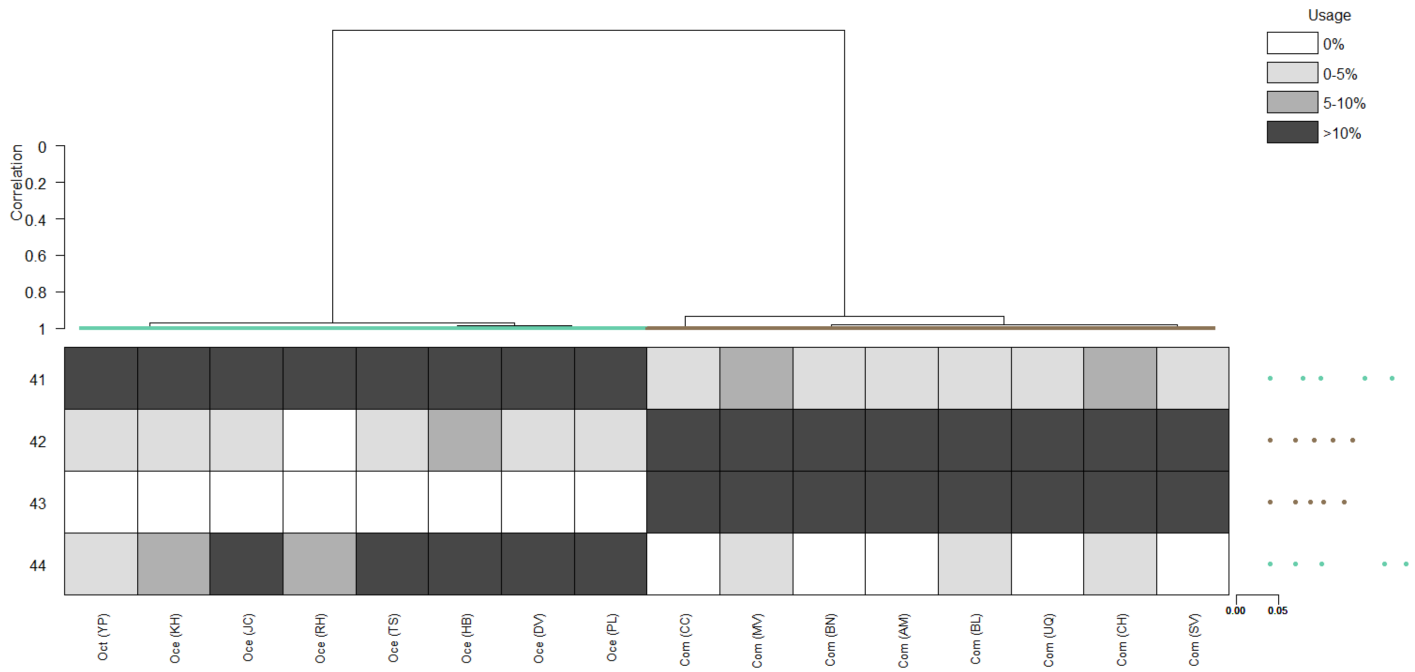
Trial 7 (linkage=complete)



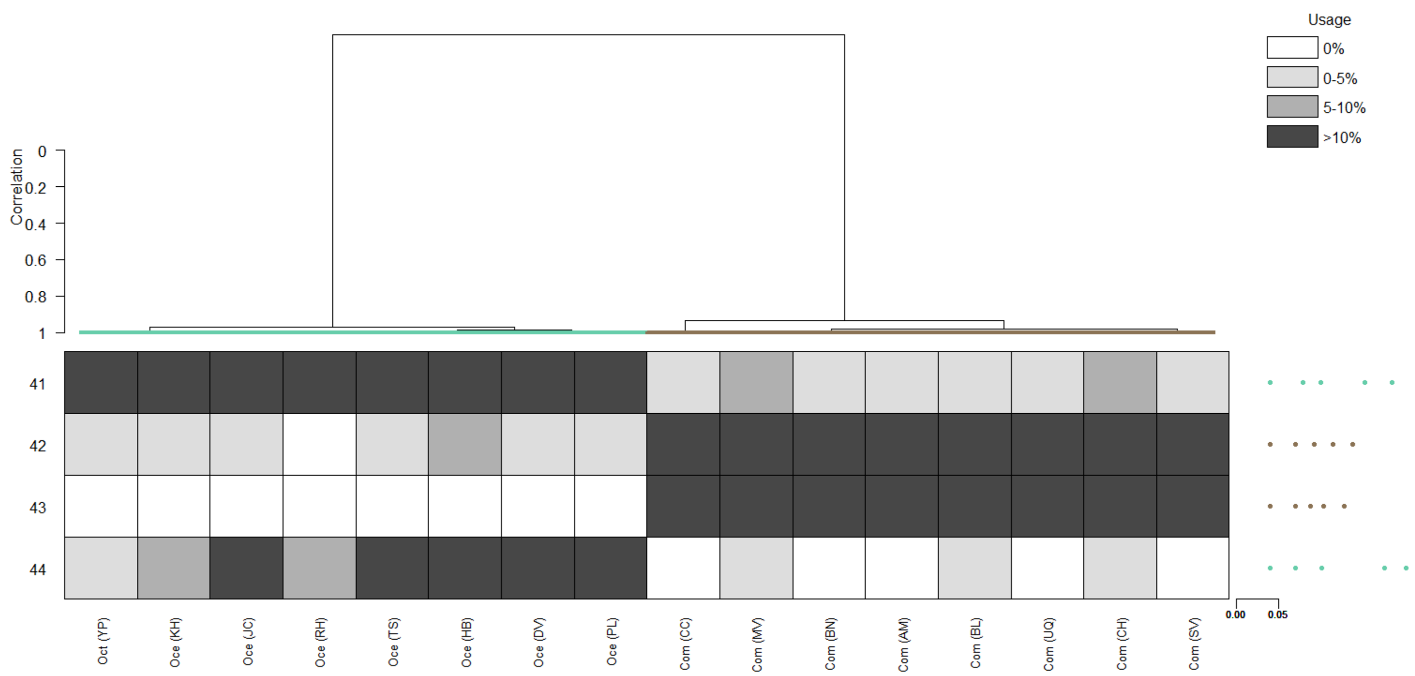
Trial 8 (critfact=6)



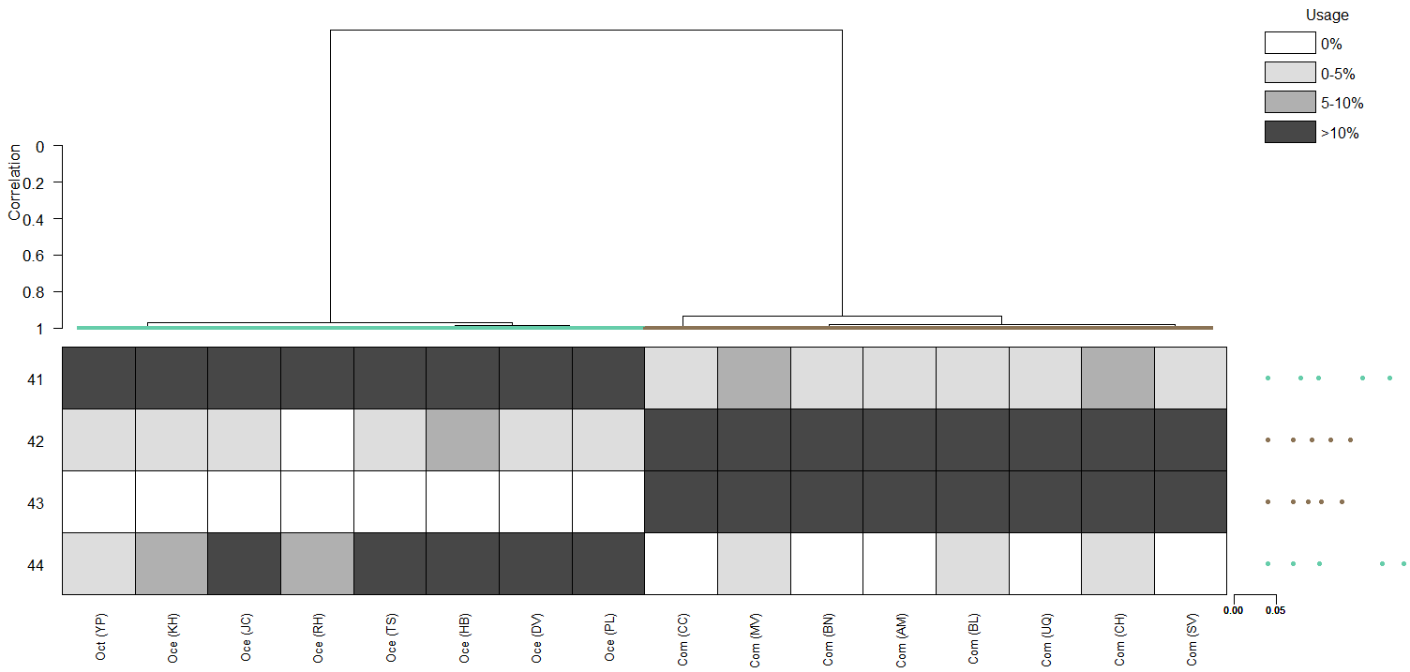
Trial 9 (critfact=10)



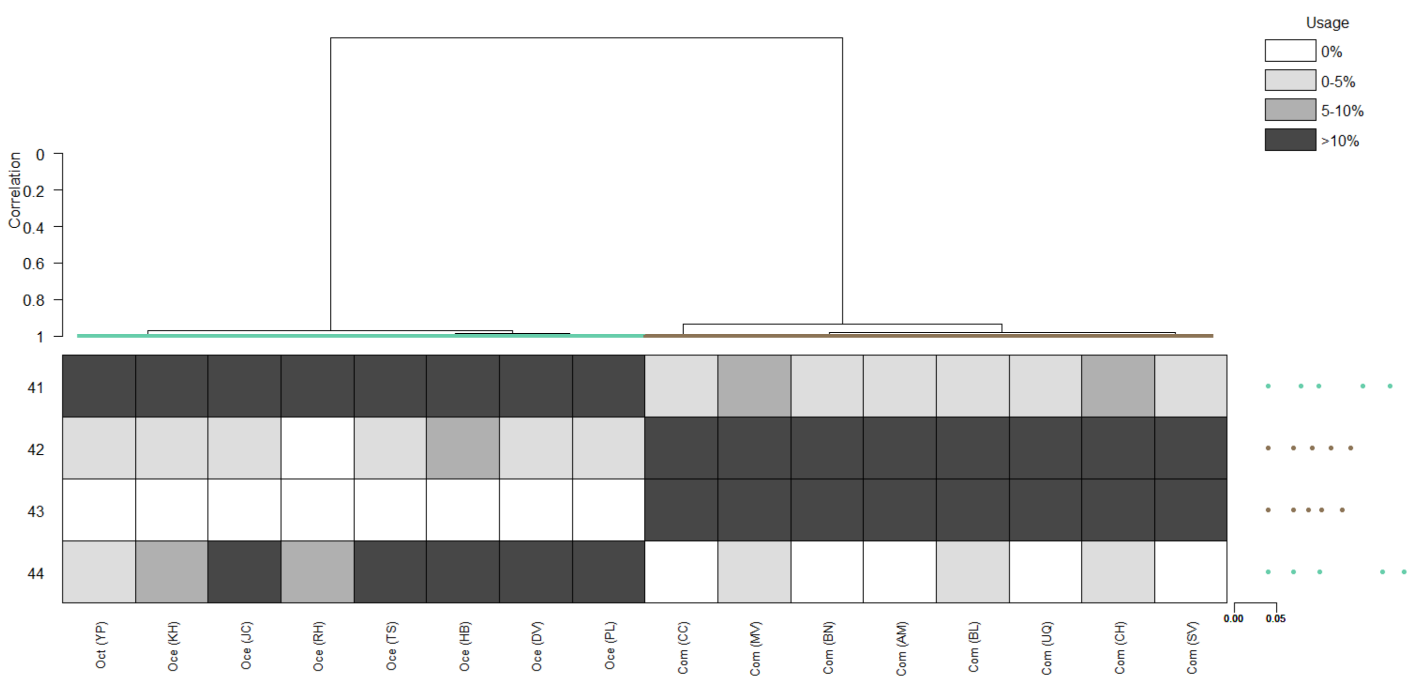
Trial 10 (critfact=18)



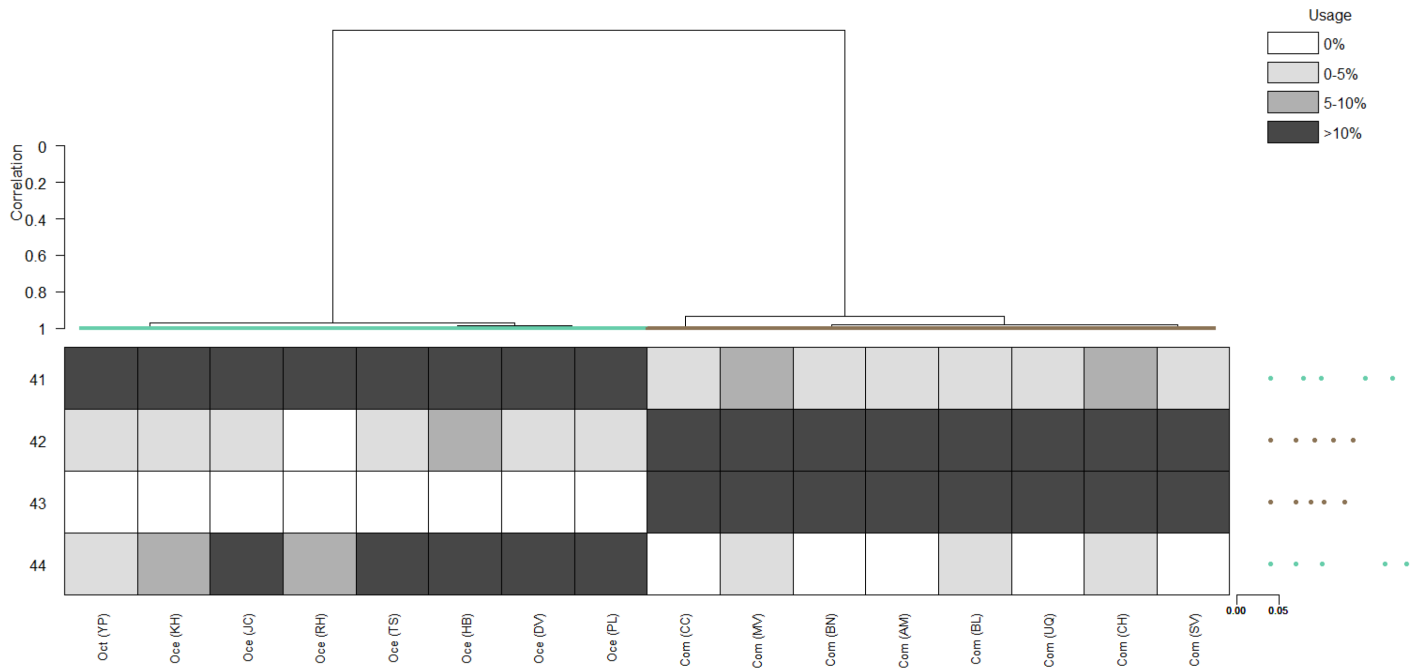
Trial 11 (critfact=22)



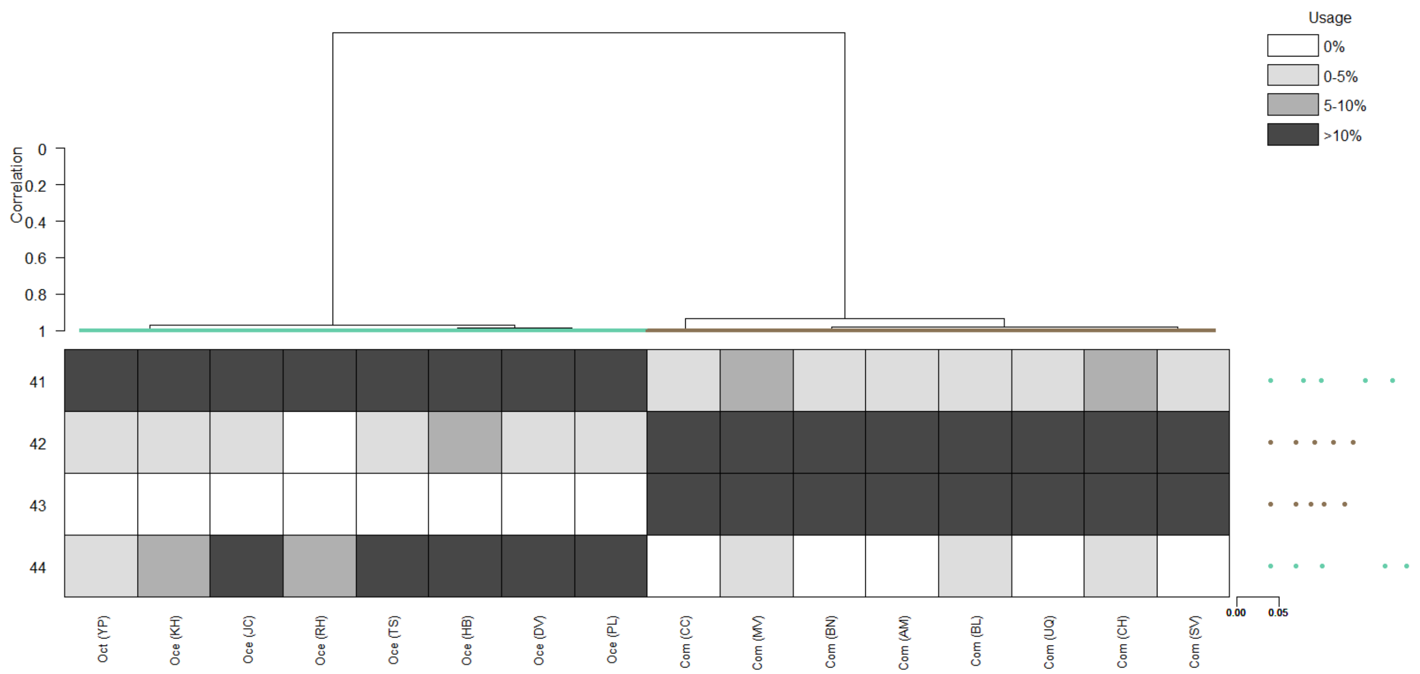
Trial 12 (critfact=26)



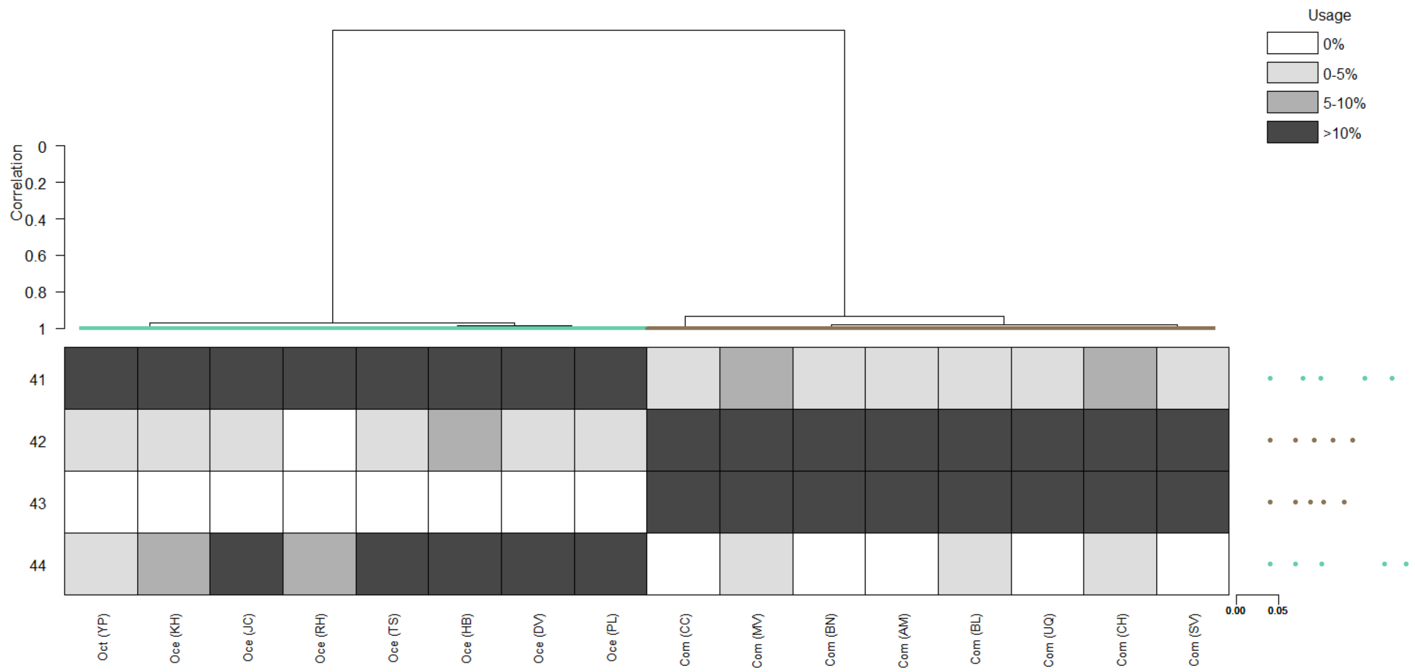
Trial 13 (minrep=3)



Trial 14 (minrep=5)



Trial 15 (minrep=7)



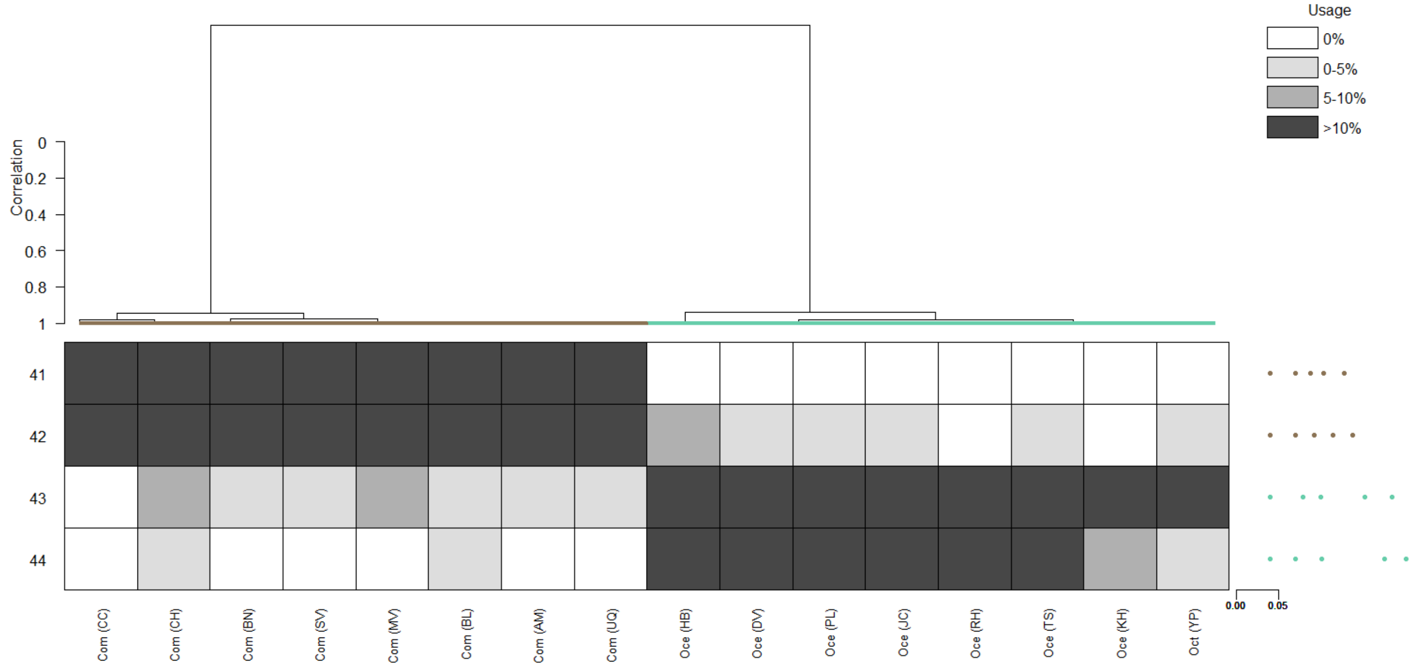
Trial 16 (minrep=9)

- No identity calls or clades delineated

Trial 17 (minrep=11)

- No identity calls or clades delineated

Trial 18 (baseline run 2)



Trial 19 (baseline run 3)